



US012403250B2

(12) **United States Patent**
Lawrence et al.

(10) **Patent No.:** **US 12,403,250 B2**

(45) **Date of Patent:** **Sep. 2, 2025**

(54) **INFUSION SET**

(71) Applicant: **Becton, Dickinson and Company**,
Franklin Lakes, NJ (US)

(72) Inventors: **Tyson Lawrence**, Cambridge, MA
(US); **James McGee**, Watertown, MA
(US); **Todd Taylor**, Cambridge, MA
(US); **Rob Colonna**, Boston, MA (US);
Eric Sugalski, Arlington, MA (US);
Zenas Lu, Cambridge, MA (US); **Chris**
K. Barmore, Arlington, MA (US);
Robert Banik, Hollywood, FL (US);
Peter Skutnik, Midland Park, NJ (US)

(73) Assignee: **Becton, Dickinson and Company**,
Franklin Lakes, NJ (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 648 days.

(21) Appl. No.: **17/167,565**

(22) Filed: **Feb. 4, 2021**

(65) **Prior Publication Data**

US 2021/0154397 A1 May 27, 2021

Related U.S. Application Data

(63) Continuation of application No. 13/138,209, filed as
application No. PCT/US2010/000145 on Jan. 21,
2010, now Pat. No. 10,933,191.

(Continued)

(51) **Int. Cl.**

A61M 39/08 (2006.01)

A61M 5/158 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **A61M 5/158** (2013.01); **A61M 5/427**
(2013.01); **A61M 39/08** (2013.01); **A61M**
5/002 (2013.01);

(Continued)

(58) **Field of Classification Search**

CPC .. **A61M 2005/1586**; **A61M 2005/1587**; **A61M**
39/0247; **A61M 2039/0258**;

(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,247,851 A 4/1966 Seibert
4,128,173 A 12/1978 Lazarus et al.

(Continued)

FOREIGN PATENT DOCUMENTS

JP 57-211353 A 12/1982
JP 61-293456 A 12/1986

(Continued)

Primary Examiner — Kevin C Sirmons

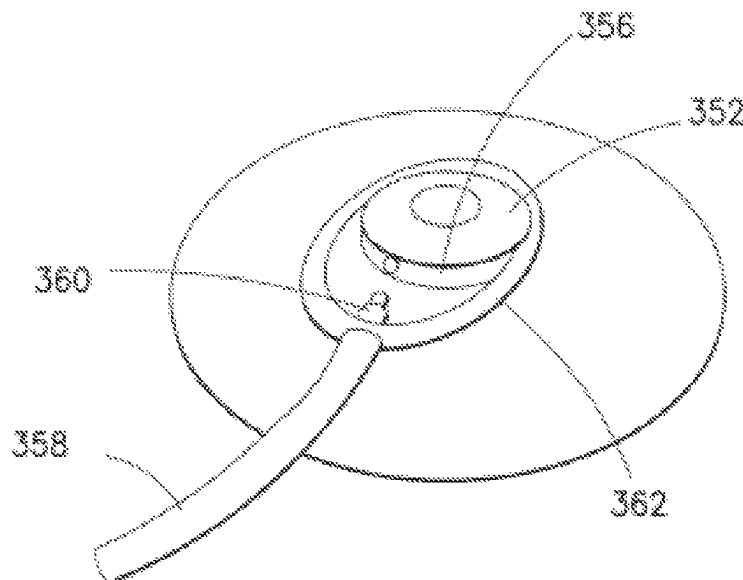
Assistant Examiner — William R Frehe

(74) *Attorney, Agent, or Firm* — Dickinson Wright PLLC

(57) **ABSTRACT**

An infusion set, including a set having a skin contacting surface, a catheter extending from the set, and a flexible ring having an integral fluid path therethrough. The flexible ring is configured to expand to a first diameter to allow placement over the set, and contract to a second diameter to securely circumscribe an outer diameter of the set. The infusion set also includes a tubing length secured to the flexible ring, and a fitting or spike disposed at an end of the tubing length and extending into the flexible ring. The fitting or spike is configured to sealably mate with an opening of the set to permit communication of a medicament therebetween.

12 Claims, 38 Drawing Sheets



Related U.S. Application Data

- (60) Provisional application No. 61/202,019, filed on Jan. 21, 2009.

(51) **Int. Cl.**

A61M 5/42 (2006.01)
A61M 5/00 (2006.01)
A61M 5/142 (2006.01)

(52) **U.S. Cl.**

CPC . *A61M 5/14244* (2013.01); *A61M 2005/1585* (2013.01); *A61M 2005/1586* (2013.01); *A61M 2005/1587* (2013.01); *A61M 2005/1588* (2013.01); *A61M 2205/586* (2013.01); *A61M 2209/088* (2013.01)

(58) **Field of Classification Search**

CPC *A61M 203/0261*; *A61M 2039/027*; *A61M 2039/0276*; *A61M 2039/0282*; *A61M 25/02*; *A61M 2025/0213*; *A61J 15/00*; *A61J 15/0015*; *A61J 15/0053*; *A61J 15/0061*

See application file for complete search history.

(56)

References Cited**U.S. PATENT DOCUMENTS**

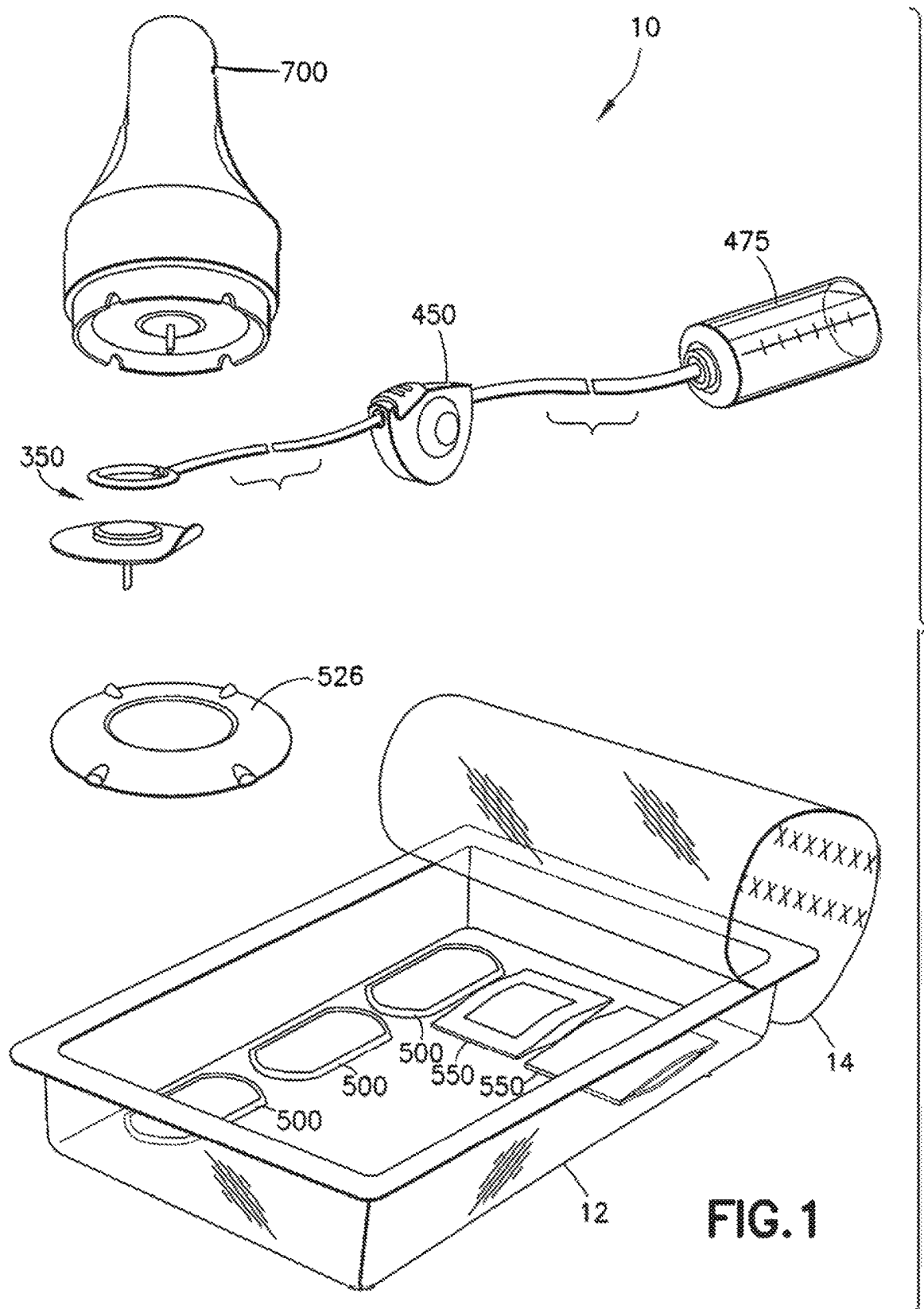
4,216,860 A 8/1980 Heimann
 4,335,770 A * 6/1982 Kulle A61J 1/10
 383/29
 4,523,679 A 6/1985 Paikoff et al.
 4,588,398 A 5/1986 Daugherty et al.
 4,605,399 A 8/1986 Weston et al.
 4,743,231 A 5/1988 Kay et al.
 4,755,173 A 7/1988 Konopka et al.
 4,769,893 A * 9/1988 Cassidy F16H 25/24
 29/418
 5,048,684 A 9/1991 Scott
 5,188,588 A * 2/1993 Schoendorfer A61M 1/302
 604/6.07
 5,226,427 A 7/1993 Buckberg et al.
 5,292,325 A 3/1994 Gurmarnik
 5,318,543 A 6/1994 Ross et al.
 5,417,671 A 5/1995 Jackson
 5,474,181 A 12/1995 Shillington et al.
 5,582,599 A 12/1996 Daneshvar
 5,620,419 A 4/1997 Lui
 5,797,882 A 8/1998 Purdy et al.
 5,797,954 A 8/1998 Shaffer et al.
 5,911,707 A 6/1999 Wolvek et al.
 5,925,017 A 7/1999 Kriesel et al.
 6,461,644 B1 10/2002 Jackson et al.
 6,475,196 B1 11/2002 Vachon

6,575,954 B1 6/2003 Ravizza
 6,736,797 B1 5/2004 Larsen et al.
 6,749,588 B1 6/2004 Howell et al.
 6,959,812 B2 11/2005 Reif
 7,207,974 B2 4/2007 Safabash
 7,981,085 B2 7/2011 Ethelfeld
 8,945,057 B2 2/2015 Gyrr
 2002/0022855 A1 2/2002 Bobroff et al.
 2002/0077599 A1 6/2002 Wojcik
 2002/0095138 A1 7/2002 Lynch et al.
 2002/0156434 A1 10/2002 Antwerp et al.
 2002/0177816 A1 * 11/2002 Brimhall A61M 25/0637
 604/174
 2004/0168944 A1 9/2004 Massengale et al.
 2005/0013957 A1 1/2005 Leschinsky
 2005/0107743 A1 5/2005 Fangrow, Jr.
 2006/0030815 A1 * 2/2006 Csincsur A61M 39/1011
 604/93.01
 2006/0065772 A1 3/2006 Grant et al.
 2006/0129090 A1 6/2006 Moberg et al.
 2006/0173410 A1 * 8/2006 Moberg A61M 25/02
 604/93.01
 2006/0282045 A1 12/2006 Wilkinson et al.
 2007/0021729 A1 1/2007 Mogensen
 2007/0299409 A1 12/2007 Whitbourne et al.
 2008/0103367 A1 5/2008 Burba
 2008/0215003 A1 9/2008 Komerup et al.
 2008/0243051 A1 10/2008 DeStefano
 2008/0243084 A1 * 10/2008 DeStefano A61M 5/158
 604/180
 2008/0243085 A1 10/2008 DeStefano
 2008/0255516 A1 10/2008 Yodfat et al.
 2008/0295960 A1 12/2008 Schallau, II et al.

FOREIGN PATENT DOCUMENTS

JP 64-005566 A 1/1989
 JP 3-240664 A 10/1991
 JP 2000-279508 10/2000
 JP 2003-527138 A 9/2003
 JP 2005-503242 A 2/2005
 JP 2005-506110 A 3/2005
 JP 2007-511325 A 5/2007
 JP 2008-501483 A 1/2008
 WO 1997-028750 A1 8/1997
 WO 97/42901 11/1997
 WO 99/33504 7/1999
 WO 02/083021 A1 10/2002
 WO 2004101071 A2 11/2004
 WO 2005079441 A2 9/2005
 WO 2006097111 A2 9/2006
 WO 2007093182 A2 8/2007
 WO 2008014791 A1 2/2008
 WO 2009070896 A1 6/2009
 WO 2010080715 A1 7/2010

* cited by examiner



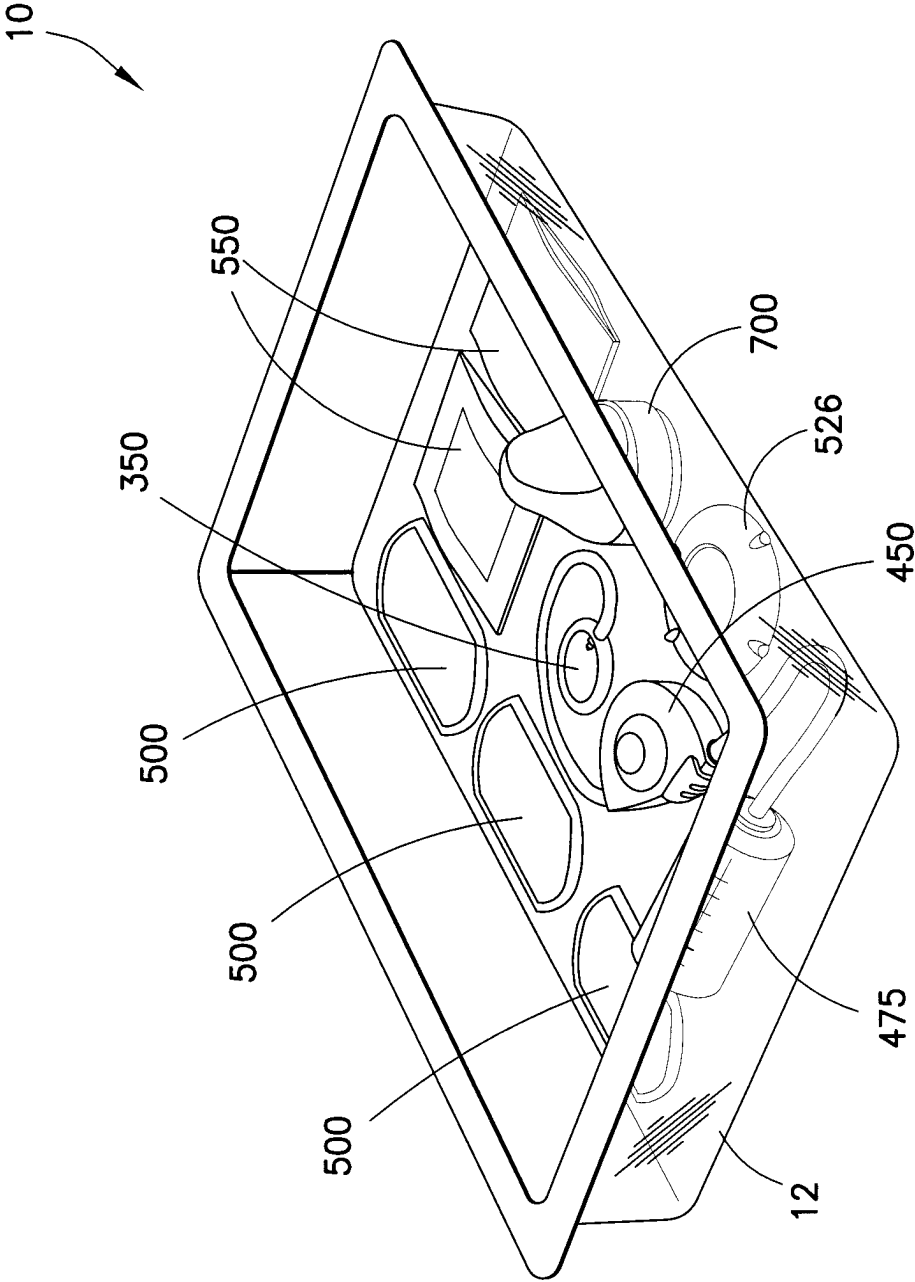


FIG. 2

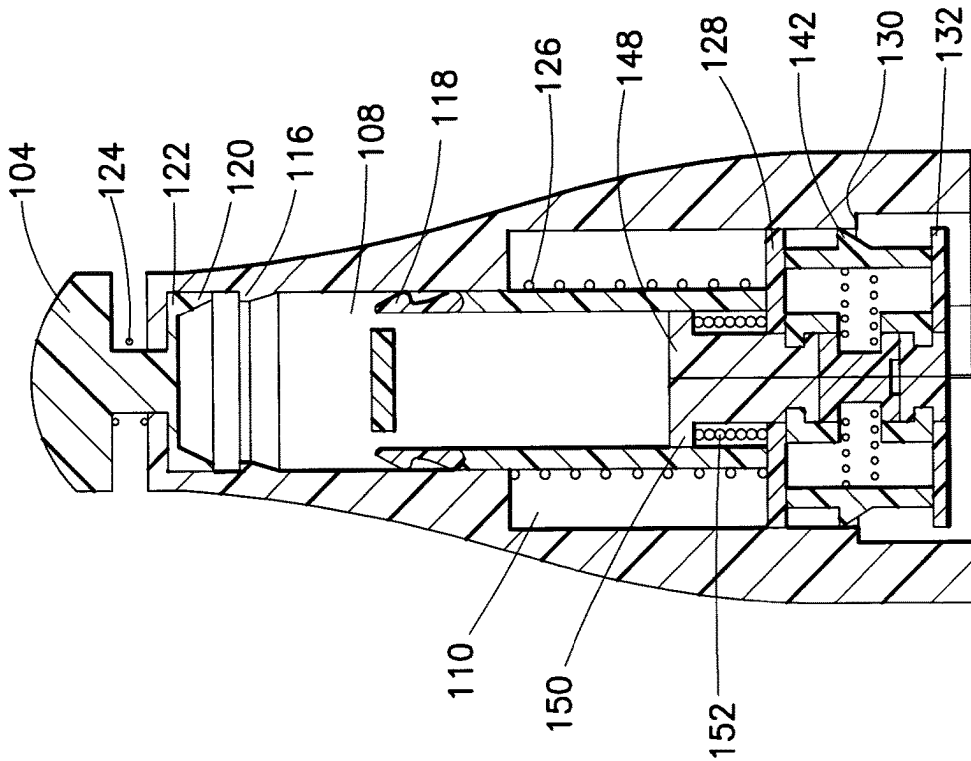


FIG. 3b

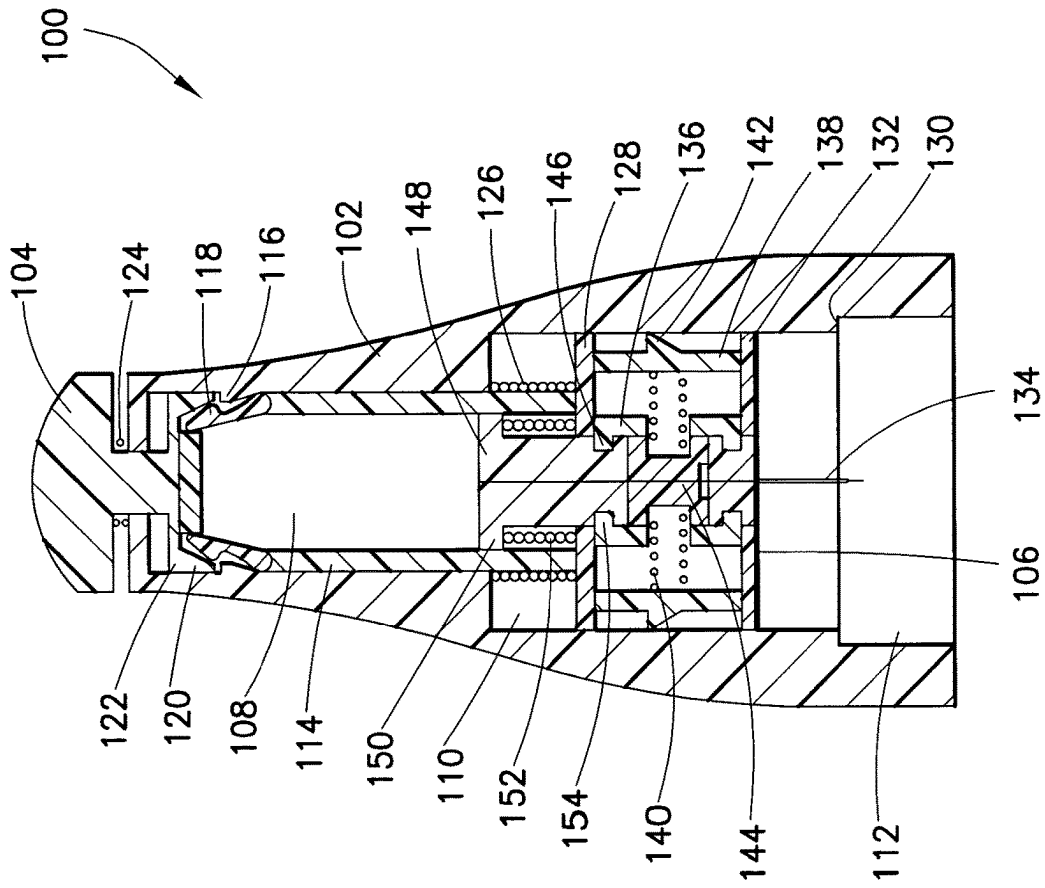


FIG. 3a

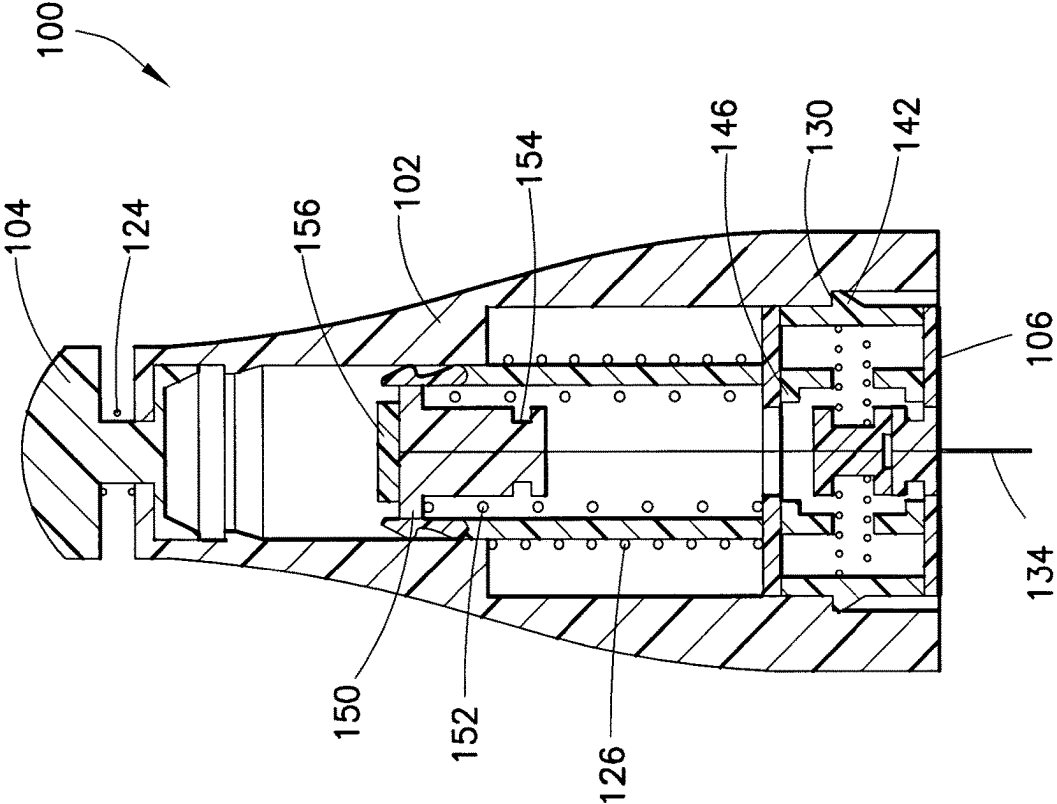


FIG. 3c

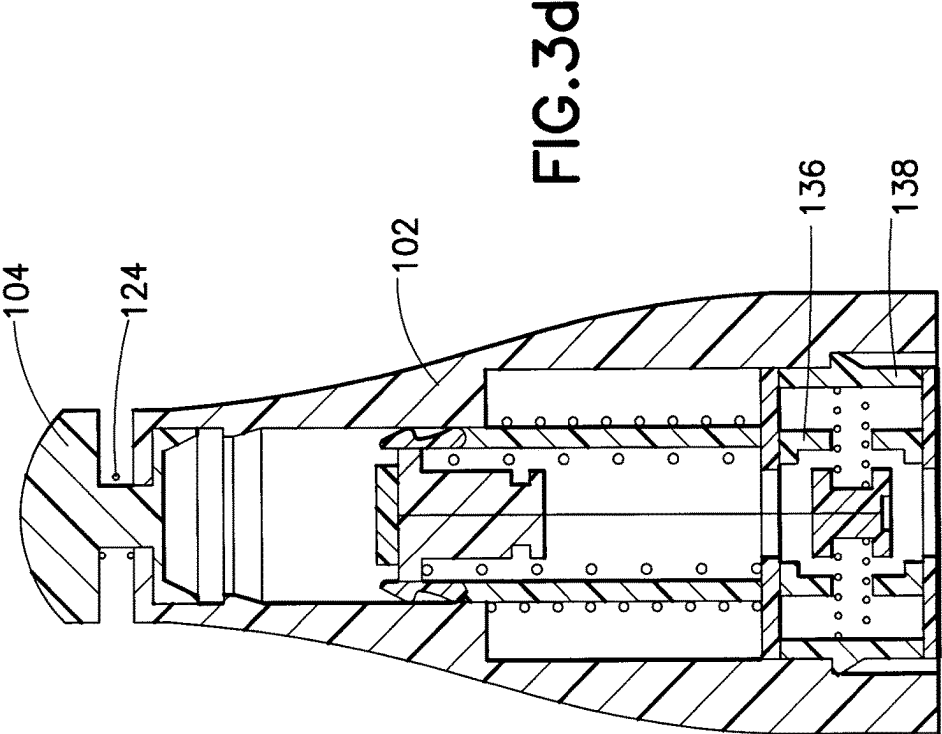


FIG. 3d

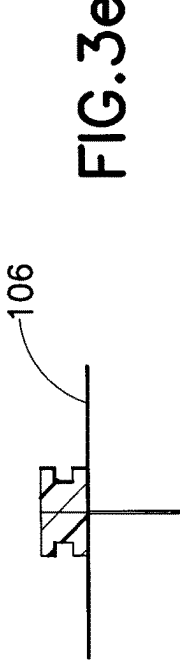


FIG. 3e

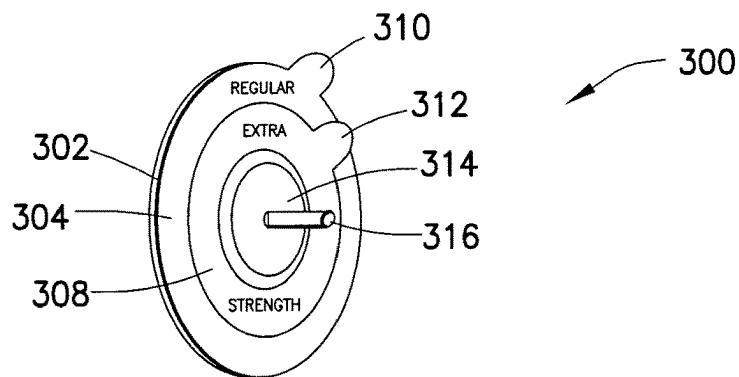


FIG. 4a

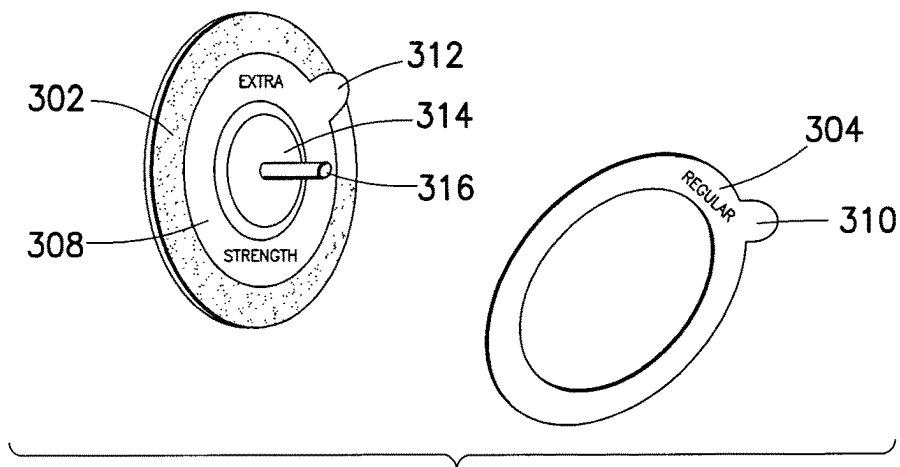


FIG. 4b

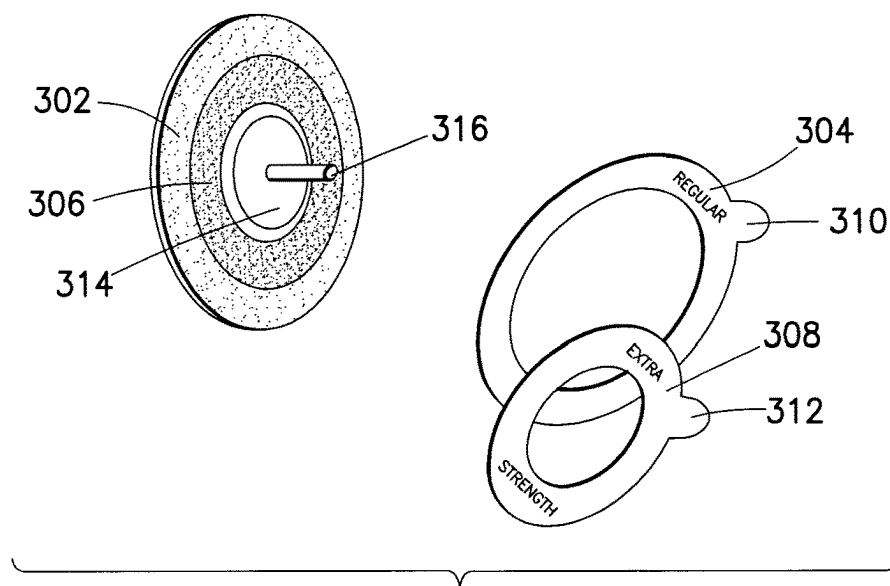


FIG. 4c

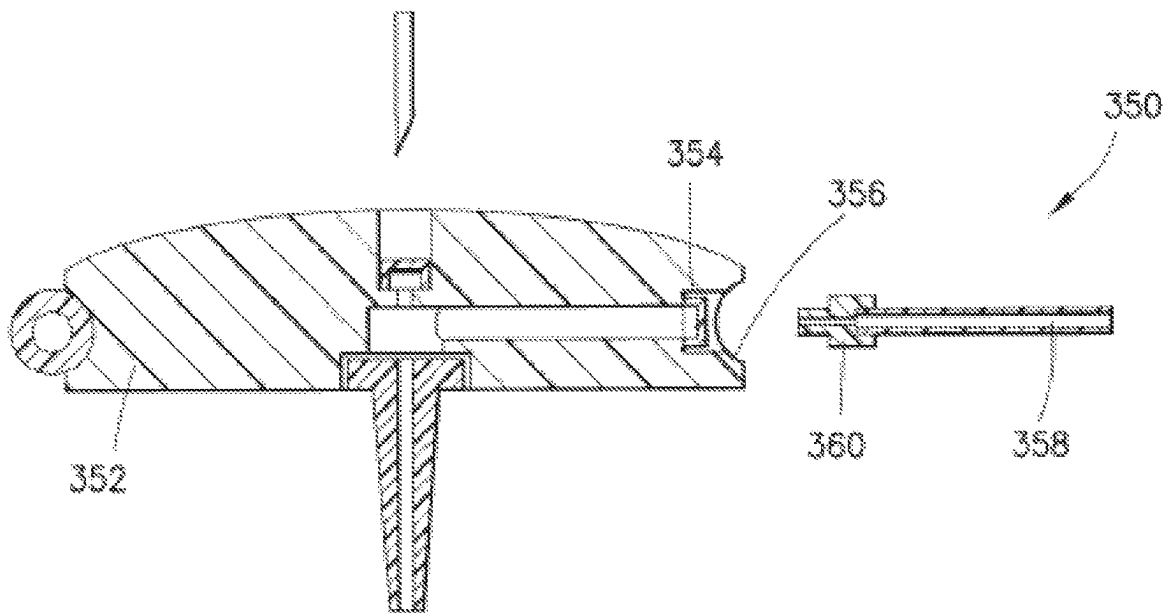


FIG. 5a

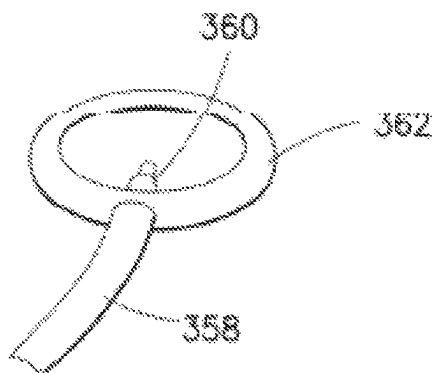


FIG. 5b

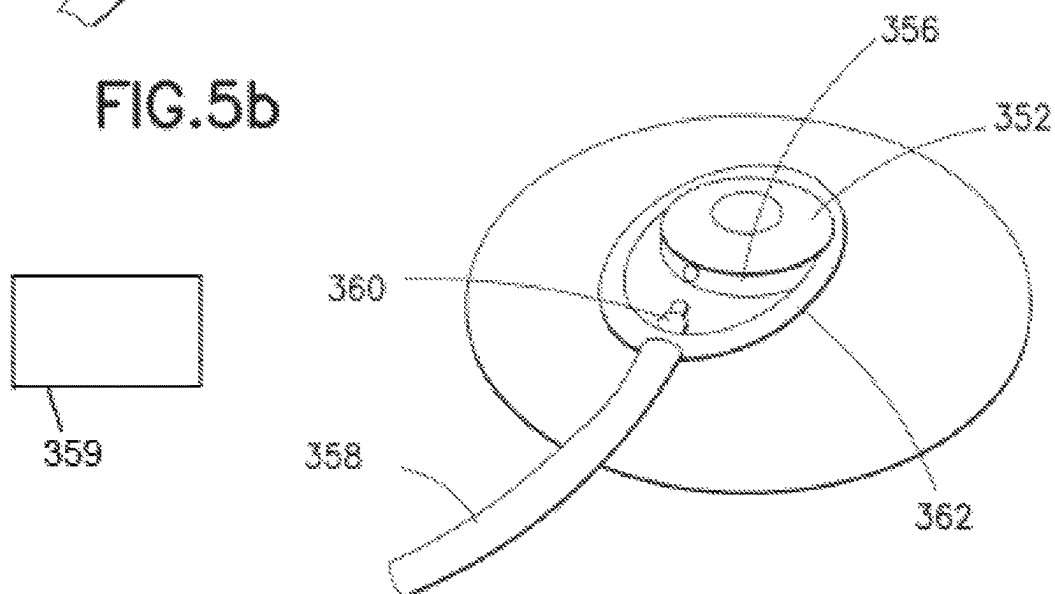


FIG. 5c

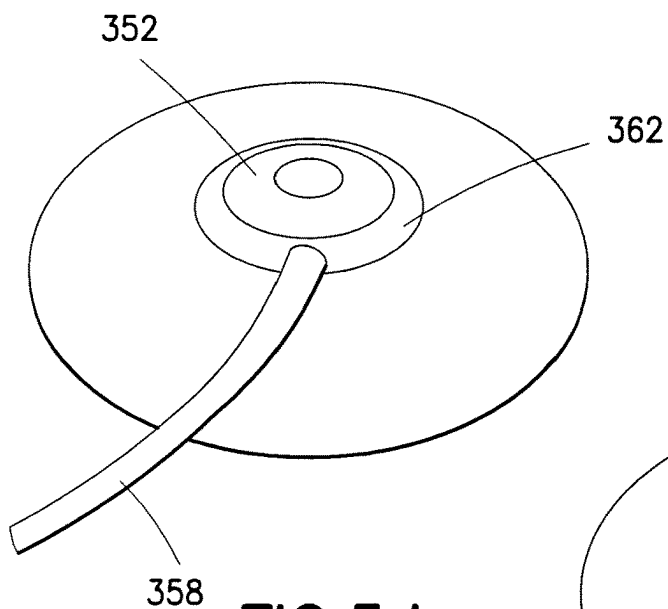


FIG. 5d

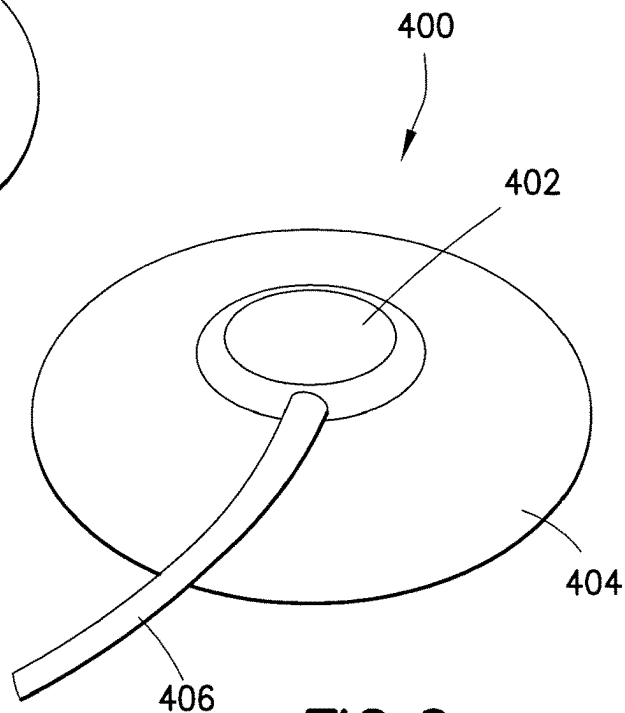


FIG. 6a

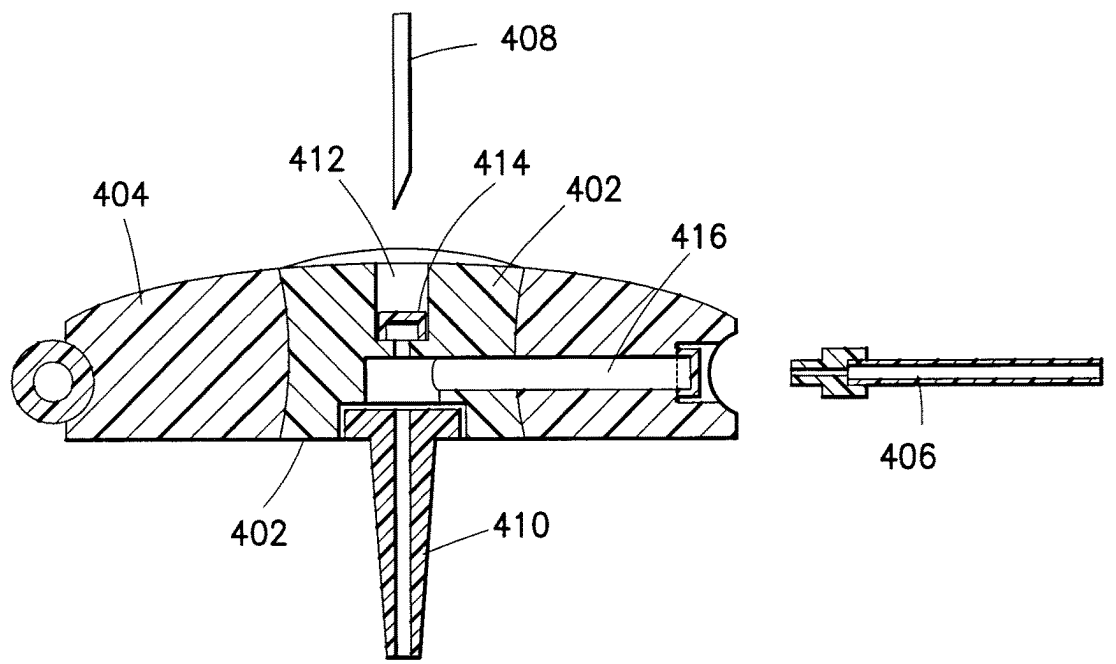


FIG. 6b

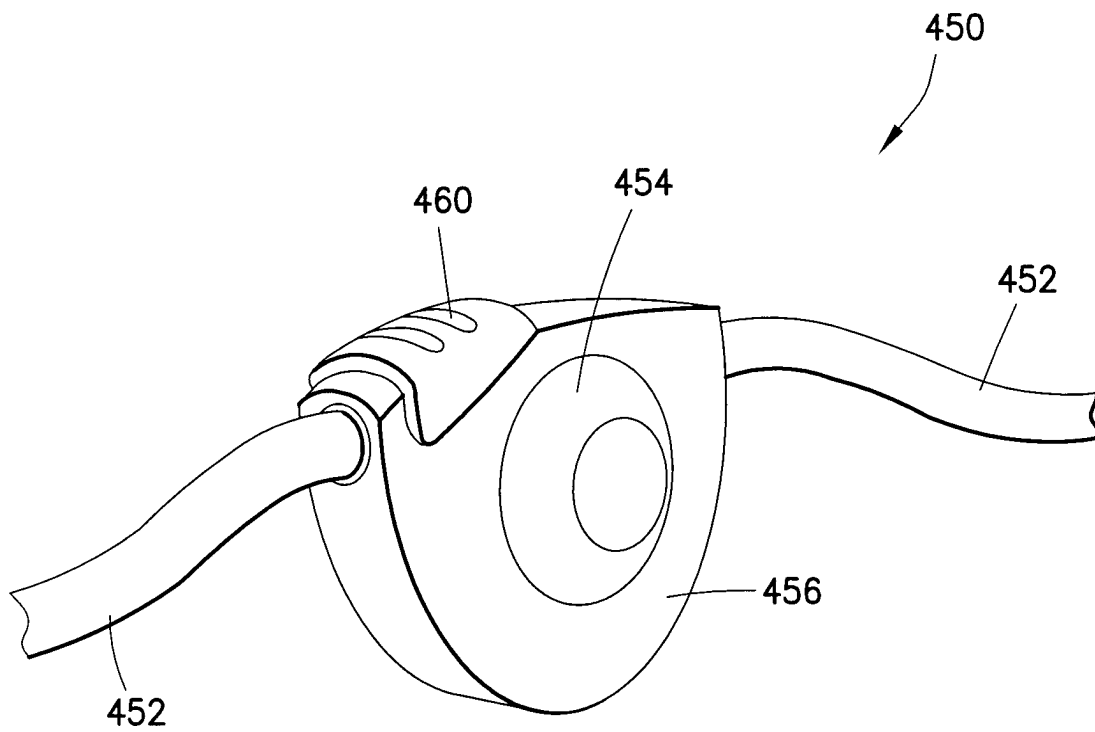


FIG. 7a

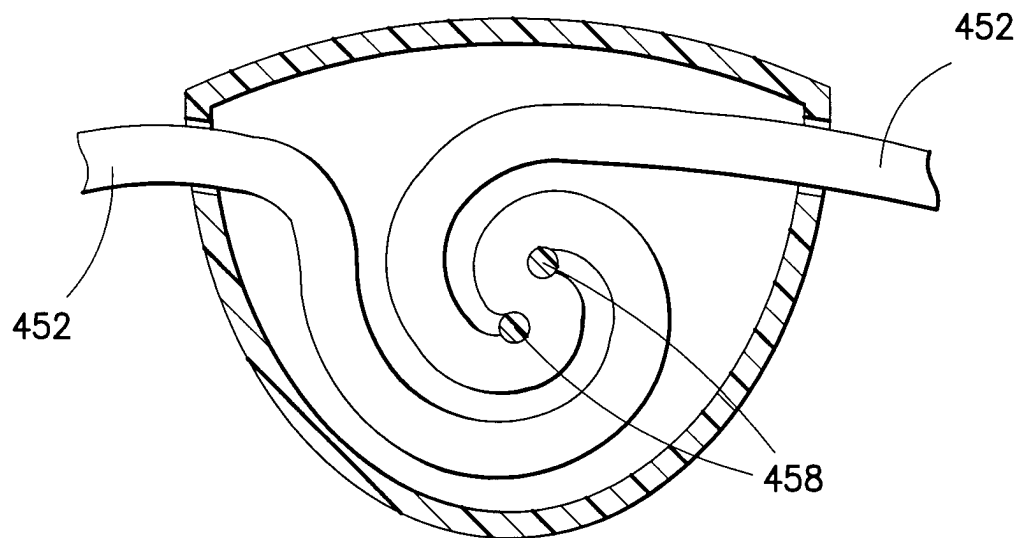


FIG. 7b

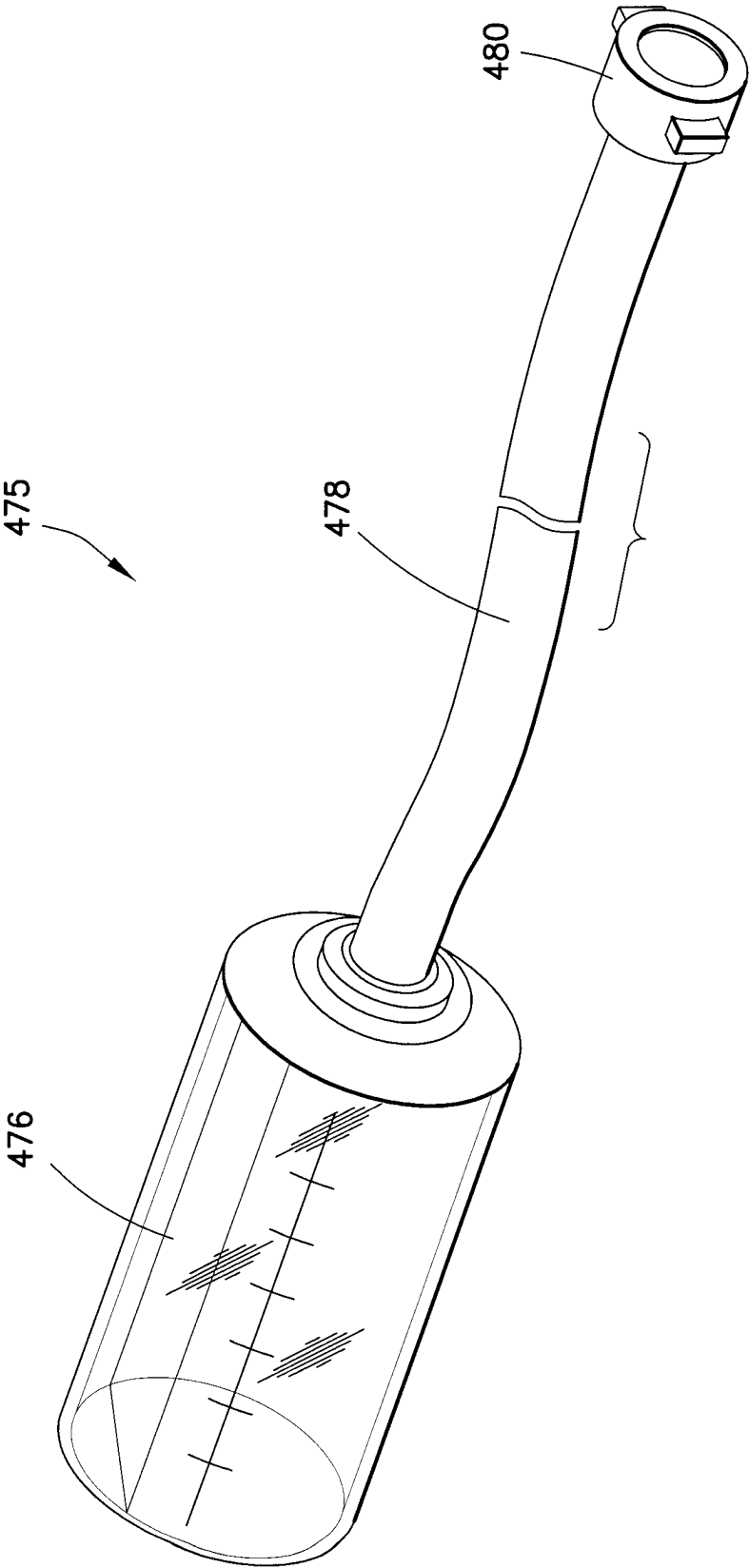


FIG. 8

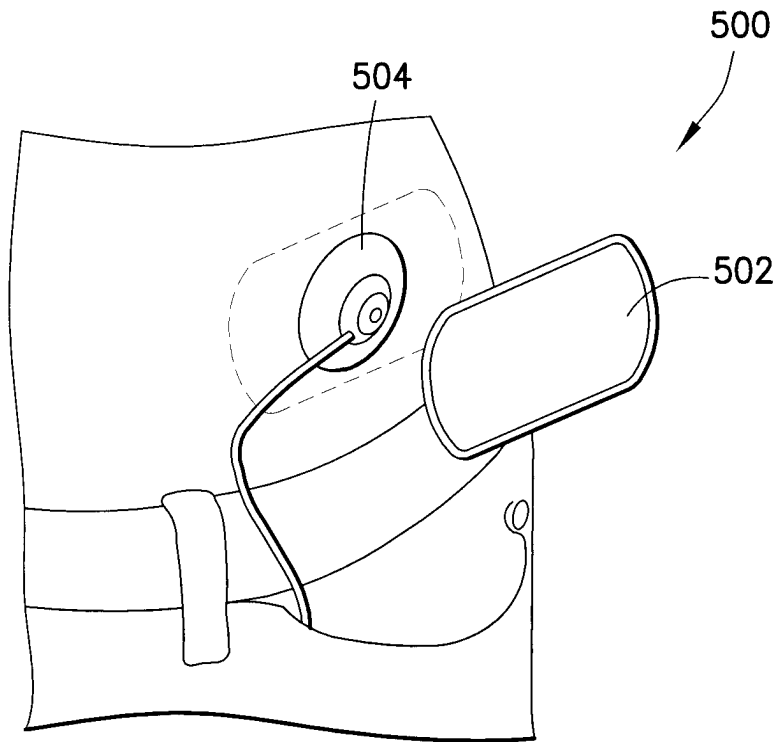


FIG. 9a

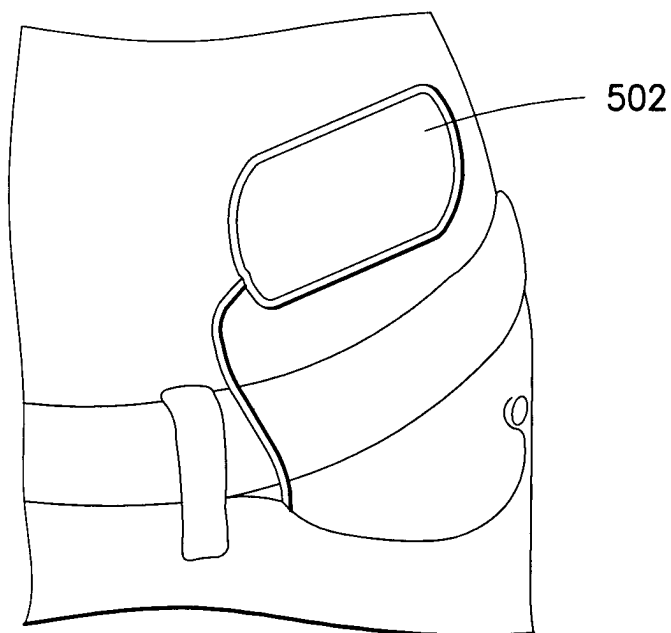


FIG. 9b

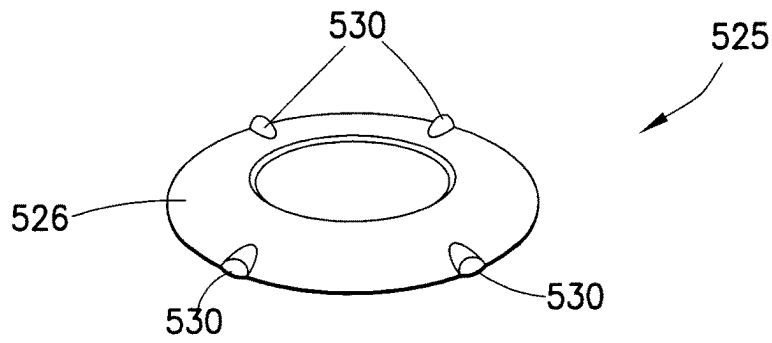


FIG. 10a

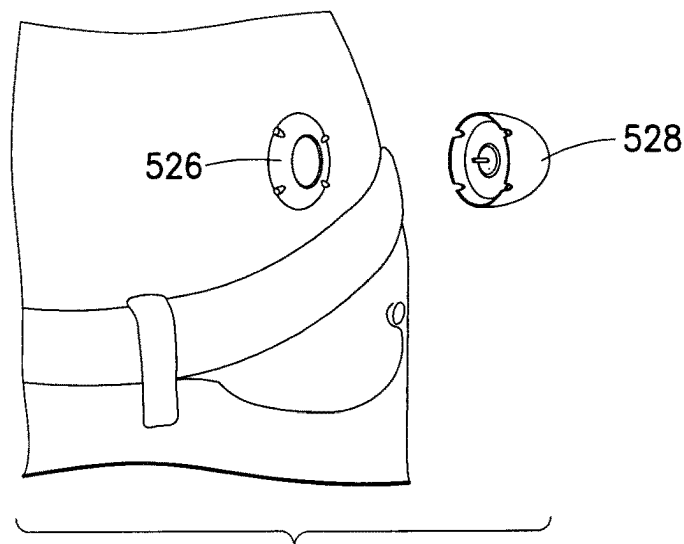


FIG. 10b

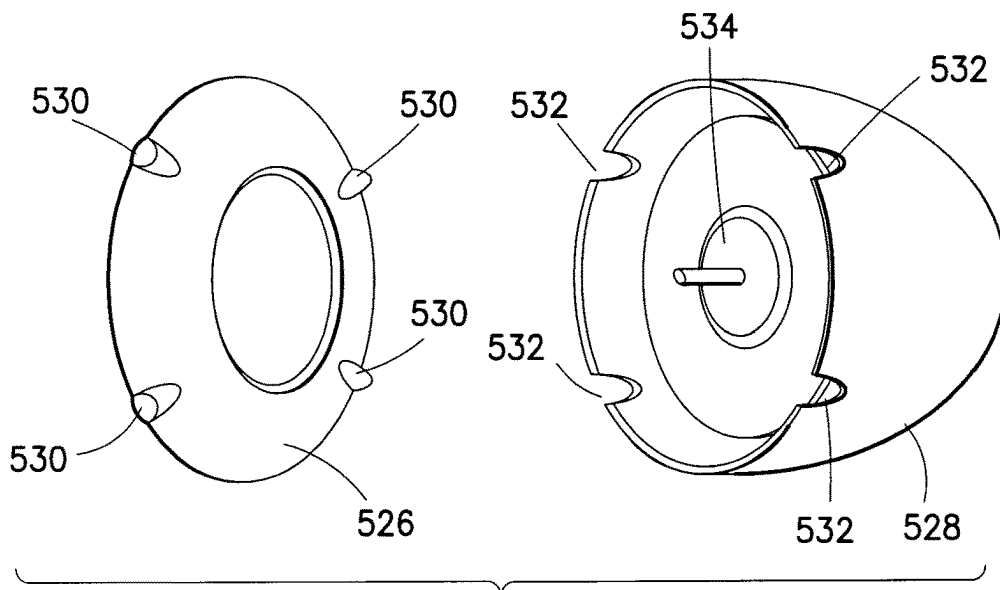


FIG. 10c

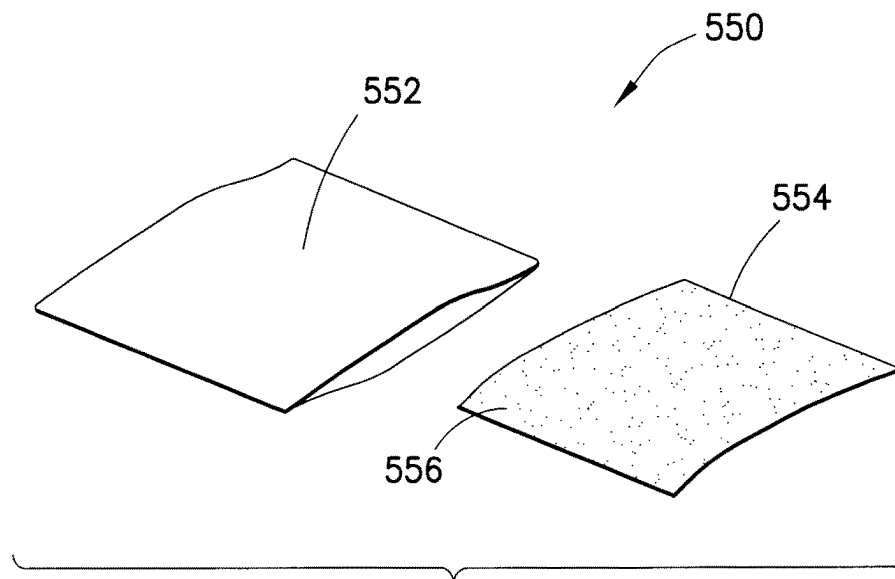


FIG. 11

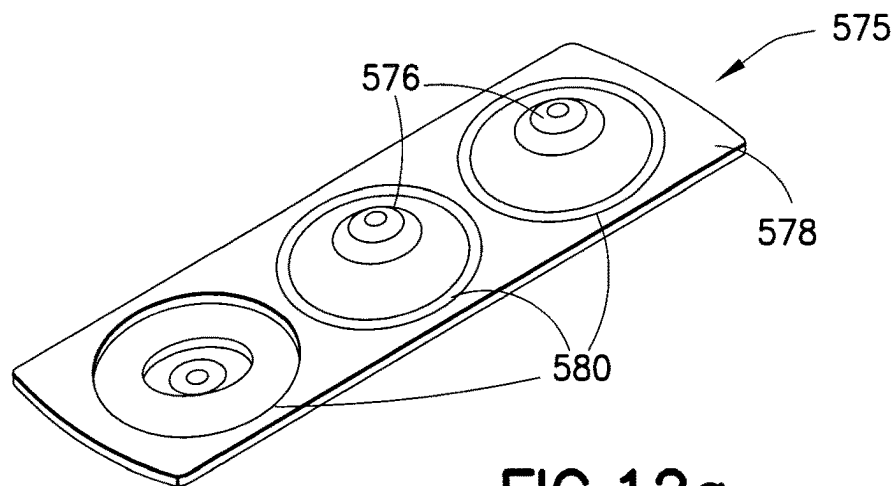


FIG. 12a

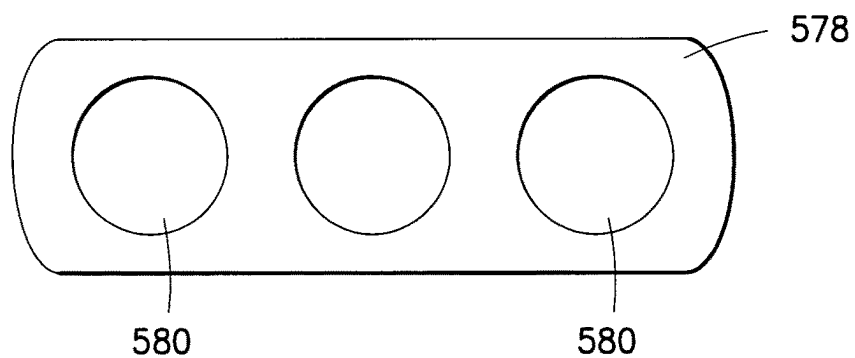


FIG. 12b

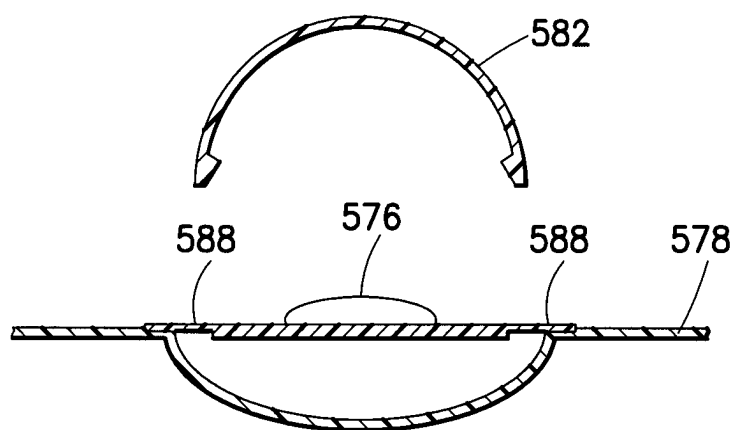


FIG. 12c

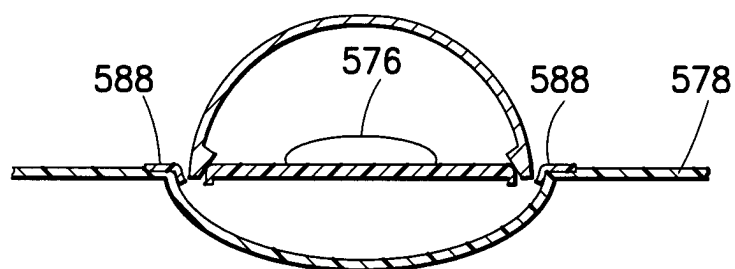


FIG. 12d

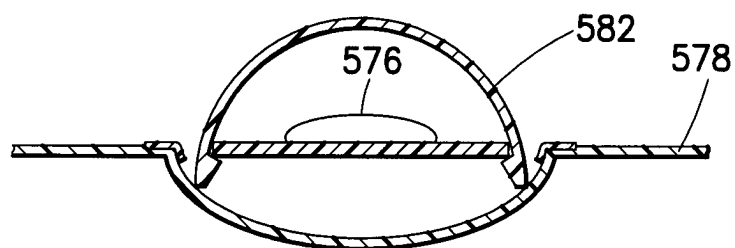


FIG. 12e

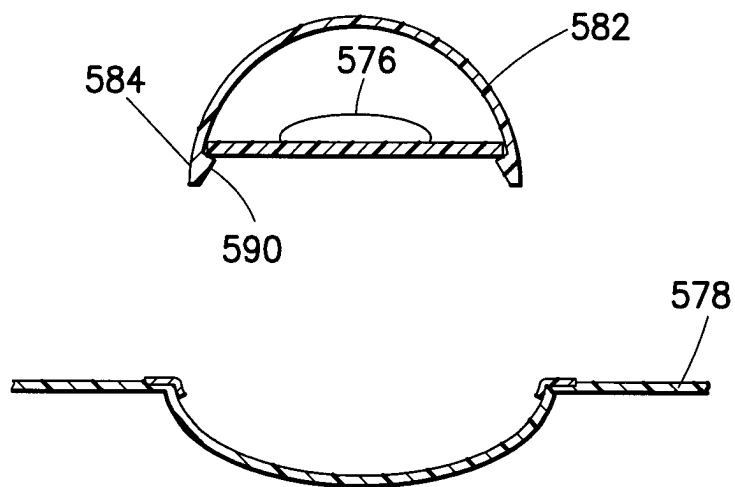


FIG. 12f

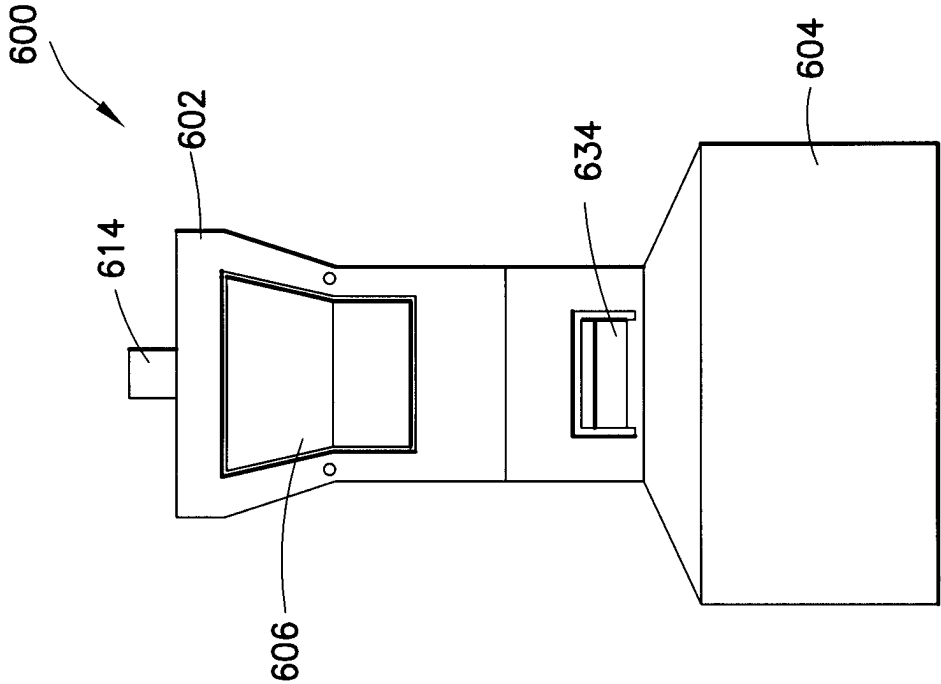


FIG. 13a

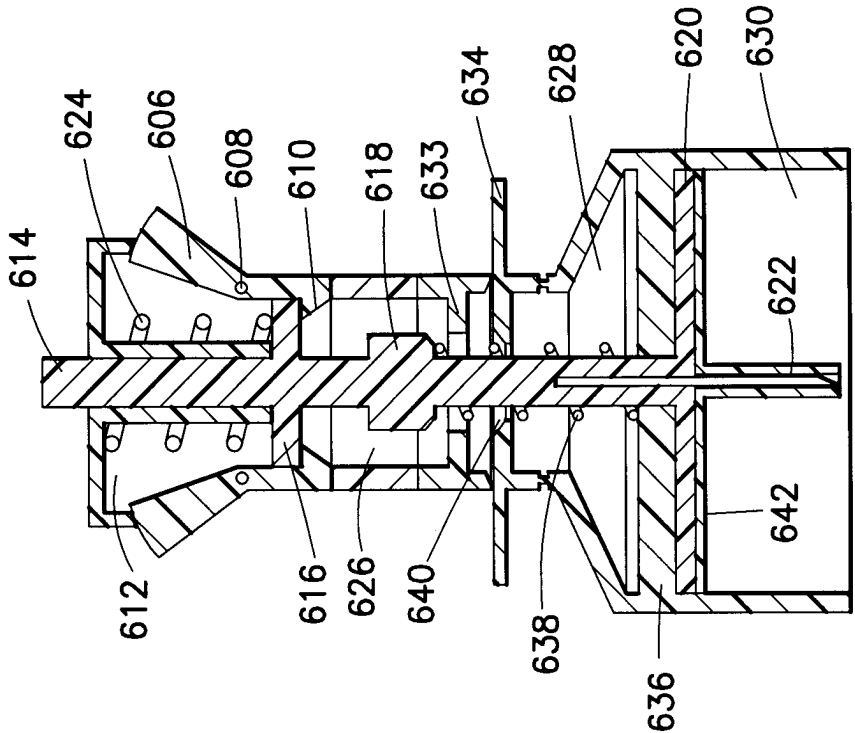


FIG. 13b

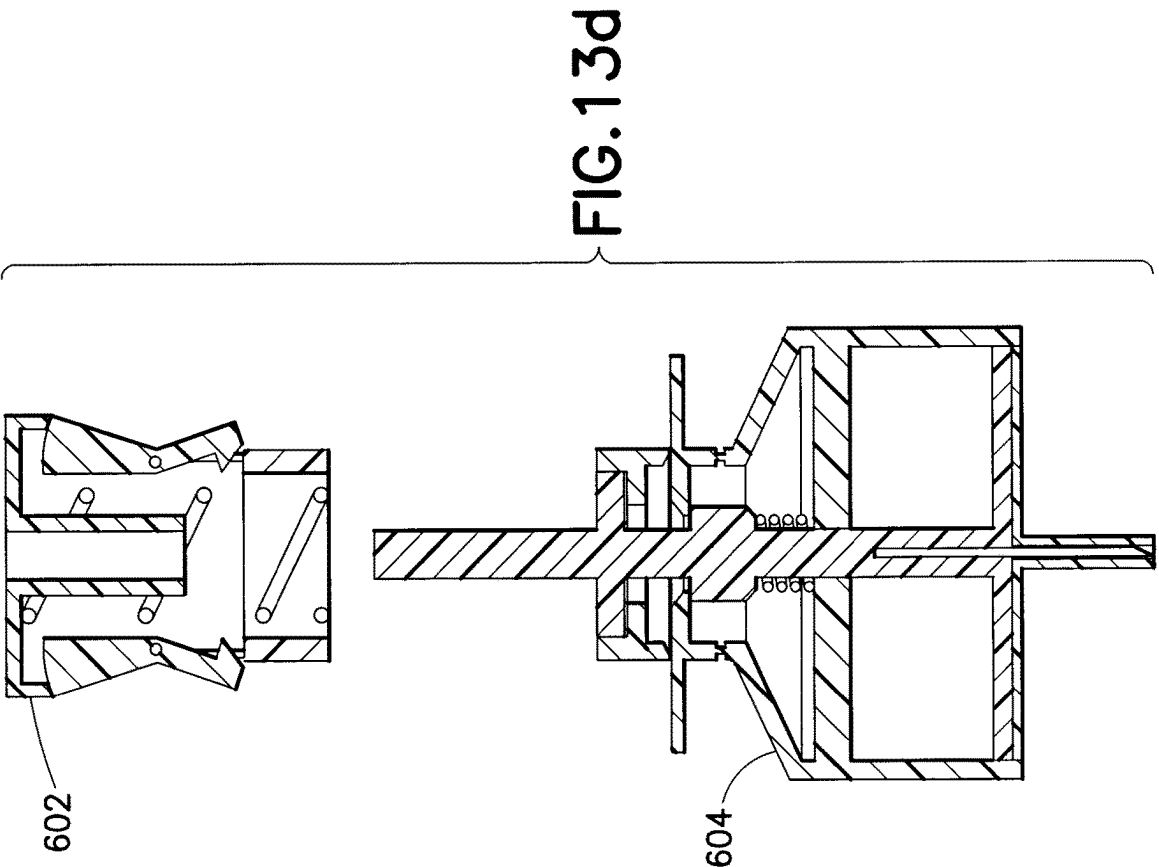


FIG. 13d

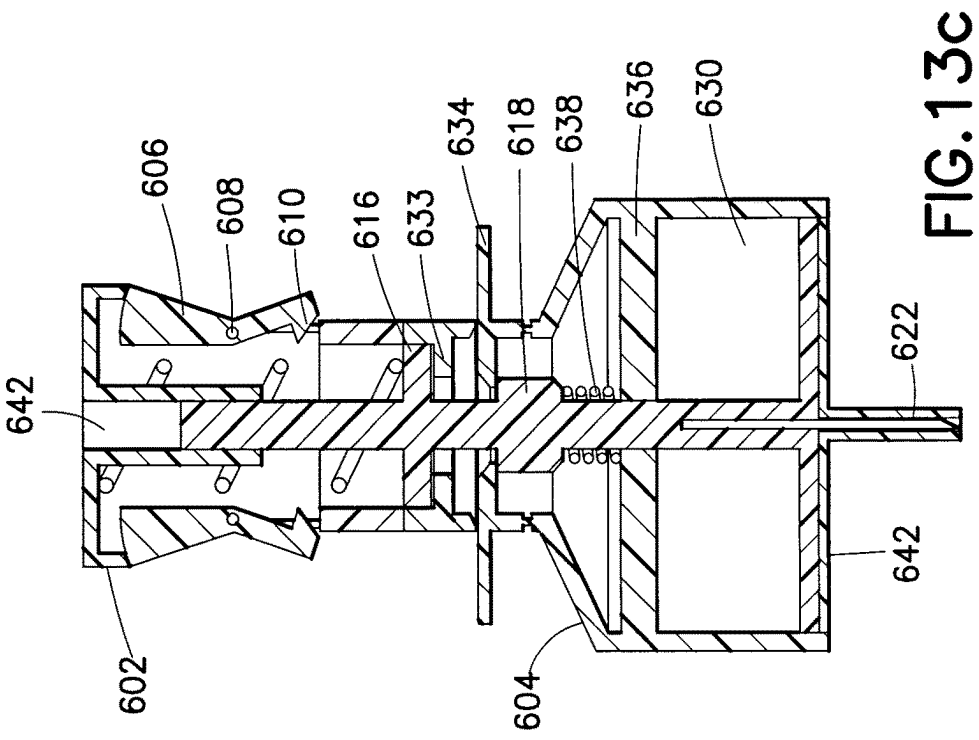


FIG. 13c

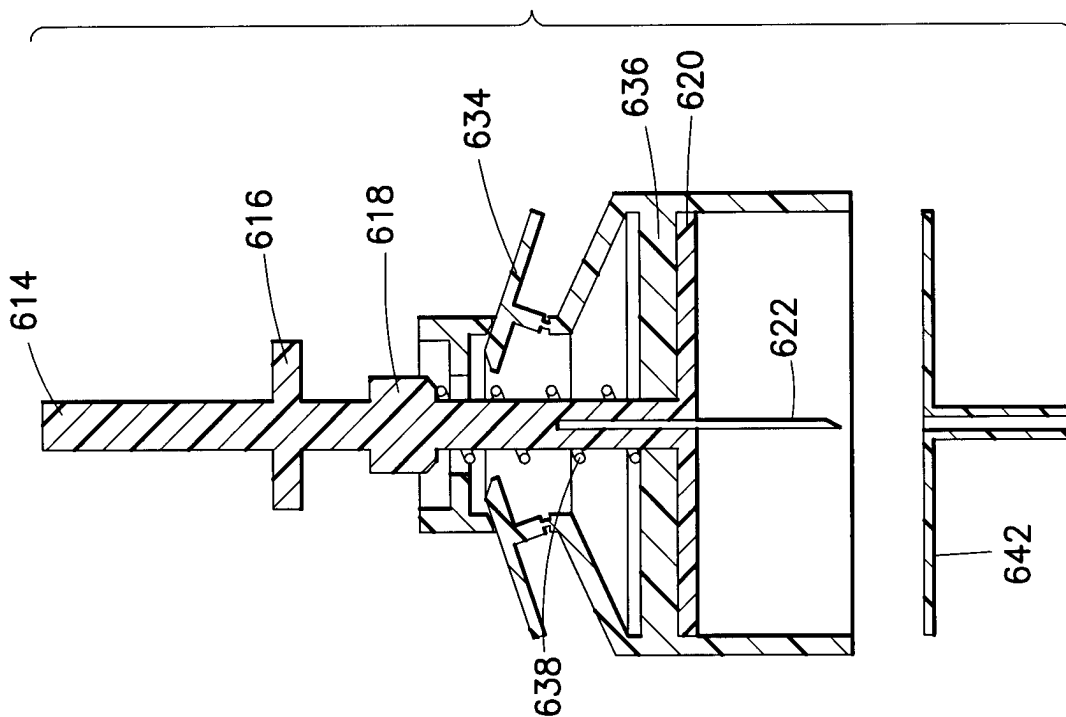


FIG. 13f

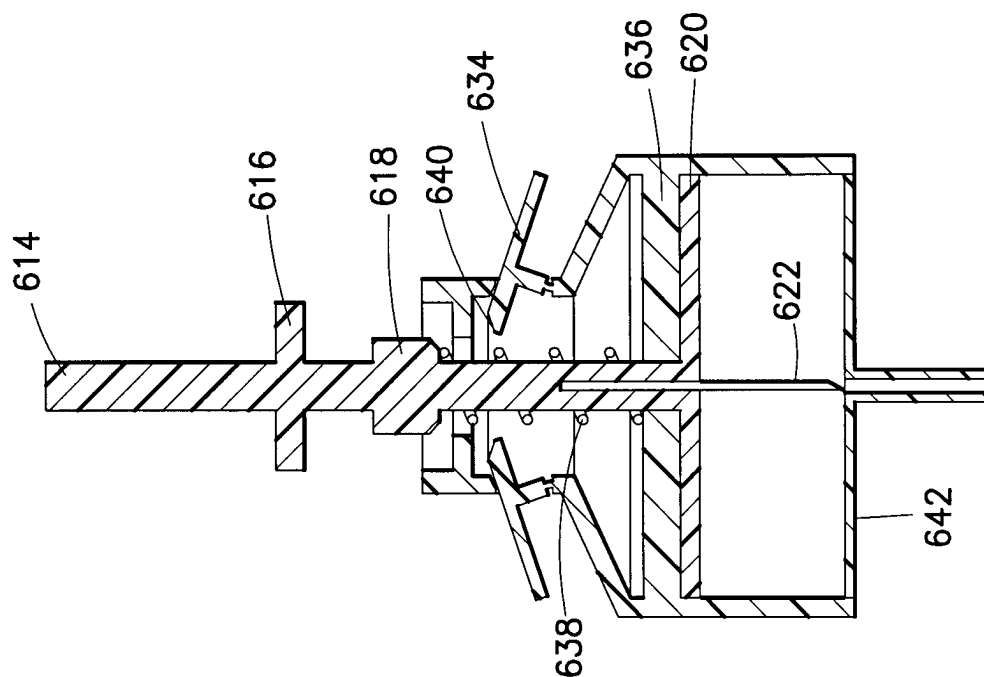


FIG. 13e

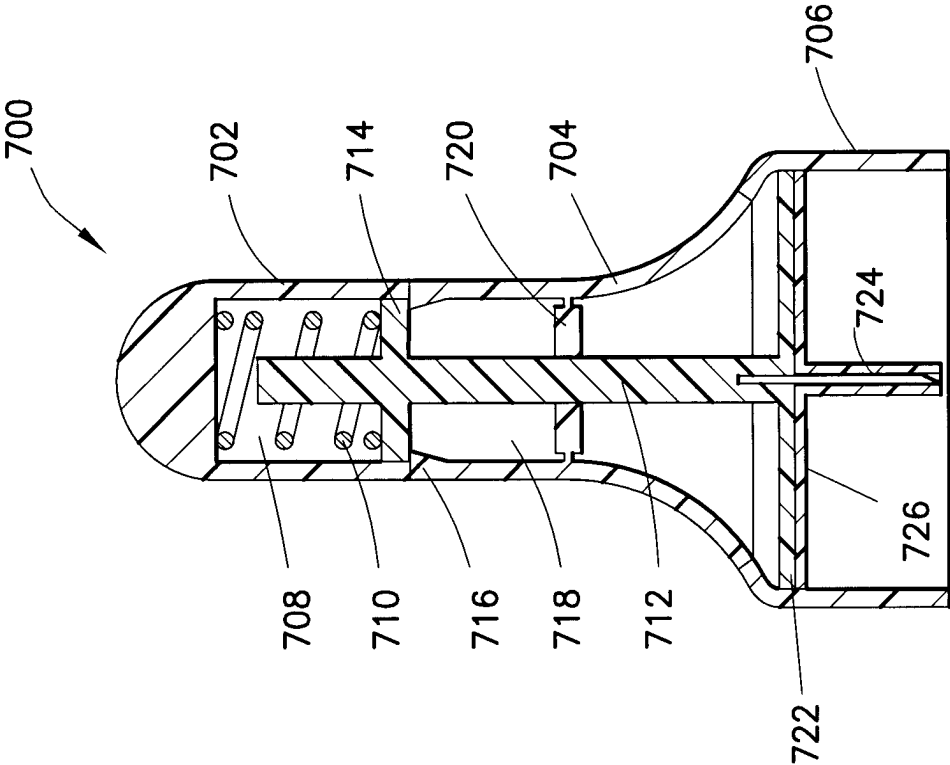


FIG. 14a

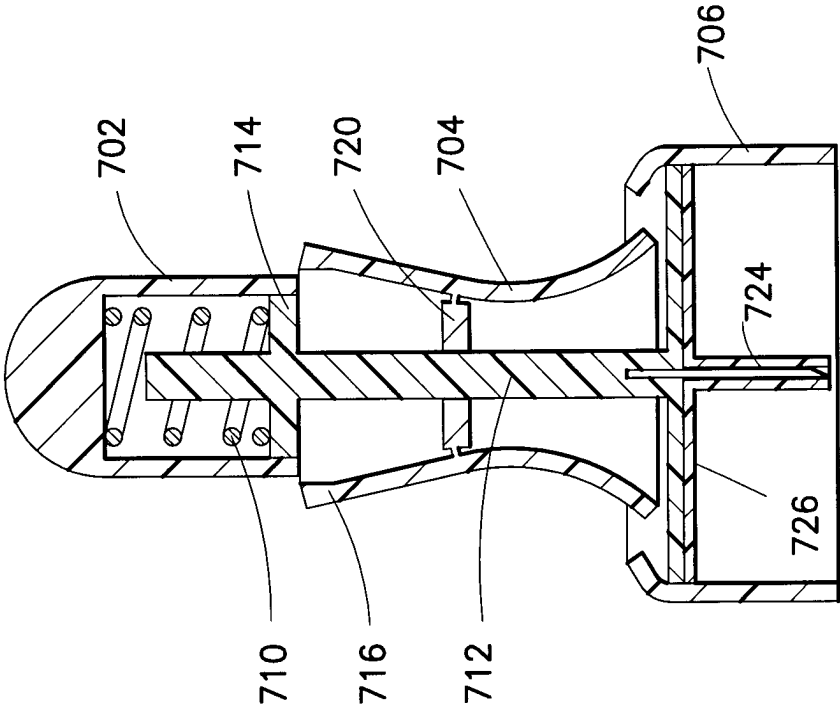
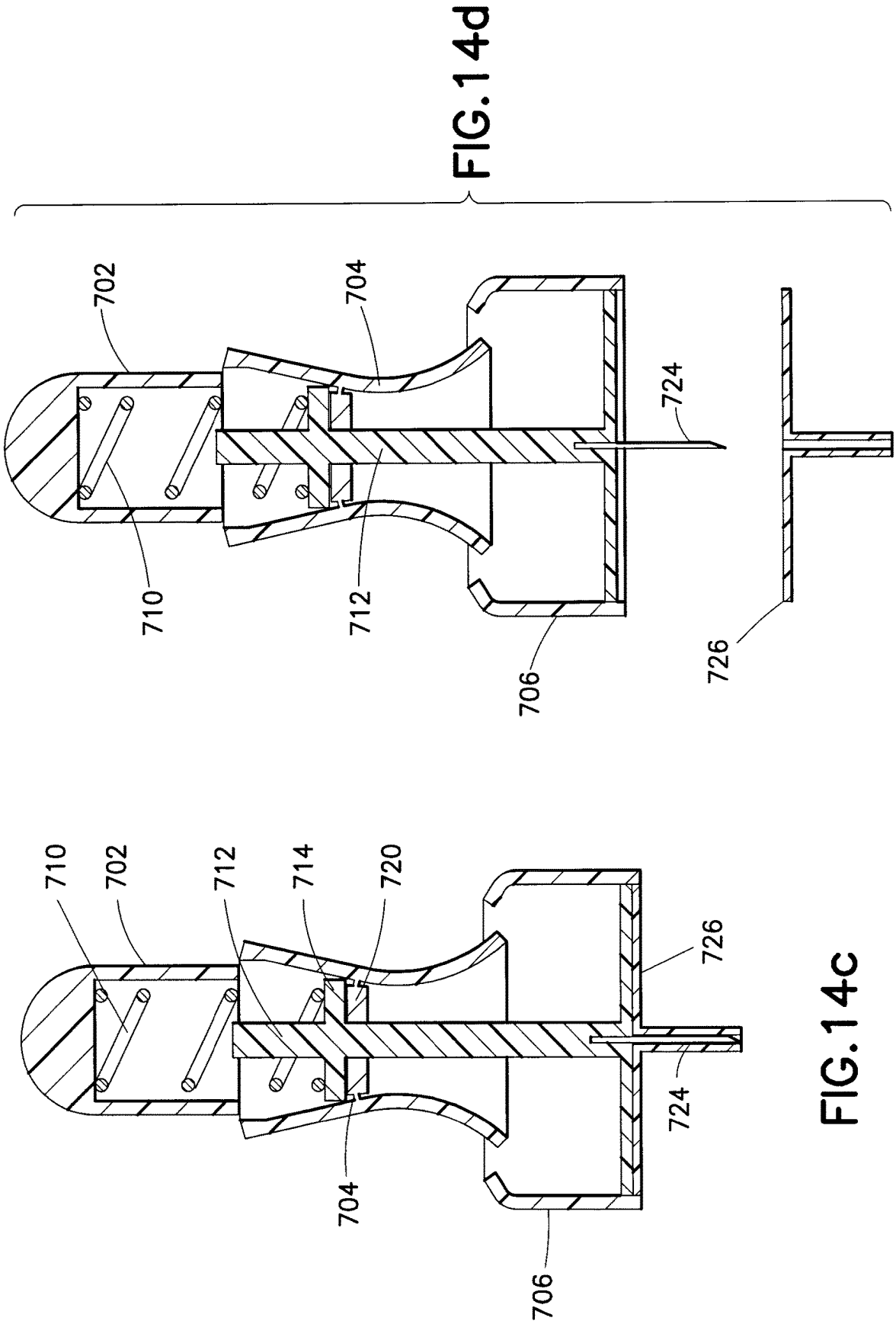


FIG. 14b



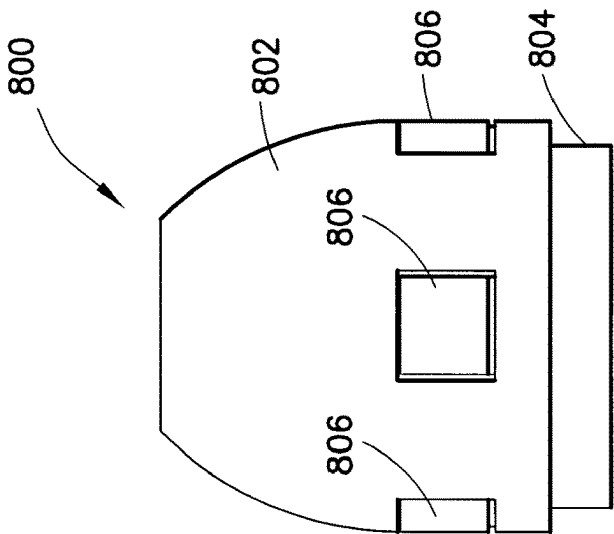


FIG. 15a

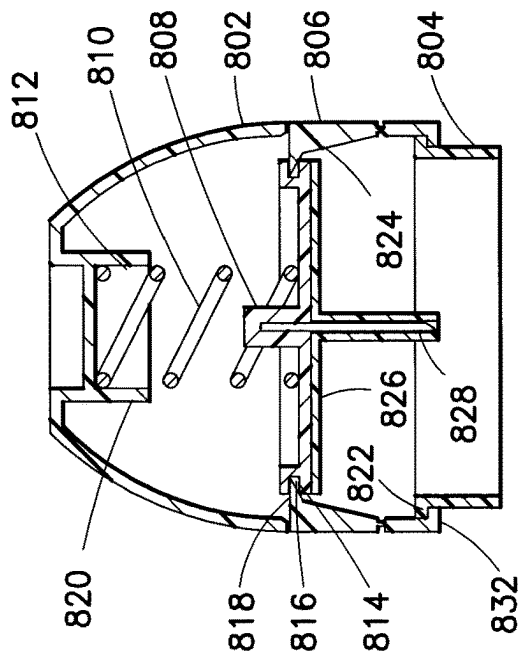


FIG. 15b

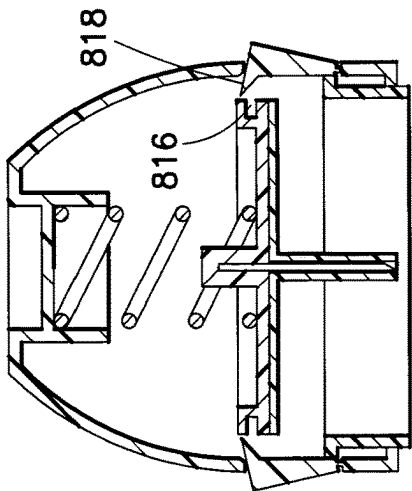


FIG. 15c

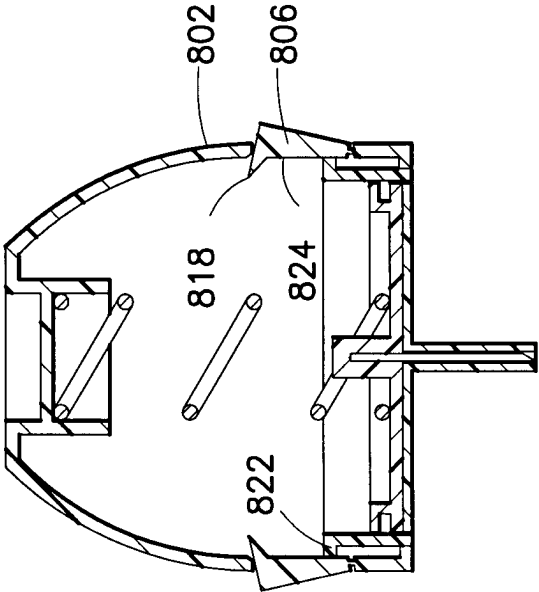


FIG. 15d

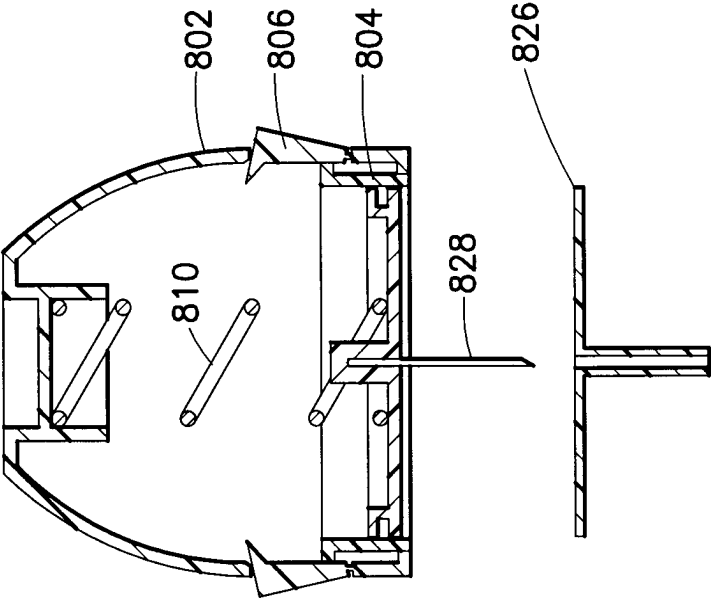


FIG. 15e

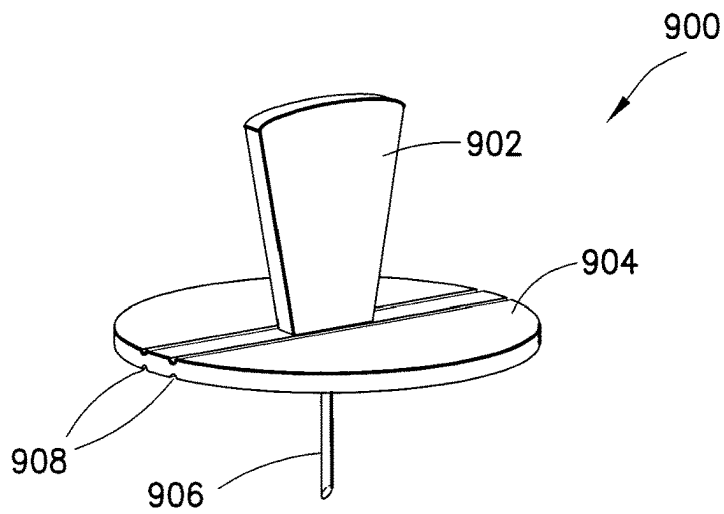


FIG. 16a

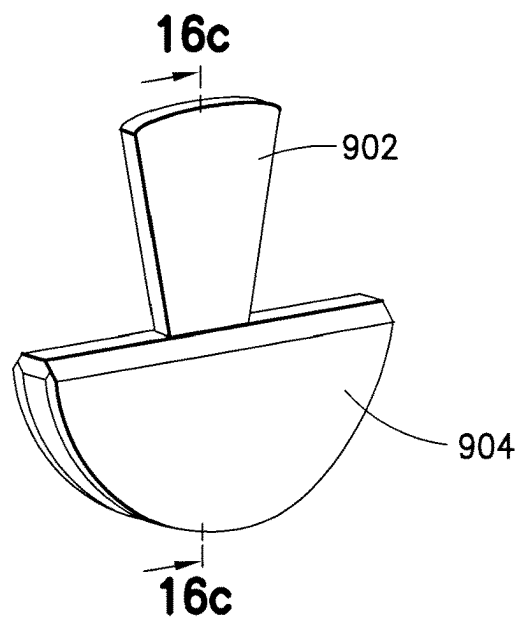


FIG. 16b

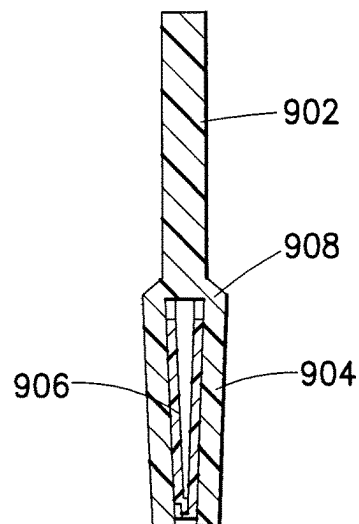


FIG. 16c

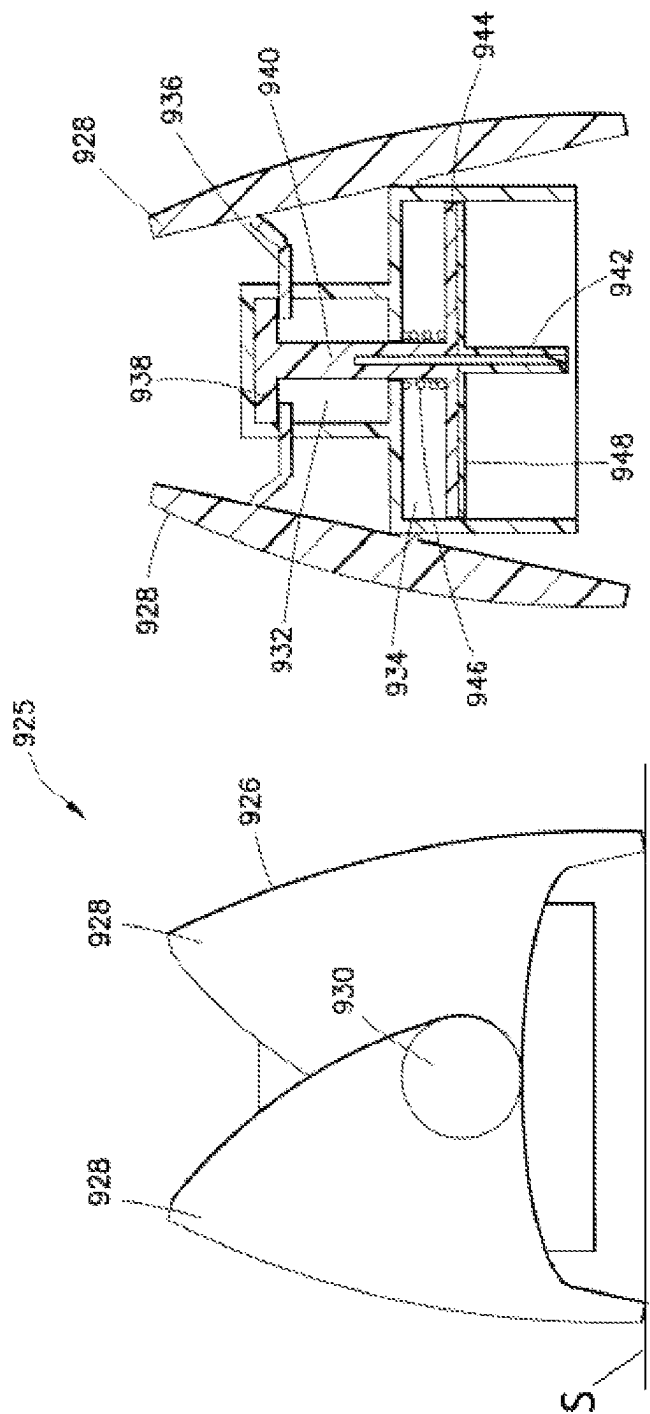


FIG.17a

FIG.17b

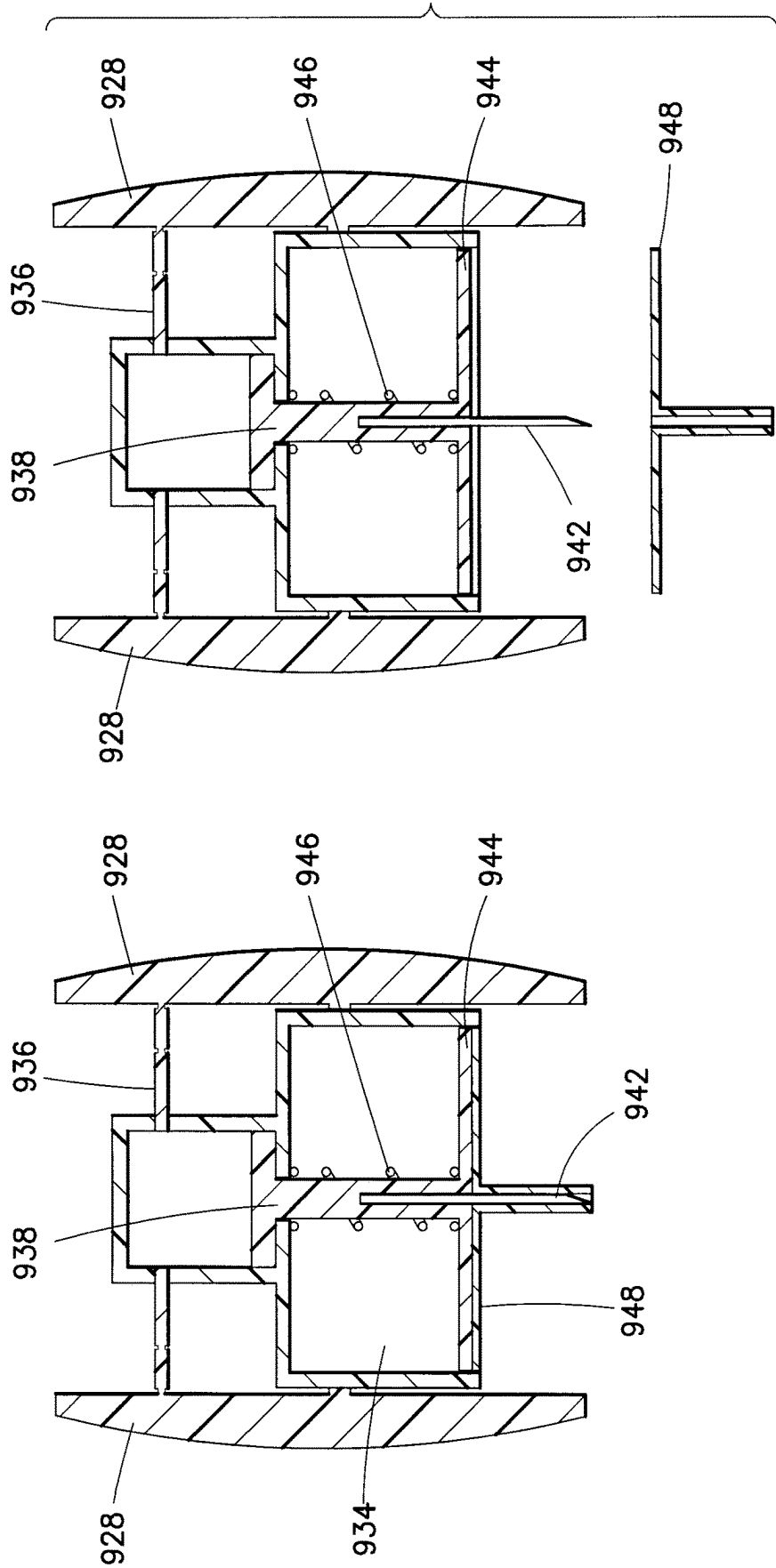


FIG.17c

FIG.17d

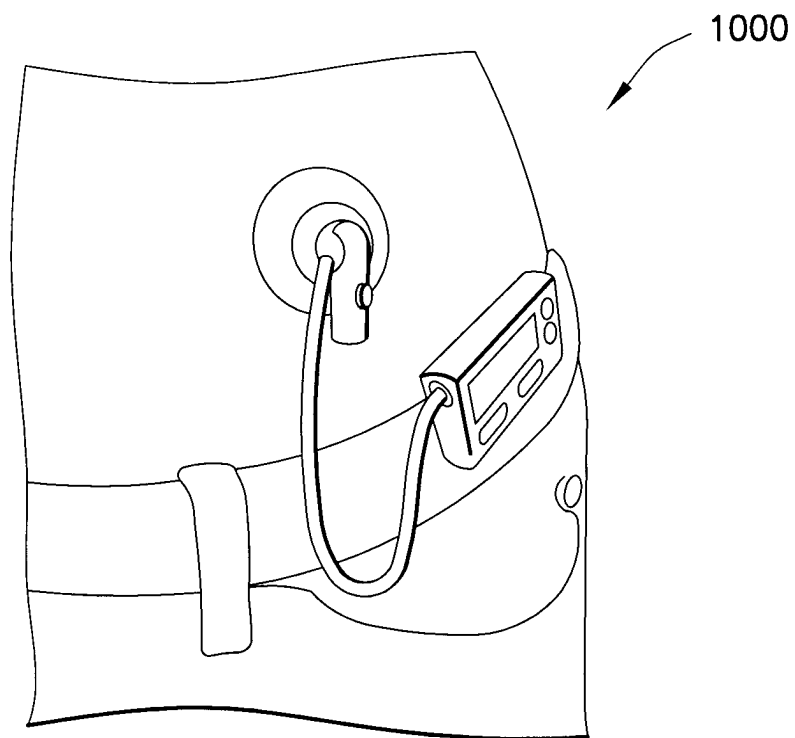


FIG. 18a

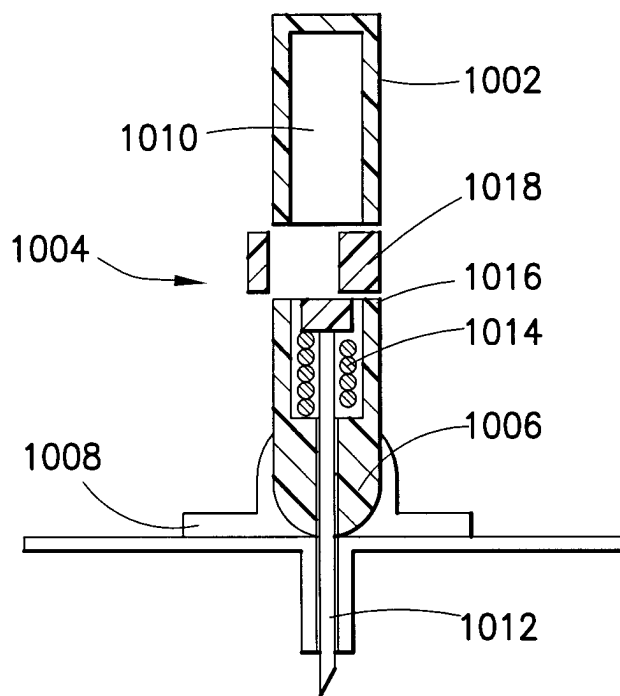


FIG. 18b

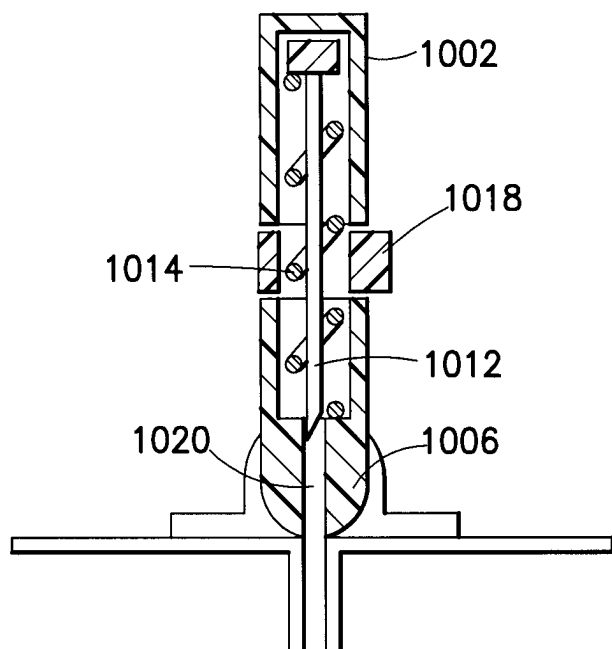


FIG. 18c

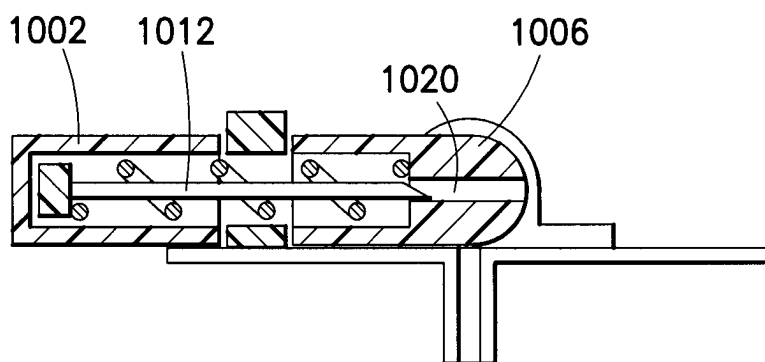


FIG. 18d

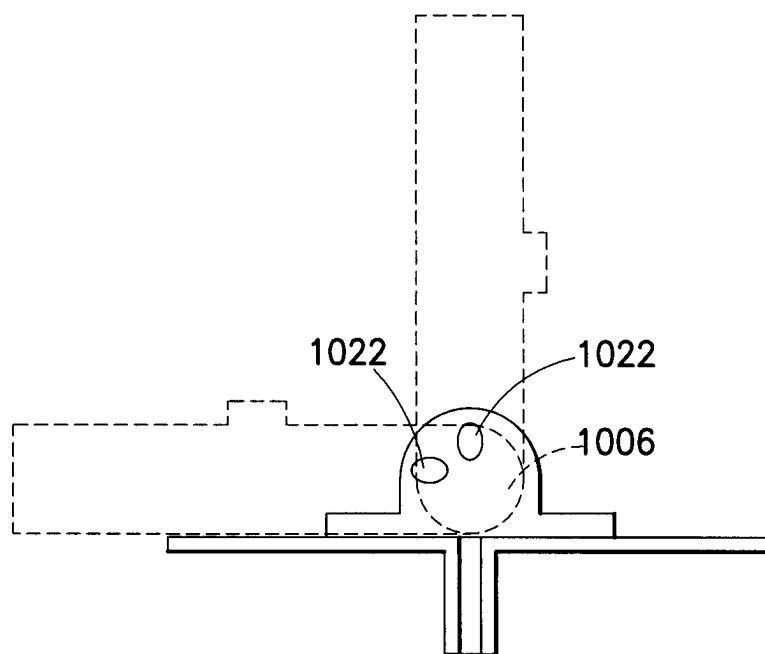


FIG. 18e

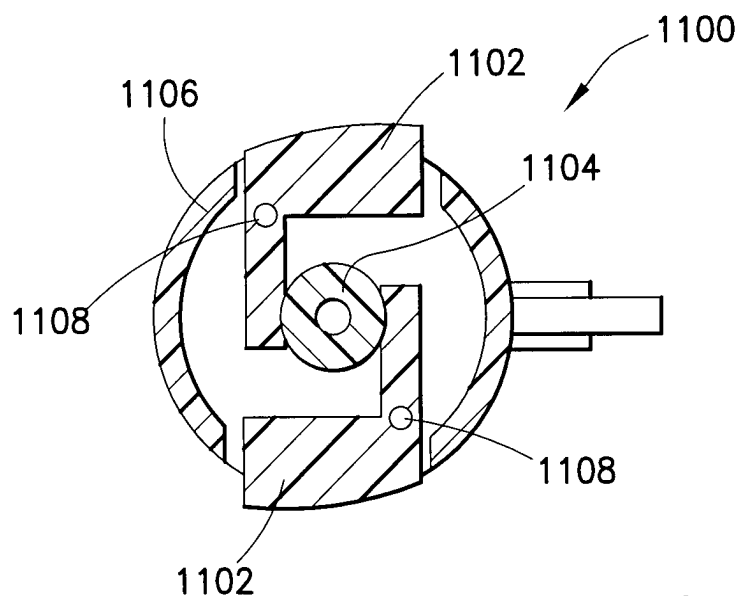


FIG. 19a

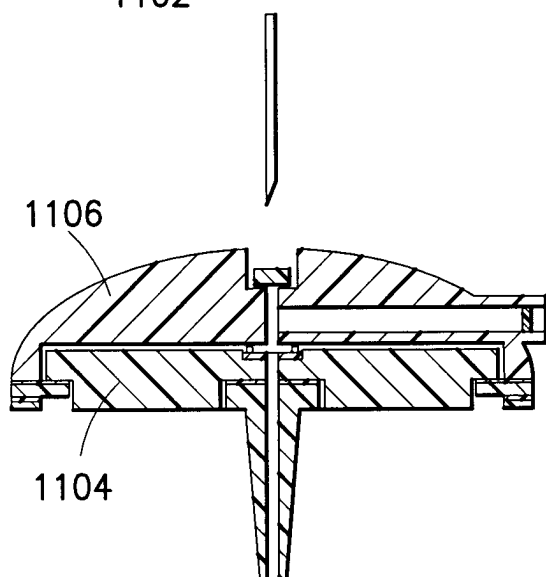


FIG. 19b

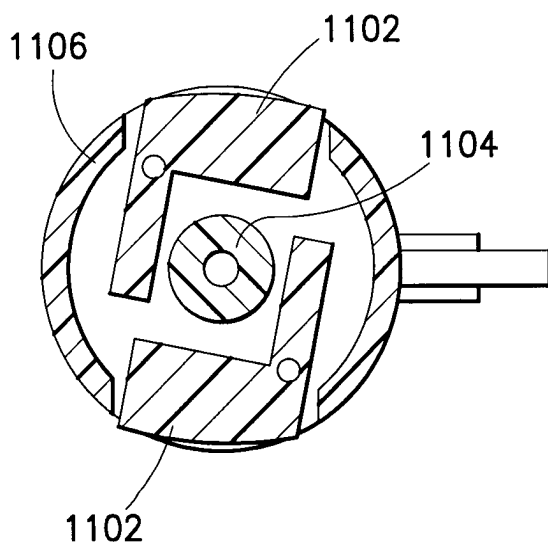
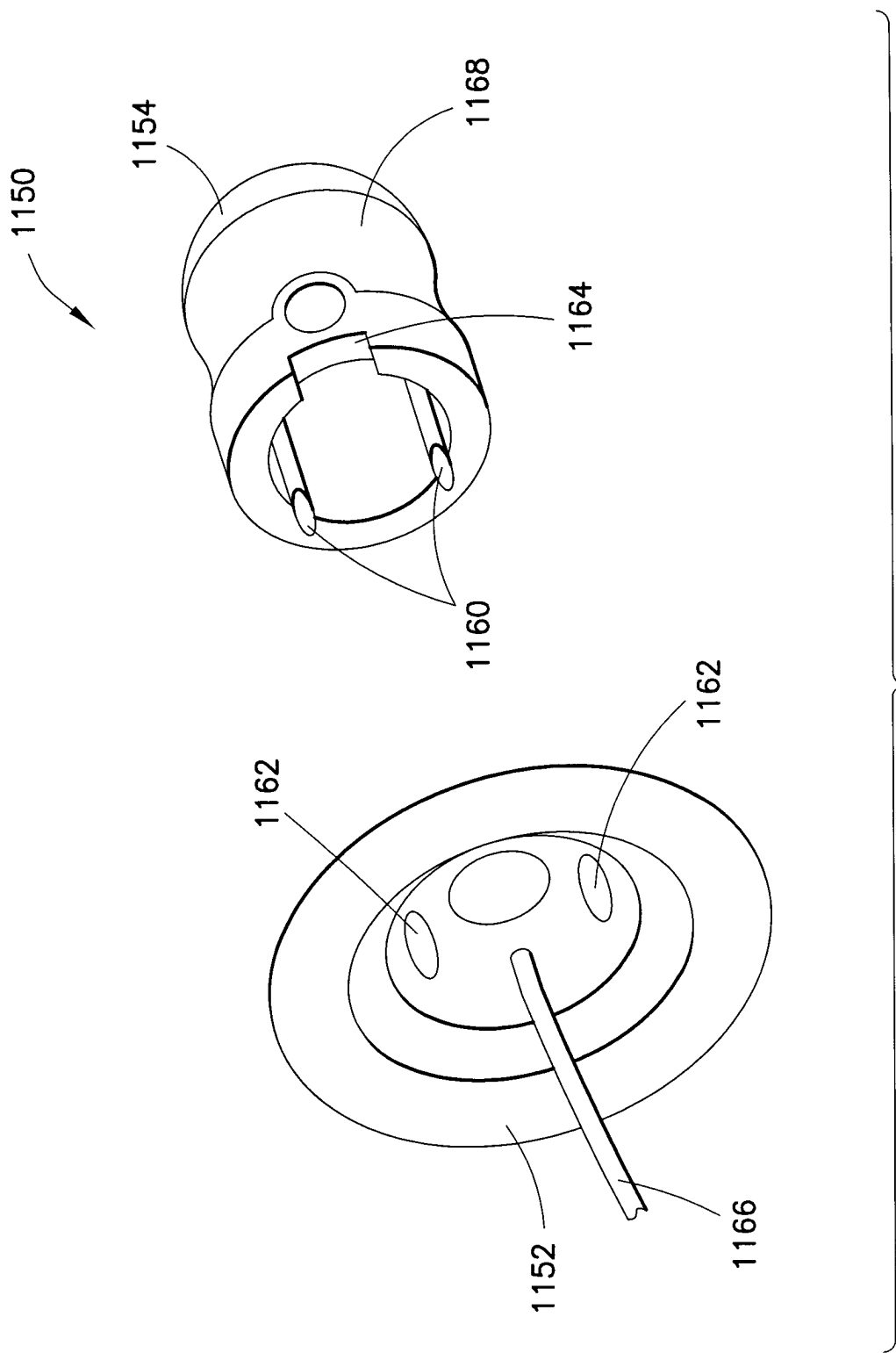
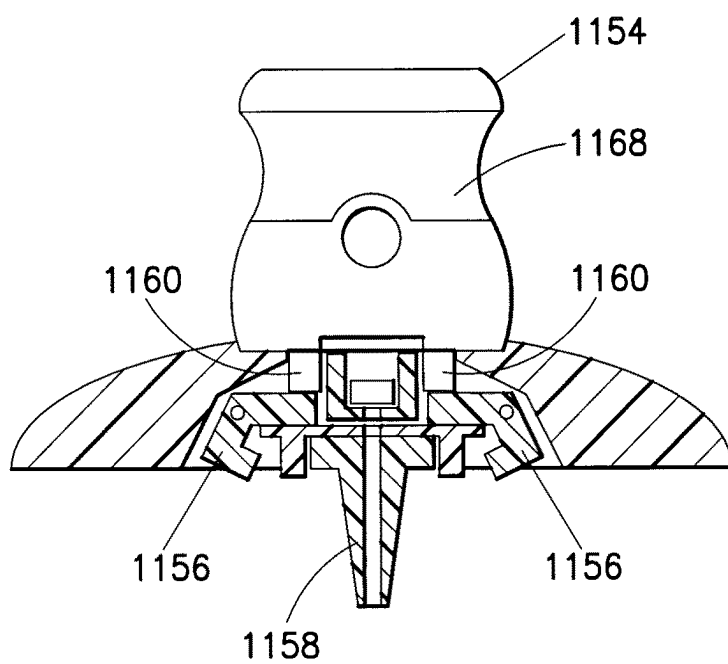
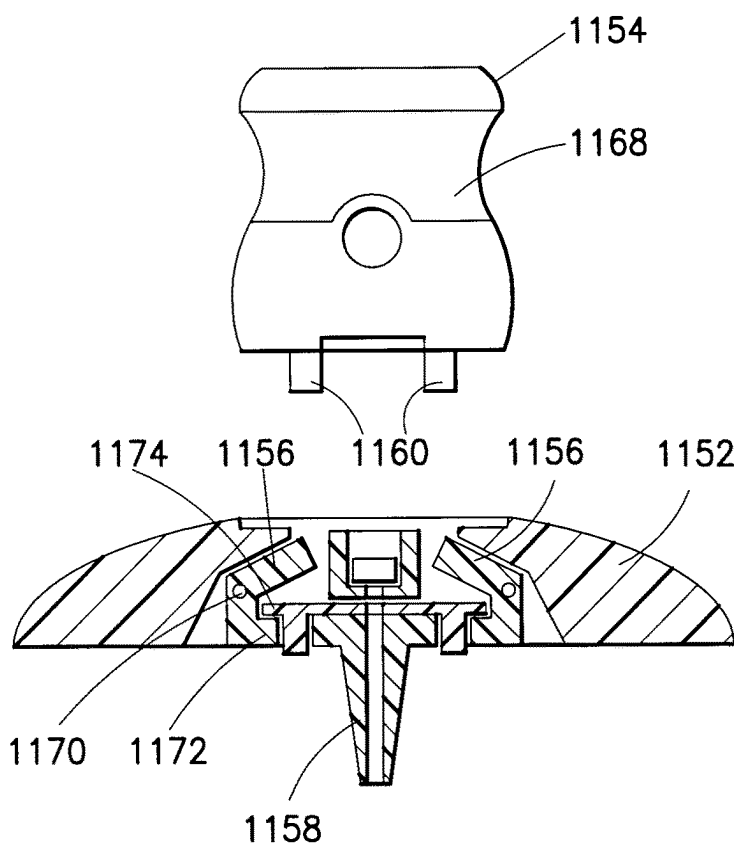
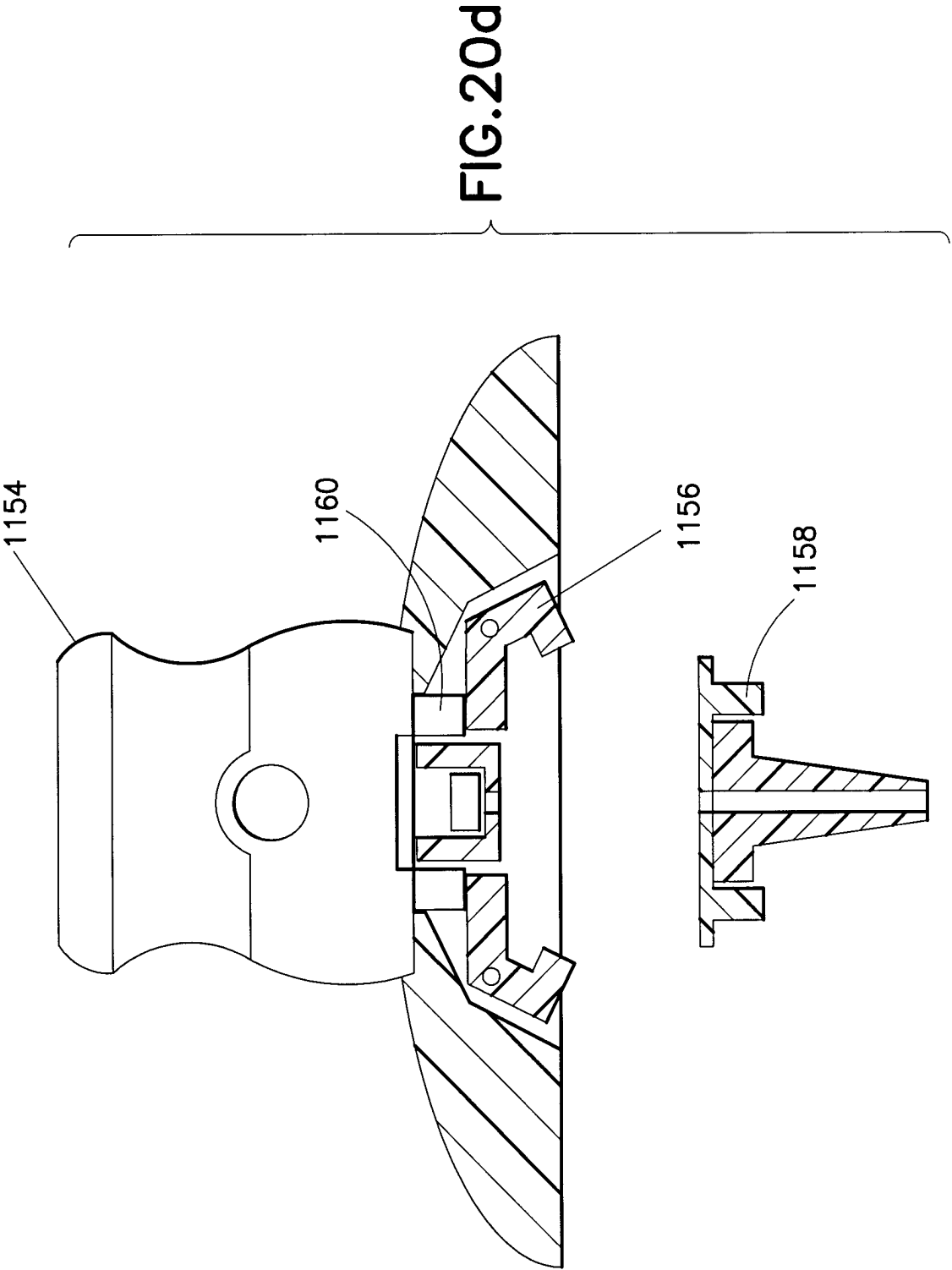
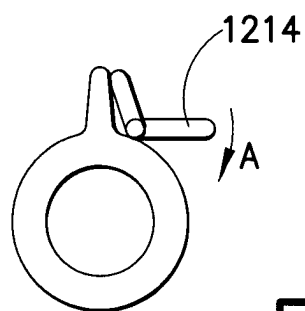
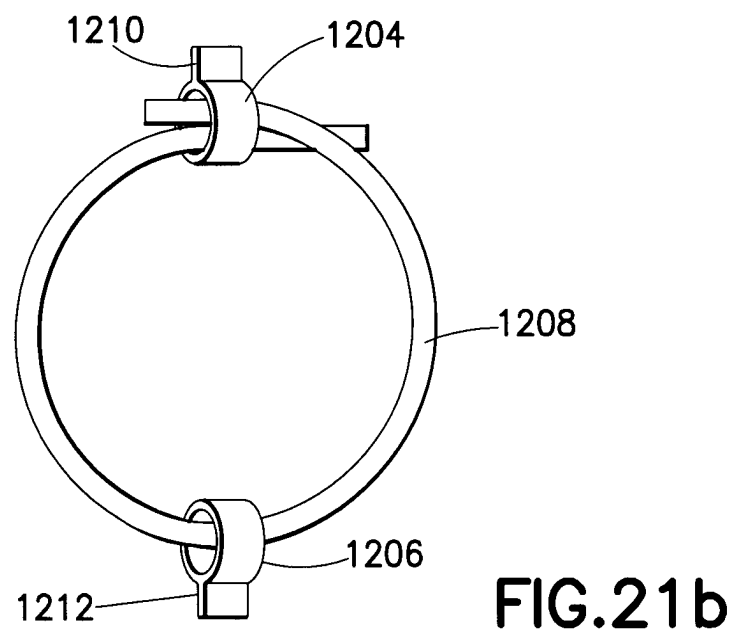
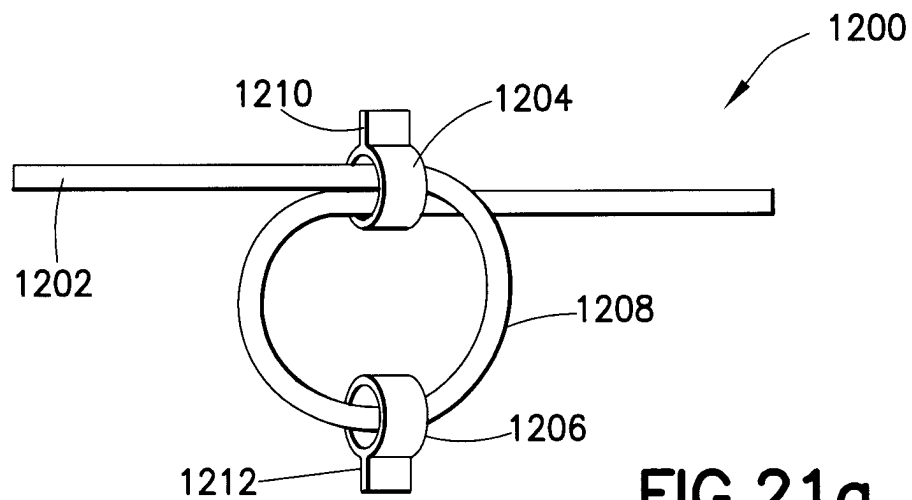


FIG. 19c









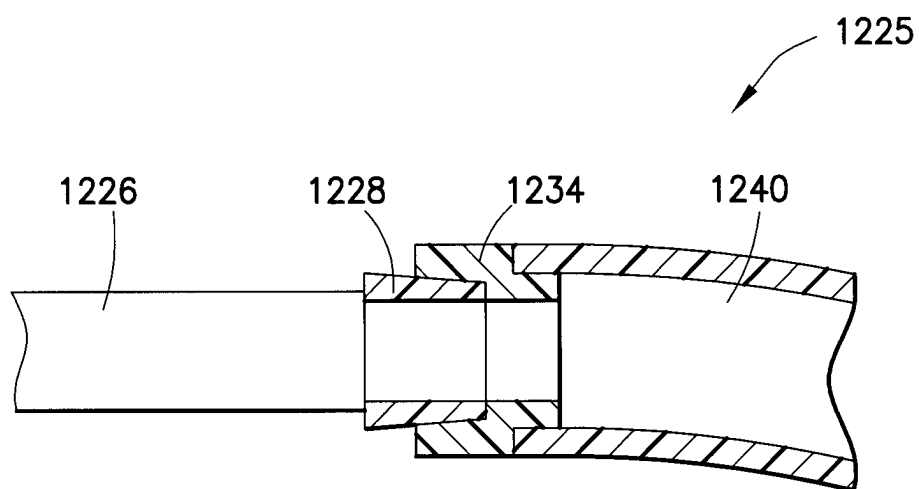


FIG. 22a

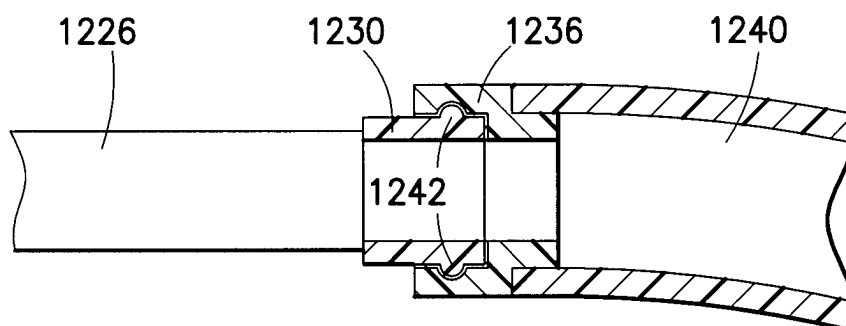


FIG. 22b

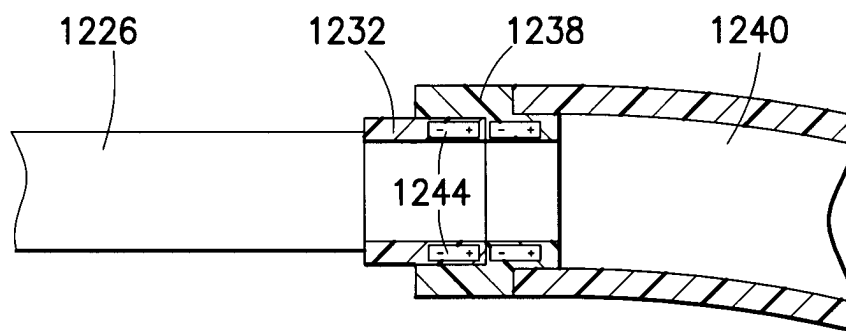
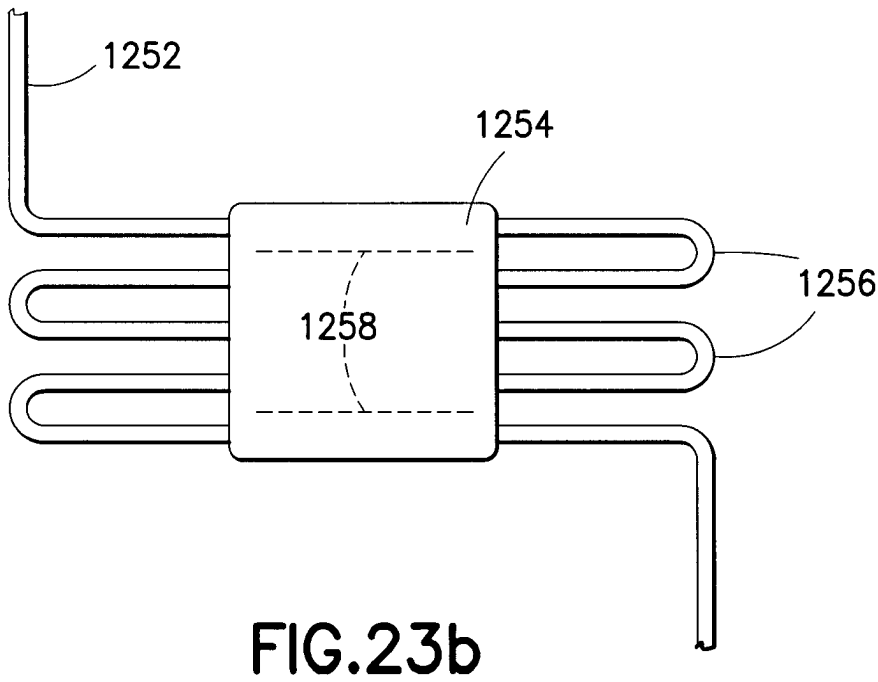
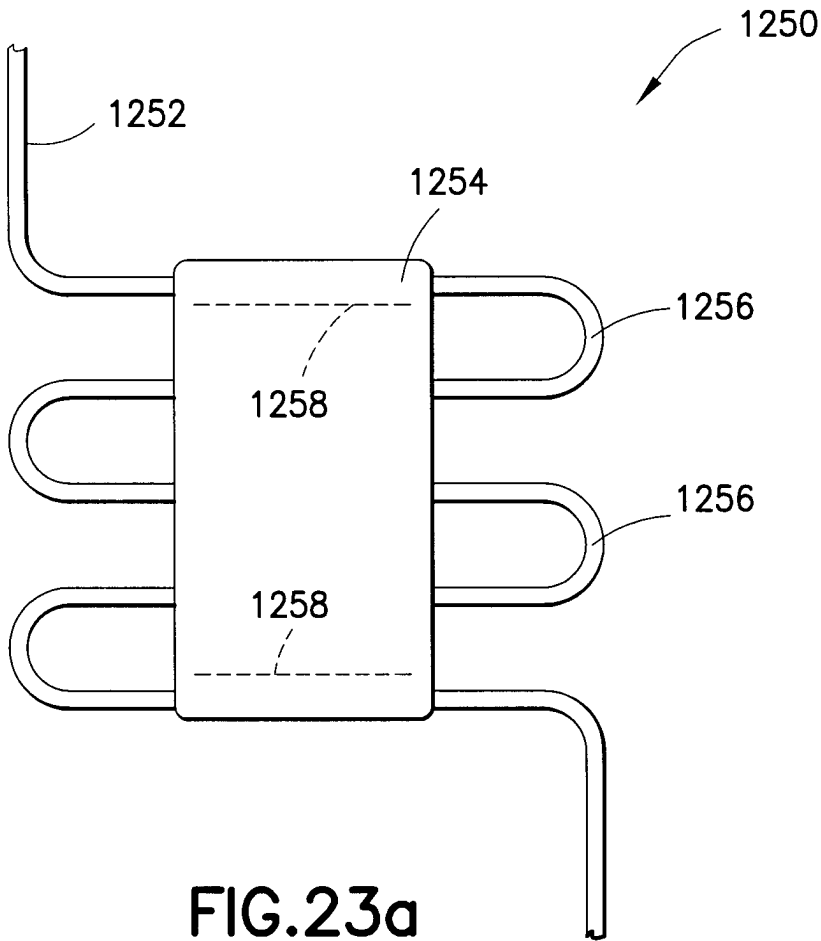
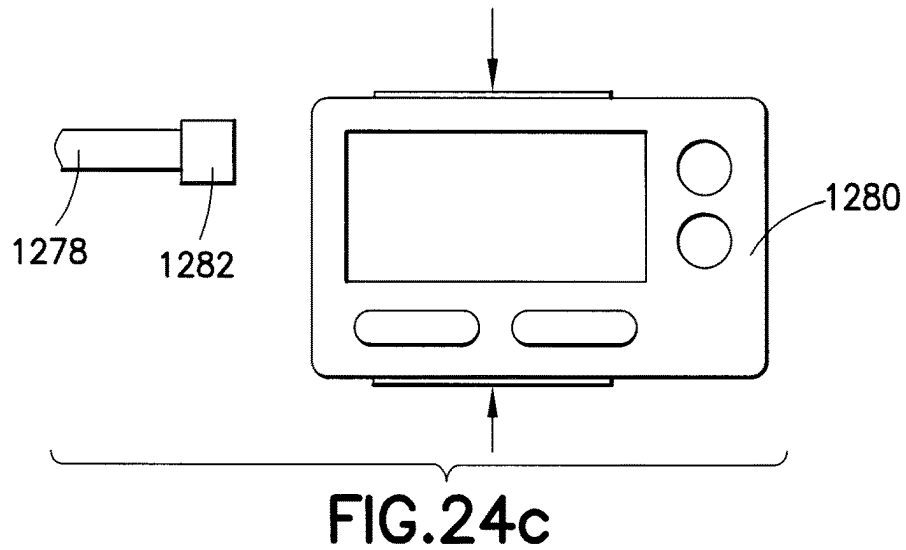
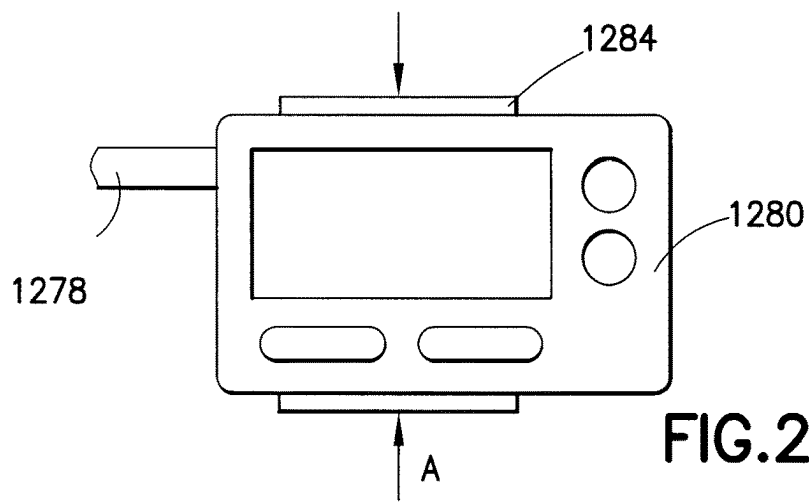
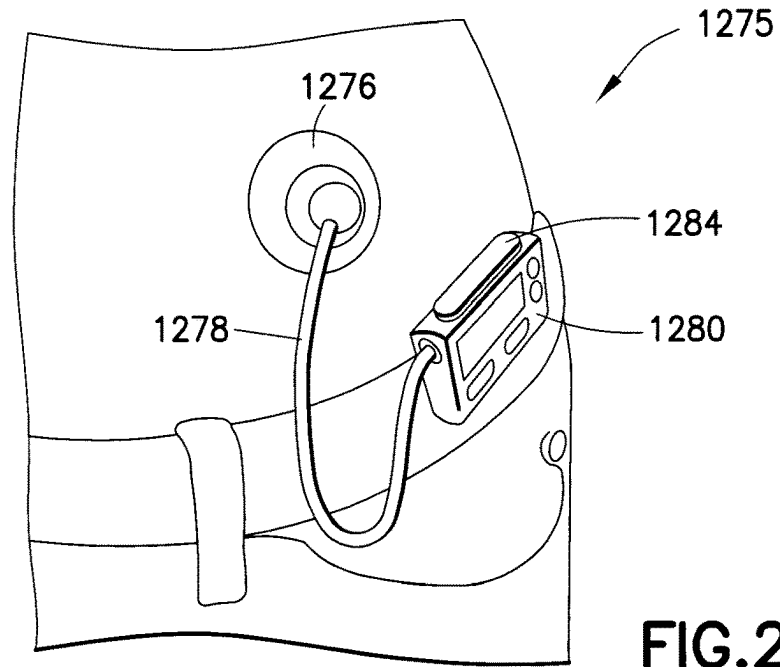


FIG. 22c





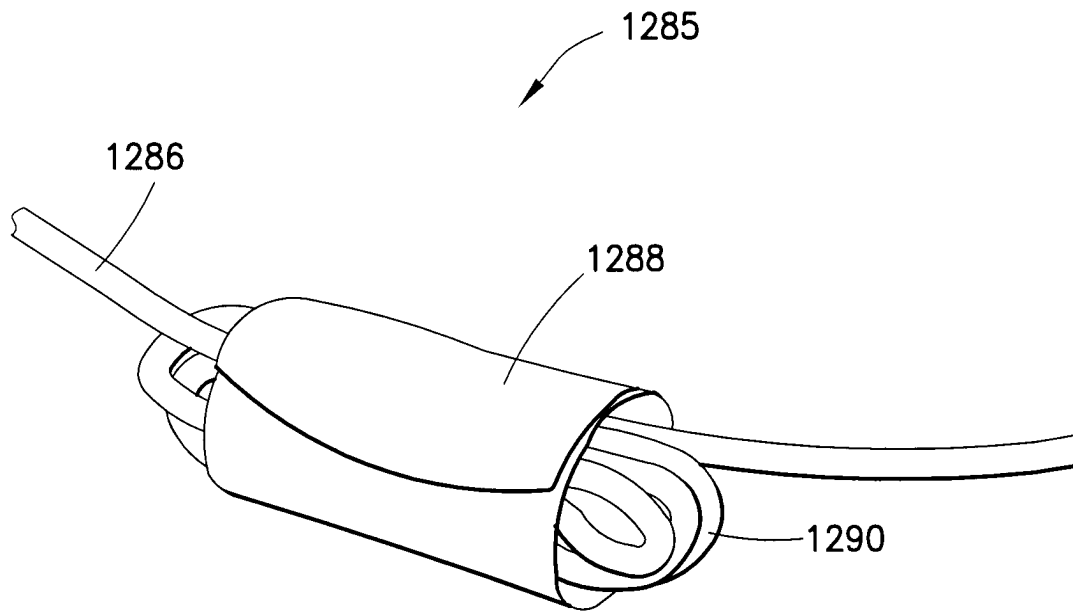


FIG.25a

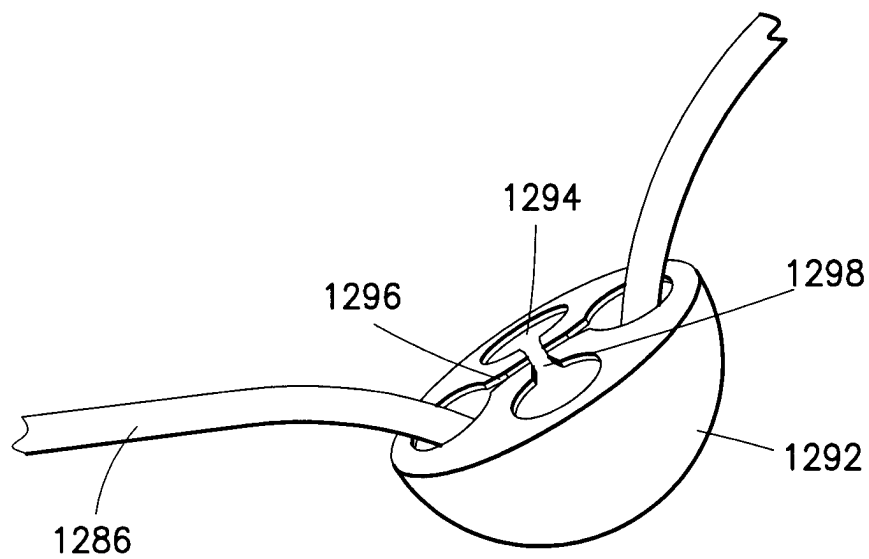


FIG.25b

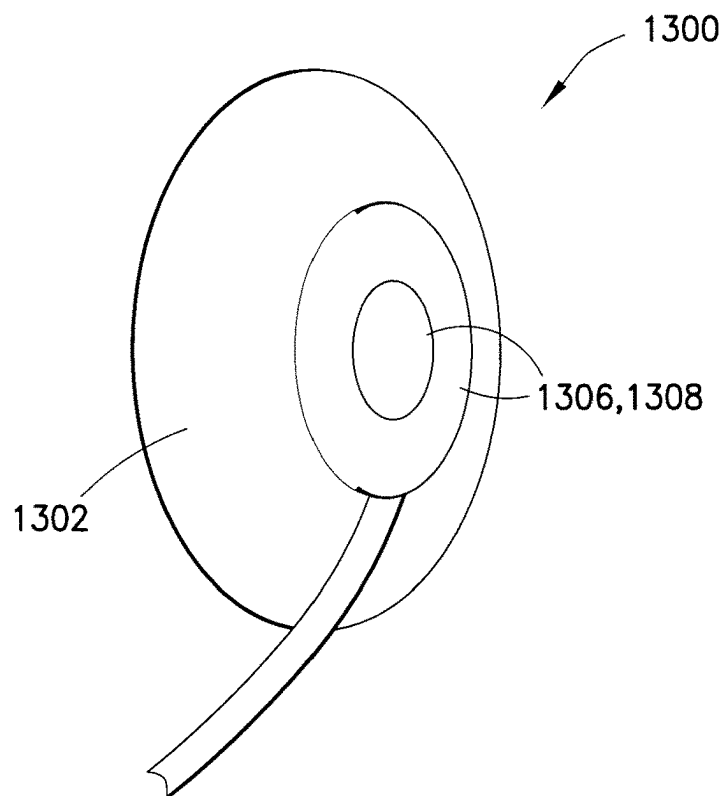


FIG. 26a

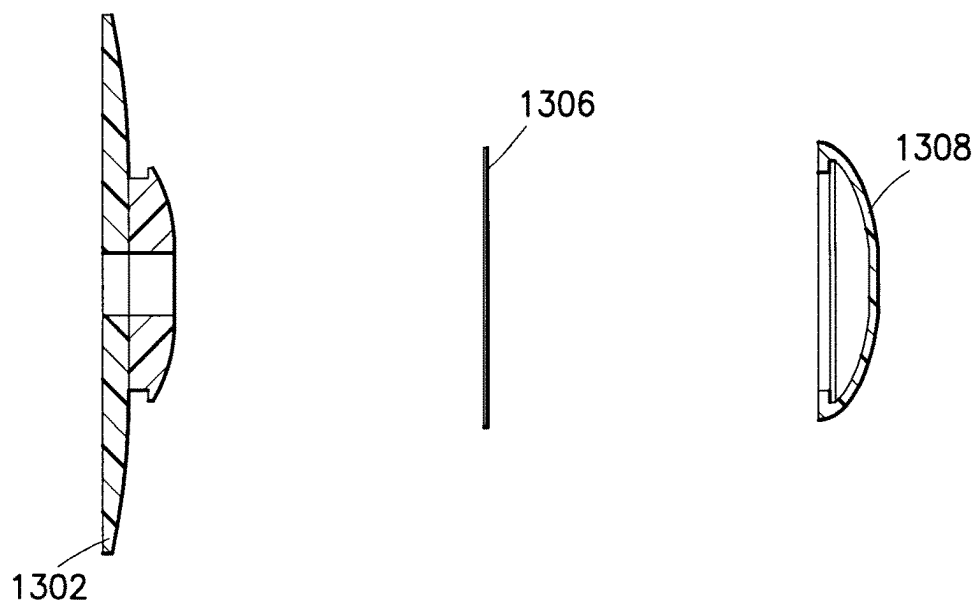
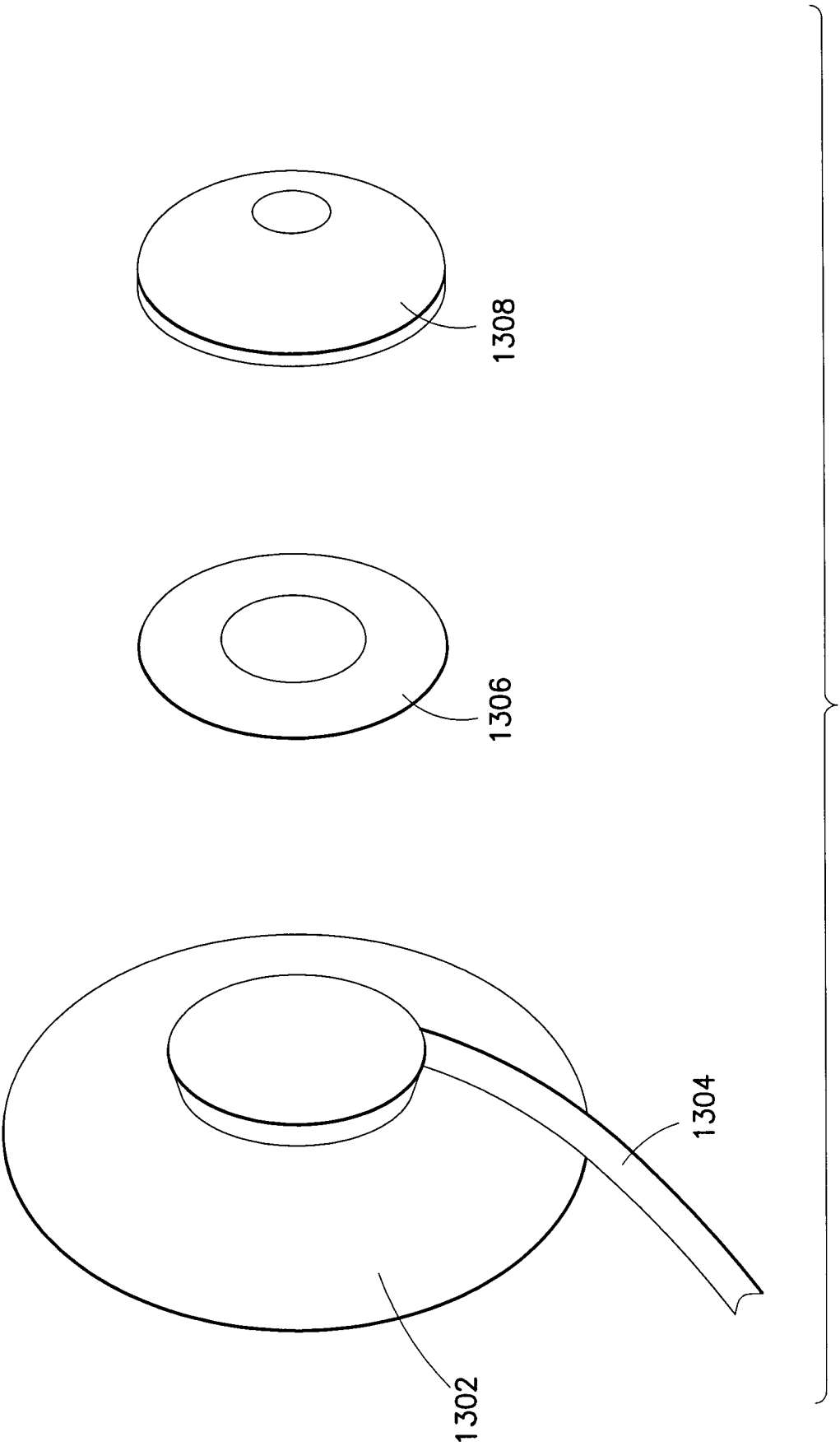


FIG. 26c



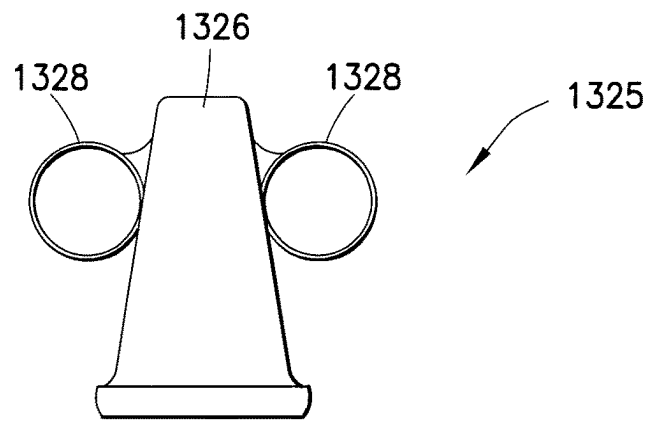


FIG. 27a

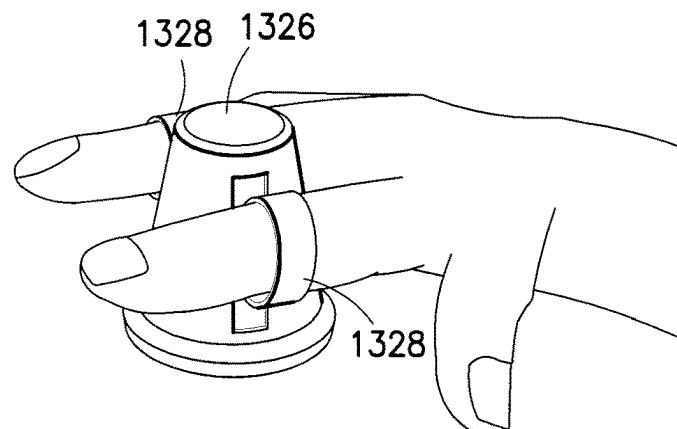


FIG. 27b

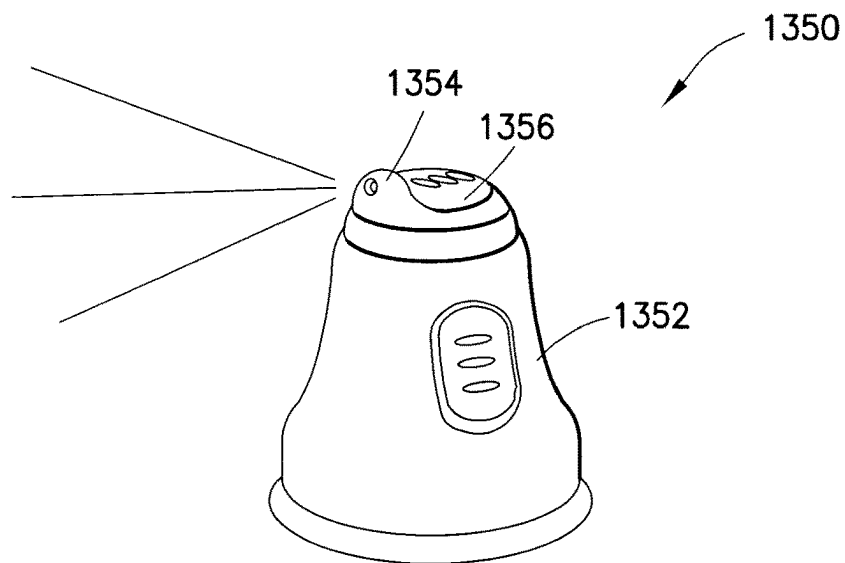
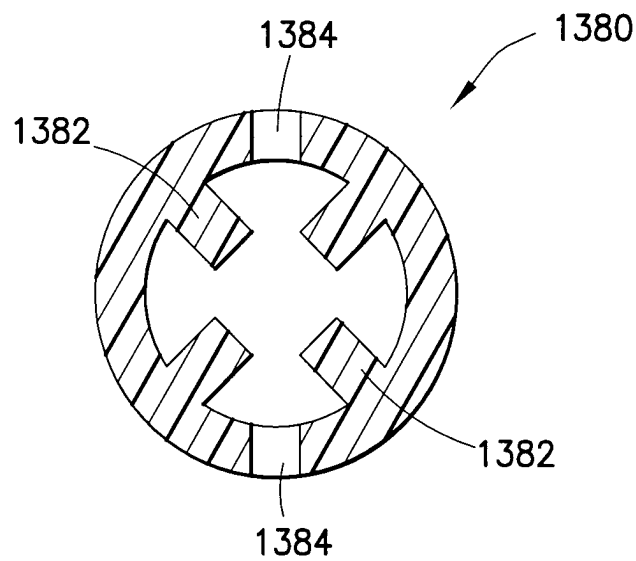
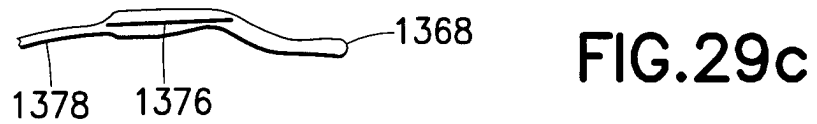
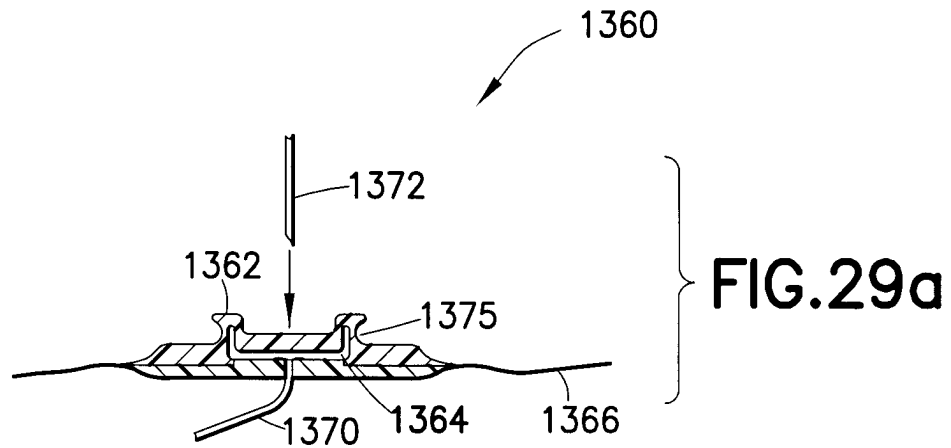


FIG. 28



1

INFUSION SET**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a continuation of U.S. patent application Ser. No. 13/138,209, filed on Feb. 1, 2012, which is the U.S. national stage of international patent application No. PCT/US10/00145, filed on Jan. 21, 2010, which claims the benefit under 35 U.S.C. § 119(a) of U.S. Provisional Application No. 61/202,019, entitled "Infusion Set", filed on Jan. 21, 2009, the entire content, disclosure and subject matter of said application being expressly incorporated herein by reference.

FIELD OF THE INVENTION

The present invention relates generally to components, elements and packaging of infusion sets, including features and elements in the areas of tube management, site management, set adhesion, set insertion, set placement and changing operations and packaging.

BACKGROUND OF THE INVENTION

A large number of people, such as those suffering from conditions such as diabetes use some form of infusion therapy, such as daily insulin infusions to maintain close control of their glucose levels. Currently, in the insulin infusion treatment example, there are two principal modes of daily insulin therapy. The first mode includes syringes and insulin pens. These devices are simple to use and are relatively low in cost, but they require a needle stick at each injection, typically three to four times per day. The second mode includes infusion pump therapy, which entails the purchase of an insulin pump that lasts for about three years. The initial cost of the pump can be significant, but from a user perspective, the overwhelming majority of patients who have used pumps prefer to remain with pumps for the rest of their lives. This is because infusion pumps, although more complex than syringes and pens, offer the advantages of continuous infusion of insulin, precision dosing and programmable delivery schedules. This results in closer blood glucose control and an improved feeling of wellness.

As patients on oral agents eventually move to insulin and their interest in intensive therapy increases, users typically look to these insulin pumps for improvements in the management of their condition. Therefore interest in better pump-related therapy is on the rise. In this and similar examples, what is needed to fully meet this increased interest are advanced, improved, and novel new components, elements and packaging of current and future insulin infusion sets, including features and elements in the areas of tube management, site management, set adhesion, set insertion, set placement and changing operations and packaging.

Accordingly, a need exists for such advanced, improved, and novel new components, elements and packaging of current and future infusion sets, that further provide simplicity in manufacture and use improvements for both insulin and non-insulin applications.

SUMMARY OF THE INVENTION

An object of the present invention is to substantially address the above and other concerns, and provide advanced, improved, and novel new components, elements and packaging of current and future infusion sets, that

2

further provide simplicity in manufacture and use improvements for both insulin and non-insulin applications.

Another object of the present invention is to provide a collection of advanced, improved, and novel new components and elements in a single package to simplify assembly and use of the infusion set by the user.

Another object of the present invention is to provide an exemplary pushbutton-type inserter, squeeze-type inserter, contact-type inserter, skin pinching-type inserter, folding retraction-type inserter, and/or multistage-type inserter having at least one reusable stage, which can be provided in the single package to simplify assembly and use of the infusion set by the user.

Another object of the present invention is to provide an exemplary adhesion means with two or more user-selectable degrees of adhesion strength which can be provided in the single package to simplify assembly and use of the infusion set by the user.

Another object of the present invention is to provide an exemplary self-sealing tube connection means which can be provided in the single package to simplify assembly and use of the infusion set by the user.

Another object of the present invention is to provide an exemplary set having one or more clear and/or magnifying lens features to view a site beneath the set which can be provided in the single package to simplify assembly and use of the infusion set by the user.

Another object of the present invention is to provide an exemplary tube management element having a spring-loaded circular tube reel, tubing pull ties, elastic accordion, pouch or shortened length to manage a tube or tube loop, which can be provided in the single package to simplify assembly and use of the infusion set by the user.

Another object of the present invention is to provide an exemplary tube connection element including a tapered connector, detent connector and/or a magnetic attraction connector, which can be provided in the single package to simplify assembly and use of the infusion set by the user.

Another object of the present invention is to provide an exemplary adhesion concealment means for concealment of the set once in position and/or a decoration means for decoration of the set once in position, which can be provided in the single package to simplify assembly and use of the infusion set by the user.

Another object of the present invention is to provide an exemplary set placement guidance ring and/or one or more finger loops on the inserter to aid in set placement, which can be provided in the single package to simplify assembly and use of the infusion set by the user.

Another object of the present invention is to provide an exemplary site preparation wipe or spray which can be provided as part of the inserter, or otherwise included in the single package to simplify assembly and use of the infusion set by the user.

Another object of the present invention is to provide an exemplary package which can hold a number of sets that can be easily released and retrieved from the tray by an inserter to simplify assembly and use of the infusion set by the user.

Another object of the present invention is to provide an exemplary insertion needle handle and shroud which can be provided in the single package to simplify assembly and use of the infusion set by the user.

Another object of the present invention is to provide an exemplary squeeze-type latch between an upper portion and a lower portion of the set, and/or a tool removable upper

portion of the set, which can be provided in the single package to simplify assembly and use of the infusion set by the user.

Another object of the present invention is to provide an annular fluid reservoir and/or fluid path in the set hub, which can significantly minimize the penetration distance of the tubeset connector into the hub while maintaining a sufficiently large tubeset connector geometry.

Another object of the present invention is to provide substantially closed or sealed annular fluid reservoir and/or fluid path in the set hub, which can reseal any insertion openings generated by the insertion needle, and allow penetration by a tubeset connector needle.

Another object of the present invention is to provide an infusion set constructed of a soft, pliable and/or elastic or similar material such that the infusion set is soft or pliable to a degree that allows the elasticity of the materials to affix a tube "ring" of the tubeset connector to the hub.

Another object of the present invention is to provide an infusion set constructed such that the tube ring of the tubeset connector can include the tubeset connector needle to pierce the hub, wherein the elasticity of the materials function to seal the insertion site of the tubeset connector needle.

Another object of the present invention is to provide an infusion set constructed such that the tubeset connector needle of the tube ring of the tubeset connector can pierce the hub at any rotational position, and wherein the elasticity of the materials function to seal the insertion site of the tubeset connector needle such that the tubeset connector needle of the tube ring of the tubeset connector can be withdrawn and the tube ring repositioned at a different rotational alignment position.

Another object of the present invention is to provide a catheter constructed of a body temperature softening polyurethane or similar material, and include one or more features including a splined lumen and holes or openings along a body length, including cross-drilled holes.

These and other objects are substantially achieved by providing a collection of advanced, improved, and novel new components and elements in a single package to simplify assembly and use of the infusion set by the user, including one or more of an exemplary pushbutton-type inserter, squeeze-type inserter, contact-type inserter, skin pinching-type inserter, folding retraction-type inserter, and/or multistage-type inserter having at least one reusable stage. One or more of the exemplary embodiments comprise a user gripping surface and a means to release a firing spring for set placement, and one or more retraction features to retract the insertion needle to avoid any dangers to the user and permit safe disposal. Further, one or more of the exemplary embodiments can comprise a set placement guidance ring to be placed on the insertion site and which mates with an end of the inserter to ensure that the inserter is properly positioned before release of the set. Further, one or more of the exemplary embodiments can comprise one or more finger loops extending from a body of the inserter to aid in set placement. Still further, one or more of the exemplary embodiments can comprise a squeeze-type latch between an upper portion and a lower portion of the set comprising one or more pushbuttons which articulate a latch between the upper and lower portions, thereby releasing the upper portion of the set from the lower portion which can remain in position. In yet other embodiments of the present invention, the upper portion can be configured to be tool-removable, such that the tool comprises one or more pins which are inserted into the lower portion and release a latch between the upper and lower portions, thereby releasing the upper

portion of the set from the lower portion which can remain in position. Still further, one or more of the exemplary embodiments can comprise an insertion needle handle and shroud comprising one or more hinged flat members which can be bent to cover the protruding needle after use.

These and other objects are also substantially achieved by providing a collection of advanced, improved, and novel new components and elements in a single package to simplify assembly and use of the infusion set by the user, including one or more of an exemplary adhesion means with two or more user-selectable degrees of adhesion strength comprising at least a first and second adhesion ring having different degrees of adhesive strength. The user can select which ring to use by removing a cover of the desired ring and leaving the remaining rings covered. A self-sealing tube connection means can be provided and comprise a set having a groove into which an elastic ring and tube connector can be positioned, such that the elastic ring secures the tube to the set and seals the connection between each. The set can comprise one or more clear and/or magnifying plastic components to view a site beneath the set, wherein the adhesive pad is provided with one or more clearances to allow the visual access.

These and other objects are also substantially achieved by providing a collection of advanced, improved, and novel new components and elements in a single package to simplify assembly and use of the infusion set by the user, including one or more tube management elements comprising a spring-loaded circular tube reel which serves to feed and retract excess tube as urged by a reel spring, one or more tubing pull ties which can be pulled to either secure or release tubing being stored in a large loop, an elastic accordion or pouch in which to store tubing, or provide shortened tube lengths between set and pump. Such tubing can further comprise one or more embodiments of a tube connection including a tapered connector in which a tight engagement provides connection, a detent connector in which projecting detents and recessed detents provide connection, and a magnetic connector in which magnetic attraction provides connection. Still further, an insulin supply comprising an insulin content, supply vial and tubing can be provided with the tubing and tubing connectors.

These and other objects are also substantially achieved by providing a collection of advanced, improved, and novel new components and elements in a single package to simplify assembly and use of the infusion set by the user, including one or more of an adhesion concealment means for concealment of the set once in position comprising an adhesive pad to be placed over a set to conceal the set, or a decorative adhesive pad to be placed over a set to enhance the set. A site preparation wipe comprising a disposable pad including a site preparation solution can be provided as well as an improved inserter body that comprises a spray mechanism and site preparation solution contents.

These and other objects are also substantially achieved by providing a collection of advanced, improved, and novel new components and elements in an exemplary package comprising a number of openings into which sets can be aligned and secured by a covering, such as foil, and which allows set removal by an inserter configured to pierce the foil and capture and remove the set from the tray.

BRIEF DESCRIPTION OF THE DRAWINGS

The various objects, advantages and novel features of the preferred embodiments of the present invention will be more

5

readily appreciated from the following detailed description when read in conjunction with the appended drawings, in which:

FIGS. 1 and 2 are perspective views of a collection of infusion set elements and associated packaging in accordance with an exemplary embodiment of the present invention;

FIGS. 3a-3e are cross-sectional views of an exemplary single-button insertion device in accordance with an exemplary embodiment of the present invention;

FIGS. 4a-4c are perspective views of exemplary set adhesion elements in accordance with an exemplary embodiment of the present invention;

FIGS. 5a-5d are views of an exemplary set connection method in accordance with an exemplary embodiment of the present invention;

FIGS. 6a-6b are views of an exemplary set site inspection element in accordance with an exemplary embodiment of the present invention;

FIGS. 7a-7b are views of an exemplary set tubing management element in accordance with an exemplary embodiment of the present invention;

FIG. 8 is a perspective view of an alternate insulin supply and associated tubing in accordance with an exemplary embodiment of the present invention;

FIGS. 9a-9b are perspective views of an exemplary set concealment element in accordance with an exemplary embodiment of the present invention;

FIGS. 10a-10c are perspective views of exemplary set placement elements in accordance with an exemplary embodiment of the present invention;

FIG. 11 is a perspective view of an exemplary set site placement preparation element in accordance with an exemplary embodiment of the present invention;

FIGS. 12a-12f are views of exemplary sets on tray packaging in accordance with an exemplary embodiment of the present invention;

FIGS. 13a-13f are views of an exemplary multistage insertion device in accordance with an exemplary embodiment of the present invention;

FIGS. 14a-14d are cross-sectional views of an exemplary "squeeze-type" inserter in accordance with an exemplary embodiment of the present invention;

FIGS. 15a-15e are views of an exemplary "contact-type" inserter in accordance with an exemplary embodiment of the present invention;

FIGS. 16a-16c are views of an exemplary needle handle and shroud in accordance with an exemplary embodiment of the present invention;

FIGS. 17a-17d are views of an exemplary "skin pinch-type" inserter in accordance with an exemplary embodiment of the present invention;

FIGS. 18a-18e are views of an exemplary "folding/retracting-type" inserter in accordance with an exemplary embodiment of the present invention;

FIGS. 19a-19c are views of an exemplary "squeeze-type" latch provided as a connection method in accordance with an exemplary embodiment of the present invention;

FIGS. 20a-20d are views of an exemplary "tool-type" latch provided as a connection method in accordance with an exemplary embodiment of the present invention;

FIGS. 21a-21c are perspective views of an exemplary pull tie, tubing management element in accordance with an exemplary embodiment of the present invention;

FIGS. 22a-22c are perspective views of exemplary tubing connecting elements in accordance with an exemplary embodiment of the present invention;

6

FIGS. 23a-23b are perspective views of an exemplary elastic accordion, tubing management element in accordance with an exemplary embodiment of the present invention;

FIGS. 24a-24c are perspective views of an exemplary short tube, tubing management element in accordance with an exemplary embodiment of the present invention;

FIGS. 25a-25b are perspective views of exemplary pouch-type, tubing management elements in accordance with an exemplary embodiment of the present invention;

FIGS. 26a-26c are perspective views of exemplary decorative, functional covering elements in accordance with an exemplary embodiment of the present invention;

FIGS. 27a-27b are perspective views of an exemplary two-finger, placement element in accordance with an exemplary embodiment of the present invention;

FIG. 28 is a perspective view of an exemplary insertion tool-mounted spray, site preparation element in accordance with an exemplary embodiment of the present invention;

FIG. 29a-29c are views of an exemplary hub including an annular fluid reservoir and/or fluid path for use with one or more exemplary embodiments of the present invention; and

FIG. 30 is a cross-sectional view of a catheter including one or more cross-drilled holes and a splined lumen in accordance with an exemplary embodiment of the present invention.

Throughout the drawings, like reference numerals will be understood to refer to like parts, components and structures.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

The embodiments of the present device described below illustrate a number of advanced, improved, and novel new components, elements and packaging of current and future infusion sets, that further provide simplicity in manufacture and use improvements for both insulin and non-insulin applications. Exemplary embodiments are presented in separate descriptions, although the individual features of these embodiments can be combined in any number of ways to meet the needs of the user.

As will be appreciated by one skilled in the art, there are numerous ways of carrying out the examples, improvements and arrangements of insulin-associated devices disclosed herein. Although reference will be made to the embodiments depicted in the drawings and the following descriptions, the embodiments disclosed herein are not meant to be exhaustive of the various alternative designs and embodiments that are encompassed by the disclosed invention.

The embodiments of the present device described below illustrate a number of features and elements in the areas of tube management, site management, set adhesion, set insertion, set placement and changing operations and packaging. A collection of exemplary embodiments of the present invention is shown by way of example in FIGS. 1 and 2, which serve to introduce elements described in greater detail below. FIG. 1 illustrates an exemplary infusion set 10 including the following features. As shown in FIG. 1, an exemplary infusion set can comprise an inserter, such as the squeeze-type inserter 700 for use with a set, such as the ring-sealed set 350. A tube and associated tube management devices, such as the circular reel 450, can be provided for communication with an insulin pump (not shown) or with an insulin supply, such as the insulin supply 475. A placement assistance element can be provided, such as the placement assistance ring 526, and the entire arrangement can be placed into a sealed tray 12 with a number of site preparation

elements, such as the pads 550, and site concealment elements, such as the pads 500. The tray can be comprised of any suitable plastic, fiber or composite material compatible with the components, and can provide compartments, padding or element securing detents or moldings. The set can be packaged in the tray 12 and sealed with a clear and/or labeled cover 14, and includes every component needed to position, connect, insert, and wear the set, as well as the insulin 475 itself as shown in FIG. 2. Each of the exemplary components, including a number of additional or alternate components, will now be described individually in greater detail.

An exemplary embodiment of the present invention can be provided with an insertion device as desired by a user. An exemplary insertion device 100 is shown in FIGS. 3a-3e. The exemplary insertion device of FIGS. 3a-3e provides an insertion device which can contain the set at an open, patient-contacting end, and provide an actuation button at an opposite end. Upon activation, the insertion device places the set and automatically retracts the insertion needle back into the insertion device.

As shown in FIG. 3a, the insertion device can comprise a substantially cylindrical housing 102 from which a spring biased projection or pushbutton 104 can extend, and in which a set 106 can be positioned for use. The housing 102 can have a first diameter at a lower portion thereof in which the set 106 can be positioned, and transition to a reduced second diameter at an opposite end to substantially equal a diameter of the pushbutton 104.

Within the body of the housing 102, a number of elements are contained which serve to fire the set 106 into position, and then retract the insertion needle. To do so, the housing 102 comprises a first, second and third chamber 108, 110 and 112 of different widths. The first chamber 108 at an uppermost portion of the housing 102 has a width sufficient to slidably receive an end of an inserter rod 114. At an upper end of the first chamber 108, an opening is provided through which the pushbutton 104 slidably enters the first chamber 108, and a lower end of the first chamber opens to create the second chamber 110 as described in greater detail below. The first chamber further comprises at least one inclined detent 116 disposed upon an inner wall of the first chamber 108 which serves to capture and secure a similar, deflectable detent 118 at an upper end of the inserter rod 114. One or more further similar, deflectable detents 120 are disposed upon an end 122 of the pushbutton 104 within the first chamber 108. The end 122 of the pushbutton 104 is provided to have a width substantially equal to the width of the first chamber 108 to align and guide the pushbutton 104 during operation. A spring 124 is disposed concentrically with the pushbutton 104 and is captured between an outer surface of the housing 102 and an expanded head of the pushbutton 104 to constantly urge the pushbutton 104 upward.

As noted above, the pushbutton 104 comprises an end 122 with one or more inclined detents 120. When pressed downward, the inclined detents 120 come into contact with the inclined detents 118 of the inserter rod 114 which are being held by the detents 116 of the first chamber 108. The contact releases the detents 118 and allows the inserter rod 114 to be urged downward by a firing spring 126 disposed within the second chamber 110 of the housing 102.

The second chamber 110 has a width wider than that of the first chamber 108, which is sufficient to slidably receive the inserter rod 114 and more specifically, a width sufficient to slidably contain planar members 128 and 132 of the inserter rod 114.

The firing spring 126 is disposed concentrically with the inserter rod 114 about an outer circumference of the inserter rod body as captured within the second chamber 110. That is, the firing spring 126 is captured within the second chamber 110 of the housing 102 between an upper wall of the second chamber 110, and the upper surface of the planar member 128 of the inserter rod 114. In doing so, the firing spring 126 constantly urges the inserter rod 114 downward.

At a lower portion of the housing 102, the third chamber 112 can be provided having a width wider than that of the second chamber 110, thereby creating a shoulder 130 therebetween. As described in greater detail below, the shoulder 130 is configured to allow downward travel of the inserter rod 114 and capture one or more detents on an outer surface of the inserter rod 114 to prevent retraction of the inserter rod 114, yet permit full retraction of a needle carrier and inserter needle.

As noted above, the inserter rod 114 is configured to slidably travel through each of the first, second and third chambers 108, 110, and 112, of the housing 102. The inserter rod 114 is substantially cylindrical and comprises a diameter at an upper portion substantially equal to the width of the first chamber 108 to be directed and guided by the first chamber 108 during use. A lower portion of the inserter rod 114 comprises the first and second planar members 128 and 132, which comprise a diameter substantially equal to the width of the second chamber 110 to be directed and guided by the second chamber during use.

As noted above, the inserter rod 114 comprises the first and second planar members 128 and 132. The inserter rod 114 further comprises an inserter needle guide 144 and at least third and fourth elements 136 and 138 extending between the first and second planar members 128 and 132. A spring 140 is captured between the needle guide 144 and fourth element 138, and passes through an opening in the third element 136, to constantly urge both the third and fourth elements 136 and 138 outward from a center axis of the insertion device, and against an inner wall of the second and third chambers 110 and 112. The fourth element 138 comprises an inclined detent 142 which is urged against the inner wall of the second and third chambers 110 and 112. In doing so, as the inserter rod 114 is urged downward and the detent 142 passes the shoulder 130, the spring 140 urges the element 138 against the inner wall of the third chamber 112 such that the detent 142 is captured by the shoulder 130 and prevents the retraction of the inserter rod 114 as shown in FIGS. 3c and 3d. The needle carrier and inserter needle are permitted to retract as described in greater detail below.

The third element 136 comprises at least one projection 146 which is configured to capture a groove 154 in the needle carrier 148 slidably disposed within an inner opening of the inserter rod 114. The needle carrier 148 secures the inserter needle 134 at a lower end, such that the inserter needle extends through the needle guide 144 and through an opening in the planar member 132 of the inserter rod 114. The needle carrier 148 further comprises the groove 154 which can be captured by the projection 146. The remainder of the needle carrier 148 extends through the planar member 128 and into the inner opening of the inserter rod 114 and terminates at a planar end 150. The planar end 150 of the needle carrier 148 has a width substantially the same as the width of the inner opening of the inserter rod 114 to align and guide the needle carrier 148 during retraction. A retraction spring 152 is captured between the planar end 150 of the needle carrier 148 and the planar member 128 of the inserter rod 114. In doing so, the retraction spring 152 constantly urges the needle carrier 148 upward.

While in the pre-use position, a large portion of the second chamber 110 remains open to the end of the device. In doing so, the set 106 can be positioned on the extended needle 134, at an opposite side of the second planar member 132 of the inserter rod 114. The set 106 can be gently held within the second chamber 110 through contact with the walls of the chamber, and/or through contact with the inserter needle 134. As noted elsewhere, the set 106 can include any number or configurations of adhesive pads (not shown) and other connection features, which can be accommodated by the insertion device 100.

As shown in FIGS. 3a-3e, the compression of the push button 104 releases the inclined detents 118 of the inserter rod 114 as shown in FIG. 3a, permitting the firing spring 126 to drive the needle 134, set 106, and adhesive pad into the region of the skin beneath the third chamber 112 of the device as shown in FIG. 3b, and also releases the spring 140 to lock the inserter rod 114 in the extended position as shown in FIG. 3c. That is, upon release, the inserter rod 114 is free to travel downward as urged by the trapped spring 126. In doing so, the inserter rod 114, including its end 132 and needle 134 travel downward through the third chamber 112, urging the set 106 downward with it. At or before reaching the travel limit of the inserter rod 114, the set 106 is positioned, retraction of the needle carrier 148 and needle 134 occurs, and the device can be removed as shown in FIG. 3d, thereby leaving the set 106 at the desired insertion site as shown in FIG. 3e.

To retract the needle carrier 148 and needle 134, no user action is required. As shown in FIGS. 3c-3d, the shoulder 130 serves to hold the inserter rod 114 in the down position. When the insertion is completed and the inserter rod 114 is captured in the down position by the outward movement of members 136 and 138, the movement simultaneously releases the needle carrier 148 and needle 134, and allows the needle carrier 148 and needle 134 to retract upward as urged by the retraction spring 152 as shown in FIGS. 3c and 3d. The needle carrier 148 and needle 134 are retracted until contacting an upper stop 156 disposed at an upper end of the inserter rod 114.

The insertion device 100 of FIGS. 3a-3e uses the single button press of the pushbutton 104 to release the firing mechanism, insert the set 106, and safely retract the needle 134 after insertion. The insertion device 100 is packaged and stored with the firing mechanism in an upright and locked position, with the firing spring 126 compressed as shown. The inserter rod 114 has the flexible wedge-shaped tabs or latches 120 at the top of it, which are locked into grooves or captured by the detents 116 at the interior top of the inserter's body 102 as shown in FIG. 3a.

Pressing the activation button 104 unlatches the firing tabs at the top of the unit as shown in FIG. 3b, and permits the firing spring 126 to extend toward its free length, propelling the needle carrier assembly 148 of the inserter rod 114 downward as shown in FIG. 3c. Upon reaching the wider region of the body interior shaft at the third chamber 112, the two spring-loaded locking halves or members 136 and 138 of the needle carrier assembly 148 move outward and lock the firing mechanism in the down position as shown in FIG. 3d, and frees the central portion of the needle carrier assembly 148 to move upward as shown in FIG. 3e, leaving the remainder of the inserter rod 114 in the down position.

By this time, the needle 134 has pierced the skin and placed the cannula and set 106, adhering it to the skin. Having been decoupled from the inserter rod 114 which has been driven downward and locked, the needle carrier assembly 148 is now free to rebound, propelled upward by the

retraction spring 152 as shown in FIG. 3e. After retracting fully upward, the needle 134 is stored permanently and inaccessibly inside the insertion device housing 102, and the device is inert.

In this and other inserter embodiments described below, the inserter body and elements can be constructed of any suitable and compatible materials such as plastic or metal. Springs can be provided as coil springs made of plastic or metal, although embodiments of the present invention are not limited thereto, and other spring or biasing means can be used, such as leaf spring or simply material resiliency. The insertion needle can comprise any suitable set insertion needle of metal or plastic, having length, thickness, and bevel dimensions suitable for set insertion.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to manage the degree of adhesion of the device in some manner as desired by a user. As shown by way of example, adhesion management elements 300 are shown in FIGS. 4a-4c. As shown in FIGS. 4a-4c, a set 314 and catheter 316 are shown encircled by concentric rings of adhesive 302 and 306. In an exemplary embodiment of the present invention, one ring can provide an adhesive with a higher degree of adhesion strength, and another ring can provide an adhesive with a lesser degree of adhesion strength, thereby allowing a user to tailor the degree of set adhesion to the user's activity plans. Although only two concentric rings of varied adhesive are shown in FIGS. 4a-4c, in yet other embodiments of the present invention, more rings can be provided, or the rings may be provided in alternate, non-circular shapes (i.e., such as oval shapes). Further, in the exemplary embodiment shown in FIGS. 4a-4c, the inner ring 306 is provided with the adhesive with a higher degree of adhesion strength and the outer ring 302 is provided with the adhesive with a lesser degree of adhesion strength, but embodiments of the present invention are not limited thereto. In yet other embodiments of the present invention, the order can be reversed or additional rings of adhesive provided.

As noted by those skilled in the art, a set typically requires a degree of adhesion to maintain proper positioning of the device. Accordingly, the exemplary embodiments of the present invention provide a set that comprises an adhesive pad or patch on the underside with a user-configurable adhesion mechanism to adapt the set to the expected environment in which it will be worn. As shown in FIG. 4a, the adhesive pad or patch can be provided in two or more, selectable strengths which can be variably exposed and utilized depending on the user's preference. A region of basic adhesive 302 having a removable cover 304 segmented from other covers, can be used for everyday needs while the region of extra-strong adhesive 306 remains covered by a similar segmented cover 308 as shown in FIG. 4b. The region of extra-strong adhesive 306 can be exposed by the removal of the cover 308 if athletic activities might be expected to stress the set adhesion as shown in FIG. 4c. To further simplify use, each cover 304 and 308 of each region can comprise a tab 310 and 312, respectively, to aid in removal of each adhesive cover. The covers 304 and 308 can also be labeled, color-coded or textured to show the adhesive strength thereunder.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide a simple but effective connection system as desired by a user. As shown by way of example, such a connection method 350 is shown in FIGS. 5a-5d.

11

In such an exemplary embodiment, a set **352**, once placed, has a port **354** that should be able to be easily connected and disconnected with a tubing **358** leading to an insulin pump (359). To do so, the set **352** can comprise the “self-sealing” connection port **354** on the outer, circular perimeter of the set **352**, within a circumferential groove **356** on the body of the set **352** as shown in FIG. **5a**. An incoming tube **358** can comprise a fitting **360** at the end designed to securely fit into and seal with the port **354** on the set **352**, using the assistance of a flexible, resilient ring **362** extending outward from it as shown in FIG. **5b**.

Connection of the tube **358** to the set **352** can then be accomplished by stretching the ring **362** around the far side of the set **352** as guided by the groove **356** to a first diameter to allow placement, then placing the connection fitting **360** into the port **354** as shown, and allowing the elastic ring **362** to contract to a second diameter to retain it securely as shown in FIGS. **5c** and **5d**. The tube, set and connectors can be constructed of any suitable material as described herein, and the ring **362** can be constructed of any compatible, resilient material which can be easily molded into the desired shape and maintain elasticity at least for an expected shelf life of the device. In a similar manner, the exemplary embodiment of the present invention shown in FIGS. **29a-29c**, described in greater detail below, comprises an infusion set constructed of a soft, pliable and/or elastic or similar material such that the infusion set is soft or pliable to a degree that allows the elasticity of the materials to affix the tube ring of the tubeset connector to the hub in any number of rotational positions and which further includes a tubeset connector needle to pierce the hub, wherein the elasticity of the materials function to seal the insertion site of the tubeset connector needle.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to allow site inspection in some manner as desired by a user. As shown by way of example, a site inspection embodiment **400** is shown in FIGS. **6a-6b**. As shown in FIGS. **6a-6b**, a set **404** and its housing can include a means to inspect the region of the skin immediately surrounding the insertion point, to ensure that the site is in good condition. In an exemplary embodiment of the present invention, the set **404** can include an element **402** extending from a top surface to a bottom surface of the set **404** at some point near the insertion site. As shown in FIGS. **6a-6b**, the element **402** completely encircles the site, but embodiments of the present invention are not limited thereto. In yet other embodiments of the present invention, the element **402** can be provided over a narrower portion, but still sufficient to view the site from above the device.

The element **402** can be constructed of any suitable material that can be easily manufactured, bonded with the remaining elements of the set **404**, provide compatibility with the contents or other materials, including the skin surface, and provide a degree of visibility between the top and bottom surfaces of the set **404**. As shown in FIG. **6b**, the sides of the element **402** can be configured, contoured or otherwise provided with features to be held in place by the body of the set **404**, and a top surface can be contoured to add a degree of magnification. For example, the housing of the set **404** or the top surface of the element **402** can include or comprise a clear plastic magnifying element that allows for even better site inspection abilities.

Further, the element **402** can be provided with an upper surface opening **412** to secure a septum **414** and for insertion of a placement needle **408** into a catheter **410**. The element **402** can further provide an opening **416** to facilitate intro-

12

duction of the infusion substance through the tubing **406**. Still further, where the lower surface of the set **404** is provided with an adhesive pad (not shown) at the base of the set **404**, the adhesive pad can include a cutaway portion to permit visibility through the element **402**.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to secure, contain, and/or conceal the tubing of the device in some manner as desired by a user. As shown by way of example, a tube management reel **450** is shown in FIGS. **7a-7b**. The tubing **452** connecting the insulin supply and pump (not shown) to the infusion set (not shown) can be packaged on a spring-loaded circular reel **454** disposed within or at one side of a reel housing **456**.

As shown in the cross-sectional view of FIG. **7b**, the tubing **452** can enter and exit the housing **456** at opposite sides near an upper surface, and wrap about a spring mechanism using pins **458**. The construction of a spring-loaded circular reel is known to those skilled in the art, so additional features of which are omitted for clarity. However, the circular reel is provided with pins **458** between which the tube **452** is secured within the housing **456** such that, feeding tubing from the housing results in the circular reel being wound tighter, and feeding tubing into the housing results in the circular reel being unwound and relaxed. Accordingly, the circular reel and pins maintain a tension on the tube **452** urging the tubing into the housing.

The circular reel can further comprise a catch/latch mechanism as known to those skilled in the art such that pulling the tube **452** a first time feeds a length of tube and a catch is provided to prevent a reverse spring-urged action. Upon pulling the tube **452** a second time, the catch can be released so that the reverse spring-urged action is released to urge the tube **452** back into the housing **456**. In doing so, the reel device allows slack tubing to be fed out precisely, with spring resistance maintaining the excess tubing rolled up and stored. The locking switch or latch can be provided to allow the user to prevent inadvertent retraction or extension once a satisfactory length of tubing has been deployed. The device can further comprise a switch **460** to actuate the tube retrieval.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide an insulin supply in some manner as desired by a user. As shown by way of example, an insulin supply **475** is shown in FIG. **8**. FIG. **8** illustrates an exemplary embodiment of an insulin container **476** and associated tubing **478**, including a tubing connection means **480**. The insulin supply **476** can be packaged in a small, sealed or sealable container that is pre-connected to a length of tubing **478**. The insulin container **476** can be integrated with a pump mechanism (not shown) externally via an installation process which the user can easily perform. The insulin container **476** and tubing **478**, once connected to a set and pump, form a system that does not need to be primed for proper function.

The insulin container **476** can be constructed of any suitable material, such as glass or plastic, to be clear to show the contents, or non-clear or opaque to protect contents from light. The container **476** can further include incremental dosage measurement marks along one or more surfaces for use during content delivery. The associated tubing **478** and connection means **480** can be constructed of any suitable material, such as rubber, to provide flexibility and compatibility with the contents. The connection means **480** can be

13

constructed in any number of ways, for example, including the connection means described in greater detail below in regard to FIGS. 22a-22c.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide set concealment in some manner as desired by a user. As shown by way of example, a concealment element **500** is shown in FIGS. 9a-9b. In FIG. 9a, an adhesive covering **502**, similar to a large adhesive bandage, can be provided to enable the user to cover the site, including the set **504**, with an inconspicuous dressing. In an exemplary embodiment of the present invention, the covering **502** comprises a flexible, skin-colored adhesive covering that can have an adhesive side and a non-adhesive side. The adhesive side can be covered with a user-removable cover (not shown) that when removed, allows the covering **502** to be secured over the site, thereby covering and to a large degree, concealing the set **504** as shown in FIG. 9b.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to aid and/or simplify placement of the device in some manner as desired by a user. As shown by way of example, a collection of placement assistance elements **525** is shown in FIGS. 10a-10c. Since the set and the corresponding insertion device can be unavoidably large in diameter and difficult to accurately locate in some circumstances, an exemplary kit including embodiments of the present invention can further include a placement ring to aid in placement of the set. An exemplary placement ring **526** is shown in FIG. 10a and can comprise a ring-shaped plastic part with one or more orienting features **530**, such as keys, on its perimeter, and a self-adhesive, covered pad on the underside (not shown). A low-profile, contoured circular plastic ring is shown, but embodiments of the present invention are not limited thereto. As the ring **526** can be provided with an adhesive pad, the ring **526** can be first gently adhered to the skin surface with the target insertion site at the center as shown in FIG. 10b. This allows careful set placement as the insertion site can now be better visualized through a center opening of the ring **526**, and the insertion device **528**, or tool, can be aligned with, and guided into final position, by the placement ring **526**.

The insertion device **528** can be provided for use with the placement ring **526**, and be constructed as described elsewhere herein and further having corresponding detents or keyways **532** to align with and receive the orienting features **530** of the placement ring **526**. When the insertion device **528** is to be placed atop the placement ring **526** as shown in FIG. 10c, the insertion device **528** self-aligns and orients for precise location of a set **534**. In the exemplary embodiment shown, the orienting features **530** are formed as raised contoured detents. Therefore, each further serves to guide, center and align the insertion device **528** upon placement. That is, such contoured elements provide a degree of self-alignment not as readily provided by square elements for example. After insertion, the set **534** is left remaining at a center of the ring **526** upon removal of the insertion device **528**. The placement ring **526** can then be removed and discarded.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional elements for the use of site preparation. Such an exemplary feature is shown in element **550** of FIG. 11. In the exemplary embodiment shown in FIG. 11, set packaging can further include a site preparation wipe **554** contained within a preservation container **552**. The packaging for the set can

14

include the wipe **554**, such as a versatile disposable wipe, paper or cloth pad that is soaked or impregnated with one or more of a disinfectant, local anesthetic or other helpful substance. The pad or wipe **554** may also be constructed having a texture, coating, or other surface feature **556** that can provide an exfoliating ability to aid in anesthetic effectiveness.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide a simple but effective means to provide packaging as a number of sets upon a tray as desired by a user. As shown by way of example, a "sets on tray" packaging arrangement **575** is shown in FIGS. 12a-12f. As shown in FIGS. 12a-12f, a number of exemplary sets **576** can be packaged upon a tray **578**. Although the exemplary tray of FIG. 12a shows the containment of three sets, any number or arrangements of sets can be included as desired by the user.

As shown in FIG. 12a, a number of disposable sets or set components **576** can be packaged in the exemplary multi-unit, foil or plastic-sealed tray **578**. In the exemplary embodiment of the present invention shown in FIG. 12a, the tray **578** can be constructed of any suitable material compatible with the set and set components to be stored therein, and can provide a number of recessed, contoured or otherwise constructed openings **580** into which the sets or set elements **576** can be positioned. The openings **580** can be configured to securely hold and protect the sets prior to use, allow easy covering of the sets and tray surface with a sealing means, such as foil or other material which can then be easily removed or punctured by the user to access the desired set and maintain protection of remaining sets as shown in FIG. 12a, and which allows such access and removal using an inserter device **582** as shown in FIGS. 12c-12f.

As illustrated in the cross-sectional views of FIGS. 12c-12f, an exemplary insertion device **582** is shown having a contoured shape into which the set **576** can be captured. To do so, the insertion device **582** can comprise one or more deflectable ends **584** which can have an inclined latch **590** to pierce the tray covering **588** as shown in FIG. 12d, deflect outward slightly due to contact with the set **576**, and then capture an outer circumference of the set **576** once the latch **590** is fully inserted. In doing so, the insertion device **582** can be used to extract a new set **576** from the package so that the user will not need to contend with opening and unsealing packaging materials. For example, an exemplary and reusable insertion device **582** for use which such a tray can comprise a hollow underside or lower surface, with the engagement features of the latches **590** oriented inward towards the set **576**, such that a user is simply required to align the insertion device **582** with a set, insert to a sufficient depth and retrieve the set **576** for use. No further user action is required in regard to handling the set **576**.

As shown in FIG. 12c, the sets **576** are aligned within each opening of the tray with sufficient clearance below the set to accommodate any elements of the set. A substantial portion of the upper surface of the set can be exposed, wherein a foil or other covering **588** can be used to secure the set **576** within the tray, and seal the contents of the tray and set **576** from contamination or other damage. In this exemplary embodiment of the present invention, the foil **588** is shown covering the limited space surrounding each device, but is not limited thereto. In yet other embodiments of the present invention the foil or covering can be more or less extensive upon the tray surface as desired.

15

As shown in FIG. 12*d*, the insertion device 582 can self-align on the blister-type package of the set 576 on the tray 578, and when pressed down, can cut through the foil or plastic seal 588 as shown in FIGS. 12*c* and 12*d*. As the insertion device 582 presses down further, it also disengages the set 576 from the packaging tray 578 with the perimeter of the inserter body. After pushing past the edge of the set 576, the locking tabs or latches 590 on the insertion device 582 engage and secure the outer circumference of the set 576, and allow the user to extract both the insertion device 582 and the set 576 from the tray 578 as shown in FIG. 12*f*. The insertion device and the set are then ready to fire in normal use. The remaining sets of the tray are left intact and ready for later use. In exemplary embodiments of the present invention, the removal of the set 576 from the tray also results in the automatic removal of any needle cover and adhesive backing, which is left with the tray 578.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide a simple but effective multistage-type inserter as desired by a user. As shown by way of example, a "multistage-type" inserter device 600 is shown in FIGS. 13*a*-13*f*. In such an exemplary embodiment, the insertion device can be constructed in such a way as to separate high-cost parts from low-cost parts, keeping the former in a reusable mechanism while allowing the latter to be safely disposed.

As shown in FIG. 13*a*, the multistage-type inserter device 600 can comprise a substantially cylindrical upper and lower element 602 and 604, respectively. The upper element 602 can have a first diameter at a lower portion thereof to seamlessly mate with the lower element 604. An upper portion of the upper element 602 can have a second diameter which is flared or expanded to provide sufficient room for operation of hinged latches 606 as described in greater detail below. The lower element 604 can also have a first diameter at an upper portion thereof to seamlessly mate with the upper element 602, and a second diameter at a lower portion which is flared or expanded to contain a set 642.

The upper element 602 can comprise at least one firing mechanism consisting of at least one hinged latch 606 rotatable about a pin 608 or other means, and which has an inclined projection 610. Each projection 610 includes an inclined lower surface to facilitate assembly with the lower element 604, and a substantially flat upper surface to restrict some portion of an inserter rod 614 as described in greater detail below. The upper element 602 further comprises a first chamber 612 in which a firing spring 624 is captured. The upper element 602 and contents thereof can comprise a reusable element that can be installed onto a disposable mechanism of the lower element 604 which can include a set, needle, adhesive pad, and a portion of the insertion mechanism.

The inserter rod 614 extends through both the upper and lower elements 602 and 604, and comprises a cross-member 616, a shoulder 618 and a planar end 620. The inserter rod 614 further comprises an inserter needle 622, which can be secured within a center opening of the inserter rod 614, and can extend downward from the rod at the end 620. Both the cross-member 616 and the lower end 620 are configured to have a width substantially equal to the width of the chamber in which each is positioned to facilitate alignment and travel of the inserter rod 614 during use. As described in greater detail below, the cross-member 616 is configured to be held in an up position by the projections 610 of the latches 606, and is configured to be blocked at a down position by the projections 633 of the lower element 604. Further, the lower

16

part of the shoulder 618 is configured to have a partially flat surface upon which the retraction spring 638 rests, and a partially inclined surface such that the shoulder can be easily forced through the opening of the partition 634 by the firing spring 624. The upper part of the shoulder 618 is configured to have a substantially flat surface to be captured by the opening of the partition 634 and prevent upward travel of the inserter rod 614 for retraction until released.

The upper portion 602 comprises the first chamber 612 in which the firing spring 624 is captured. The firing spring 624 is disposed concentrically about the inserter rod 614 and is captured between an upper wall of the first chamber 612 and the cross-member 616 of the inserter rod 614. In doing so, the firing spring 624 is configured to constantly urge the inserter rod 614 downward. Prior to use, the inserter rod 614 is held in an up position by one or more of the projections 610 of the hinged latch 606. Specifically, an inner surface of the hinged latch 606 comprises one or more of the projections 610 which extend a slight distance from the inner surface of the hinged latch 606, and which block the travel of the cross-member 616 of the inserter rod 614. In such a position, the firing spring 624 is compressed and the latches 606 capture the upper portion of the needle assembly as shown in FIG. 13*b*. The capture of the upper portion of the inserter rod 614 by the projections 610 of the latches 606, and the force applied by the firing spring 624 while in the pre-use position, also serves to secure the upper housing 602 to the bottom housing 604 prior to use. Once the latches 606 are released from the inserter rod 614, the upper housing 602 is free of the lower housing 604 and can be lifted away as shown in FIGS. 13*c*-13*d*.

The upper housing 602 can further comprise an opening 642 which can serve to support the firing spring 624 in position, and serve to guide the inserter rod 614 during use. The opening 642 can further reveal an extended portion of the inserter rod 614 as shown in FIG. 13*b* such that a user can confirm visually or by touch that the elements are all present and are properly assembled and ready for use.

The lower portion 604 comprises a second, third and fourth chamber 626, 628 and 630. The second chamber 626 is substantially open at an upper surface to slidably receive the inserter rod 614 as guided by the cross-member 616 as urged downward by the firing spring 624 when released. The second chamber 626 comprises at least one projection 633 which extends inward from an inner surface of the second chamber 626. In doing so, the projection 633 provides a downward travel limit of the inserter rod 614 through the contact between the cross-member 616 and the projection 633. As noted above, the width of the second chamber and the cross-member 616 are configured such that the inserter rod 614 is centered and guided by each.

The second and third chambers 626 and 628 are separated by a partition 634 having an opening 640 through which the inserter rod 614 extends. The opening 640 of the partition 634 is configured to have an inclined upper opening surface through which the inclined lower surface of the shoulder 618 can more easily pass as urged downward by the firing spring 624. The lower surface of the opening 640 of the partition 634 is configured to be substantially flat such that the flat upper surface of the shoulder 618 cannot pass back through the partition 634 until released for retraction as described in greater detail below. Further, the partition 634 comprises at least one segment extending some distance from an outer surface of the lower element 604 (i.e., an extended user lever) such that the partition 634 can be deflected by the user for retraction as described in greater detail below.

17

The third and fourth chambers are also separated by a partition 636, which also includes an opening through which the inserter rod 614 extends. The third chamber 628 further comprises the retraction spring 638. The retraction spring is positioned concentric with the inserter rod 614, and is captured within the third chamber 628 between the partition 636 and the shoulder 618 of the inserter rod 614. In doing so, the retraction spring 638 is configured to constantly urge the inserter rod 614 upward.

Prior to use, the firing spring 624 in the upper portion 602 is compressed and the retraction spring 638 in the lower portion 604 is relaxed as shown in FIG. 13b. During use, the release of the firing spring 624 operates the inserter as described in greater detail below, and further serves to compress the retraction spring 638 as shown in FIG. 13c. After use, the inserter rod 614 is held in position by the contact between the shoulder 618 and the partition 634 as shown in FIG. 13d. To retract the inserter rod 614, the user then presses on the extended portions of the partition 634. As shown in FIGS. 13d and 13e, the restrictive opening 640 of the partition 634 serves to hold the inserter rod 614 in the down position. When the insertion is completed and the upper portion 602 is removed, the user can press the extended portions of the partition 634, and the opening 640 is enlarged and allows the shoulder 618 to retract upward as urged by the retraction spring 638 as shown in FIG. 13e.

While in the pre-use position, a large portion of the lower portion 604 remains open to the end of the device. In doing so, the set 642 can be positioned on the extended needle 622, at an opposite side of the end 620 of the inserter rod 614. The set 642 can be gently held within the portion 604 through contact with the walls of the portion, and/or through contact with the inserter needle 622. As noted elsewhere, the set can include any number or configurations of adhesive pads (not shown) and other connection features, which can be accommodated by the two-part inserter.

As shown in FIGS. 13a-13f, the compression of the latches 606 releases the projection latch on the needle assembly of the inserter rod 614, permitting the firing spring 624 to drive the needle 622, set 642, and adhesive pad into the region of the skin beneath the portion 604, and also compress the retraction spring 638. That is, upon release, the inserter rod 614 is free to travel downward as urged by the trapped spring 624. In doing so, the inserter rod 614, including its end 620 and needle 622 travel downward through the portion 604, urging the set 642 downward with it. At or before reaching the travel limit of the inserter rod 614, the set is positioned, and the device can be removed as shown in FIG. 13f, thereby leaving the set 642 at the desired insertion site.

Then, to retract the inserter rod 614, including its end 620 and needle 622, the user then presses on the extended portions of the partition 634. As shown in FIGS. 13d and 13e, the restrictive opening 640 of the partition 634 serves to hold the inserter rod 614 in the down position. When the insertion is completed and the upper portion 602 is removed, the user can press the extended portions of the partition 634, and the opening 640 is enlarged and allows the shoulder 618 to retract upward as urged by the retraction spring 638 as shown in FIG. 13e. The inserter rod 614 is prevented from completely exiting the lower portion 604 by the contact between the lower end 620 and the partition 636.

As noted above, the exemplary embodiment of the present invention illustrates an insertion device 600 that can be constructed in such a way as to separate high-cost parts from low-cost parts, keeping the former in a reusable mechanism while allowing the latter to be safely disposed as shown in

18

FIG. 13a. A firing mechanism consisting of the hinged latches 606 and an extended, large spring is installed onto a disposable mechanism which includes the set 642, the needle 622, the adhesive pad, and a portion of the insertion mechanism. As shown, the firing spring 624 is compressed and the latches 606 capture the upper portion of the needle assembly or inserter rod 614 as shown in FIG. 13b. The device is now ready to be placed and fired.

Once the user squeezes the latches 606 on the upper part of the device, the needle assembly or inserter rod 614 becomes free to move, and is driven downward by the firing spring 624, piercing the skin, inserting and adhering the set 642 as shown in FIG. 13c. The downward motion of the needle assembly or inserter rod 614 also drives two secondary latches on the lower portion of the device outward to capture the inserter rod 614 within the opening 640 and compresses the return or retraction spring 638. After removal of the reusable part 602 of the device 600, the lower part of the device 604 remains in place and the needle will still be inserted in the skin as shown in FIG. 13d. To remove the needle 622, the latches or the member 634 can be deflected as shown in FIG. 13e to allow the return or retraction spring 638 to extract the needle 622 and render the lower part 604 of the assembly 600 inert and disposable. The inserter rod 614 is prevented from completely exiting the lower portion 604 by the contact between the lower end 620 and the partition 636.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide a simple but effective "squeeze-type" inserter as desired by a user. As shown by way of example, a "squeeze-type" inserter device 700 is shown in FIGS. 14a-14d. As shown in FIGS. 14a-14d, an exemplary embodiment of the present invention can be activated by pressing the device against the targeted portion of the skin and then squeezing a portion of the inserter body.

An exemplary construction of the squeeze-type insertion device 700 can comprise a body having at least three portions 702, 704 and 706. The portions 702 and 706 can comprise rigid structures, having a flexible user-deflectable portion 704 disposed therebetween. The upper portion 702 can comprise a substantially cylindrical outer surface having a rounded and closed end, and an opposite end which seamlessly mates with the outer surface of the user-deflectable portion 704. The lower portion 706 can have a larger diameter, substantially cylindrical outer surface, with a contour configured to seamlessly mate with the outer surface of the user-deflectable portion 704. Accordingly, the user-deflectable portion 704 can have a substantially cylindrical outer surface having a contour to provide seamless mating with the outer surface of the upper and lower portions 702 and 706. The portions 702, 704 and 706 can be constructed as a single element wherein each section is separated from the adjacent section by one or more cuts, which allow the segments to maintain assembly, but allow for the deflection of the user-deflectable portion 704 as described in greater detail below. Further, an inserter rod 712 is slidably captured within the device 700 and extends through each portion as guided by an inserter rod cross-member 714. The inserter rod 712 further comprises a lower planar end 722 and an inserter needle 724, which can be secured within a center opening of the inserter rod 708, and can extend downward from the inserter rod 712 at the end 722. Both the cross-member 714 and the lower end 722 are configured to have a width substantially equal to the width of the chamber in which each is positioned to facilitate alignment and travel of the inserter rod 712 during use.

19

As shown in FIGS. 14a-14c, the upper portion 702 provides a first chamber 708 in which a firing spring 710 is captured. Specifically, the firing spring 710 is positioned concentric with the inserter rod 712, and is captured at one end by an upper wall of the first chamber 708, and at an opposite end by the inserter rod cross-member 714. In doing so, the firing spring 710 is configured to constantly urge the inserter rod 712 downward.

Prior to use, the inserter rod 712 is held in position by one or more projections 716. Specifically, an inner circumference of the user-deflectable portion 704 comprises one or more of the projections 716 which extend a slight distance from the inner circumference of the user-deflectable portion 704, and which block the downward travel of the cross-member 714 of the inserter rod 712. The projections 716 are provided at an uppermost point of the portion 704 as this point undergoes the greatest degree of deflection during user deflection as described in greater detail below. Accordingly, in such a position, the degree of deflection of the projections 716 is maximized as shown in FIG. 14b to simplify the release of the inserter rod 712.

The user-deflectable portion 704 provides a second chamber 718 through which the inserter rod 712 is positioned and which includes a travel limit element 720. Although any part of the user-deflectable portion 704 can be compressed by a user, an exemplary embodiment of the present invention can provide one or more designations on an outer surface of the user-deflectable portion 704 to identify each as a preferred "squeeze button" area. Once the user-deflectable portion 704 is compressed by the user as shown in FIG. 14b, the projections 716 release the cross-member 714 of the inserter rod 712 and the firing spring 710 urges the inserter rod 712 downward through the second chamber 718 until the cross-member 714 is restricted from further downward travel by the travel limit element 720 as shown in FIG. 14c. As noted above, the portions 702, 704 and 706 can be constructed as a single element wherein each section is separated from the adjacent section by one or more cuts, which allow the segments to maintain assembly but allow for the deflection of the user-deflectable portion 704. Accordingly, the engagement between the portions 704 and 708 can be deflected in a manner similar to that of the deflection between the portions 702 and 704 described above in regard to the release of the inserter rod 712.

While in the pre-use position, a large portion of the lower portion 706 remains open to the end of the device. In doing so, a set 726 can be positioned on the extended needle 724, at an opposite side of the end 722 of the inserter rod 712. The set 726 can be gently held within the portion 706 through contact with the walls of the portion, and/or through contact with the inserter needle 724. As noted elsewhere, the set 726 can include any number or configurations of adhesive pads (not shown) and other connection features, which can be accommodated by the squeeze-type inserter.

As shown in FIGS. 14a-14d, the compression of the user-deflectable portion 704 releases the projection latch 716 on the needle assembly of the inserter rod 712, permitting the firing spring 710 to drive the needle 724, set 726, and adhesive pad into the region of the skin beneath the portion 706. That is, upon release, the inserter rod 712 is free to travel downward as urged by the trapped spring 710. In doing so, the inserter rod 712, including its end 722 and needle 724 travel downward through the portion 706, urging the set 726 downward with it. At or before reaching the travel limit of the inserter rod 712, the set 726 is positioned, and the device 700 can be removed as shown in FIG. 14d, thereby leaving the set 726 at the desired insertion site.

20

Such an exemplary embodiment of the present invention provides a more convenient means of actuating an insertion device by simply using "squeeze" buttons to initiate the process as shown in FIG. 14a. As shown in FIG. 14b, buttons on the side of the device at the user-deflectable portion 704 can be pressed inward, unlatching the needle assembly or inserter rod 712 and permitting it to be propelled downward by the firing spring 710 as shown in FIGS. 14b and 14c. The spring-loaded needle delivers the set 726 and adhesive pad to the skin. The device 700 can then be removed as shown in FIG. 14d, thereby leaving the set 726 at the desired insertion site.

In contrast to the other inserter devices described herein, the materials of the squeeze-type inserter 700 allow for the squeeze action of the one or more portions, or the construction of one or more portions having thinner, thereby deflectable parts.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide a simple but effective skin contact-type inserter as desired by a user. As shown by way of example, a "skin contact-type" inserter device 800 is shown in FIGS. 15a-15e. As shown in FIGS. 15a-15e, an exemplary embodiment of the present invention is shown wherein the insertion device can be activated by pressing the device up against the targeted portion of the skin.

An exemplary construction of the insertion device 800 can comprise an upper and lower portion 802 and 804, respectively. The upper portion 802 can be comprised of a dome-shaped element into which a slidably engaged lower portion 804 can be captured. Specifically, the upper portion 802 can comprise a substantially round, dome-shaped element having a flattened portion at an uppermost point. The upper portion 802 can further comprise one or more deflectable tabs 806, or cut portions, about a body circumference as described in greater detail below. The inner surface of the tabs 806 are provided with an incline 824 and a projection 818 at the uppermost point of each incline.

Within the upper portion 802, exemplary embodiments of the present invention comprise an inserter rod 808 slidably disposed within the upper portion and which is held in place by the tabs 806. As shown in FIGS. 15b-15e, the inserter rod 808 comprises a planar end 814 wherein a notch 816 is proved at each side of the planar end 814, which engages the projections 818 of the tabs 806 that extend into the upper portion 802. Accordingly, prior to use, the projections 818 secure the notches 816 of the planar end 814 of the inserter rod 808, preventing movement of the inserter rod 808. The inserter rod 808 further comprises an inserter needle 828, which can be secured within a center opening of the inserter rod 808, and can extend downward from the rod at the planar end 814.

The upper portion 802 further comprises a firing spring 810 that is disposed concentrically with the inserter rod 808 and which is captured between the end 814 of the inserter rod 808 and one or more features of the upper flattened surface of the upper portion 802. In the exemplary embodiment shown, the upper flattened surface of the upper portion 802 can comprise a molding 820 having a cup-shaped opening 812 that is sized to capture and hold an end of the firing spring 810. In this position, the firing spring 810 serves to constantly urge the inserter rod 808 downward. However, as noted above, prior to use, the inserter rod 808 is held in place by the tabs 806.

Accordingly, the device 800 further comprises the slidably engaged lower portion 804 which is slidably captured

21

at an end of the upper portion **802** by one or more contacting tabs **822** of portion **804**, and tabs **832** of the upper portion **802**. The lower portion **804** has a diameter slightly less than an inner diameter of the upper portion **802** such that the lower portion **804** is gently held in place prior to use but upon contact with a skin surface, can be easily slid upward into the upper portion **802**. In doing so, the tabs **822** of the lower portion **804** travel against an inner circumference of the upper portion **802**, and onto the tabs **806**. Specifically, the tabs **822** of the lower portion **804** travel against the incline **824** of the tabs **806**, which forces the tabs **806** outward and releases the projections **818** from the notches **816** of the planar end **814** of the inserter rod **808** as shown in FIG. **15c**. Once released in such a manner, the inserter rod **808** is free to travel downward as urged by the firing spring **810**.

While in the pre-use position, a large portion of both the upper and lower portions **802** and **804** remain open to the end of the device **800**. In doing so, a set **826** can be positioned on the extended needle **828**, at an opposite side of the planar end **814** of the inserter rod **808**. The set **826** can be gently held within the upper portion **802** through contact with the walls of the upper or lower portion, and/or through contact with the inserter needle **828**. As noted elsewhere, the set **826** can include any number or configurations of adhesive pads (not shown) and other connection features, which can be accommodated by the skin contact inserter.

As shown in FIGS. **15b-15d**, the slidable motion of the lower portion **804** into the upper portion **802** releases the pin latch **818** on the needle assembly of the inserter rod **808**, permitting the firing spring **810** to drive the needle **828**, set **826**, and adhesive pad into the region of the skin beneath the lower portion **804**. That is, upon release, the inserter rod **808** is free to travel downward as urged by the trapped firing spring **810**. In doing so, the inserter rod **810**, including its end **814** and needle **828** travel downward through the lower portion **804**, urging the set **826** downward with it. At or before reaching the travel limit of the inserter rod **808**, the set **826** is positioned, and the device **800** can be removed as shown in FIG. **15e**, thereby leaving the set **826** at the desired insertion site.

As described above, the lower portion **804** of the device becomes a movable actuation mechanism. In this case, a circumference or ring around the lower perimeter of the lower portion **804** of the device, surrounding the set **826** and needle **828**, is placed against the skin surface (not shown). The entirety of the device **800** is then pressed firmly downward. The lower ring of the lower portion **804** telescopes into the main body of the upper portion **802** of the device and as it travels inward/upward, it deflects the multiple latches around the perimeter provided by the tabs **806**, which permit the firing spring **810** to drive the needle **828**, set **826**, and adhesive pad into the skin. The device **800** can be removed as shown in FIG. **15e**, thereby leaving the set **826** at the desired insertion site.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide a needle handle and/or shroud in some manner as desired by a user. As shown by way of example, a needle handle and shroud **900** is shown in FIGS. **16a-16c**. In doing so, an enhancement can be provided to current manually inserted needles for set insertion by implementing a plastic handle **902** and a substantially circular and hinged shield **904** for use with the needle **906** and set (not shown). The handle **902** permits a sure grip on the needle **906** for insertion, while the circular shield **904**, as shown, helps to firmly press the set and set

22

adhesive into place as shown in FIG. **16a**. After placement, when the needle **906** is withdrawn, the hinged portions of the shield **904** can be folded down at each of hinges **908**, as shown in FIG. **16b** and the cross-sectional view of FIG. **16c**, to allow for safer needle disposal.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide a simple but effective skin pinch-type inserter as desired by a user. As shown by way of example, a "skin pinch-type" inserter device **925** is shown in FIGS. **17a-17d**. As shown in FIGS. **17a-17d**, an exemplary set can be packaged with a skin pinch-type inserter **925** with which a user can grasp a portion of the user's skin **S** (see, e.g., FIG. **17a**) to improve insertion of the set. The exemplary insertion device **925** can be integrated into a mechanism as described in greater detail below that incorporates features to do this.

An exemplary construction of the insertion device **925** can comprise a large clothespin-like clamp **926**, having opposite sides or legs **928** which rotate about a pin or other securing means **930**. Such an exemplary clamp **926** can further provide a space therein in which a number of remaining components can be centered. The large clothespin-like clamp **926** can include the wide, manually-actuated legs **928** as shown in FIGS. **17a-17b**. The legs **928** can be configured to be at slight angles while at rest such that a width between the legs **928** at the skin contact surface is wider than a width between the legs **928** at opposite ends. When activated by a user, the user places the entire device **925** against the skin surface such that the wider opening contacts the skin surface. The user then grasps the legs **928** at a point near the skin surface and "pinches" the legs **928** into a substantially parallel position, such that the legs **928** extend vertically from the skin surface as shown in FIG. **17c**. That is, the device can be placed on the skin surface, surrounding the target area, and the pinching legs **928** can be squeezed inwards. When pinched, the legs **928** in contact with the skin surface serve to prepare the skin surface in a number of different ways for set placement, including, but not limited to, stretching the skin at the site, leveling the skin at the site and/or raising the skin at the site.

The squeezing of the legs **928** is also tied to a latching mechanism in the central section of the device **925**. That is, when a desired degree of skin pinch is achieved, the device automatically begins the process of releasing and placing the set. To do so, the device **925** can further comprise a first and second chamber therein. The first chamber **932** is secured to the second chamber **934**, which is secured at opposite corners to an inner surface of the legs **928**. The legs **928** further comprise at least one articulated pin **936** which is inserted into the first chamber **932** when the device is at rest as shown in FIG. **17b**. The articulated pin **936** can include one or more joints or segments along its length, such that there is no interference with the movement of the legs **928** during operation.

The articulated pin **936** extends from an inner surface of the legs **928**, through an opening in the side of the first chamber **932** and secures a planar end **938** of an inserter rod **940**. As shown in FIG. **17b**, in doing so, the pin **936** holds the inserter rod **940** in an up and retracted position. As the legs **928** are pinched, the articulated pin **936** is pulled clear of the planar end **938** which allows the inserter rod **940** to move forward as urged by a firing spring **946**.

As shown in FIGS. **17b-17c**, the firing spring **946** is captured in the second chamber **934**. Specifically, the first and second chambers are in communication via an opening therebetween through which the inserter rod **940** extends.

23

The planar end **938** of the inserter rod **940** is captured in the first chamber, and can be held at an upper and retracted position in the first chamber by the articulated pin **936**. The remainder of the inserter rod extends into the second chamber **934** and terminates at an opposite planar end **944**, and further comprises an inserter needle **942**. The inserter needle **942** can be secured within a center opening of the inserter rod **940**, and can extend from the rod at the planar end **944** into the second chamber **934**. Both planar ends **938** and **944** are configured to have a width substantially equal to the width of the chamber in which each is positioned to facilitate alignment and travel of the inserter rod **940** during use.

The firing spring **946** is placed concentrically about the inserter rod **940** and is captured by the inserter rod **940** within the second chamber **934**, between the end **944** and an upper wall of the second chamber **934**, and is configured to be in a compressed state prior to use, and upon release of the articulated pin **936**, is further configured to urge the inserter rod **940** downward toward the insertion site. As shown in FIG. **17b**, while in the pre-use position, a large portion of the second chamber **934** at an opposite side of the planar end **944** remains open to the end of the device. In doing so, a set **948** can be positioned on the extended needle **942** within the second chamber **934**, at an opposite side of the planar end **944** of the inserter rod **940**. The set **948** can be gently held within the second chamber **934** through contact with the walls of the chamber, and/or through contact with the inserter needle **942**. As noted elsewhere, the set **948** can include any number or configurations of adhesive pads (not shown) and other connection features, which can be accommodated by the skin pinch inserter.

As shown in FIGS. **17b-17c**, the motion of the legs **928** releases the pin **936** latch on the needle assembly of the inserter rod **940**, permitting the firing spring **946** to drive the needle **942**, set **948**, and adhesive pad into the pinched region of the skin. That is, upon release of the articulated pin **936** from the first chamber **932**, the inserter rod **940** is free to travel downward as urged by the trapped firing spring **946**. In doing so, the inserter rod **940**, including its planar end **944** and needle **942** travel downward through the second chamber **934**, urging the set **948** downward with it. At or before reaching the travel limit of the inserter rod **940**, the set **948** is positioned, and the device **925** can be removed as shown in FIG. **17d**, thereby leaving the set **948** at the desired insertion site.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide a simple but effective folding/retractable inserter as desired by a user. As shown by way of example, a "folding/retractable" inserter device **1000** is shown in FIGS. **18a-18e**. As shown in FIGS. **18a-18e**, an exemplary set can be packaged with a hinged, vertical protruding handle **1002** which improves the ability of the user to place it and insert it as shown in FIGS. **18b-18c**, but which can be folded out of the way as shown in FIGS. **18a** and **18d**. To do so, the inserter **1004** can be constructed having a hinge, pivot pin or pivot point, or other flexible-type element **1006** which allows the handle to extend from the set **1008** at a number of angles, especially when the set **1008** is adhered to a skin surface.

As shown the cross-sectional views of FIGS. **18b-18d**, the folding/retractable inserter **1000** can comprise the handle **1002** in which a chamber **1010** is provided to contain an insertion needle **1012** and retraction spring **1014**. The handle **1002** further comprises at least one opening **1016** through which a user-accessible button **1018** can extend. As shown in FIG. **18b**, the opening **1016** and user-accessible button

24

1018 restrict travel of an end of the needle **1012** such that the retraction spring **1014** is compressed and held in the state shown in FIG. **18b**. In such a position, the inserter can be used in a conventional manner. Once the set is in place, the user can press the button **1018** such that the needle **1012** is released and the retraction spring **1014** can retract the needle **1012** from the set and into a protected and covered position within the chamber **1010** of the handle **1002** as shown in FIG. **18c**.

As more clearly shown in FIG. **18c**, the handle **1002**, hinge **1006**, and chamber **1010** can further comprise an opening **1020** through which the needle **1012** can be extended and retracted. The opening **1020** is aligned when the handle **1002** is in the upright and perpendicular position. Once the needle **1012** is retracted into the chamber **1010**, the handle **1002** can be rotated about the hinge **1006** to any number of positions, but preferably is rotated to a down and substantially parallel position as shown in FIG. **18d**. Once in this position, the inserter is left with the set at the site, but maintains a low profile due to the rotation of the handle **1002**. Further, no danger is presented by the needle **1012** of the inserter as it is fully retracted into the chamber **1010** of the handle **1002**, and is blocked from further escape as the opening **1020** is now blocked.

As noted above, the device **1000** can include the user-accessible button **1018** at any convenient location along the handle **1002**, such that the button **1018** locks the needle **1012** in an extended position for insertion as shown in FIG. **18b**. After insertion, the button **1018** can be activated or toggled, allowing the internal retraction spring **1014** to retract the needle **1012** vertically and automatically to a safe position within the chamber **1010** of the handle **1002** as shown in FIG. **18c**. With the needle **1012** fully retracted into the handle body, the handle can be folded down by way of the hinge **1006** as shown in FIG. **18d**. Such a hinge **1006** can further comprise a detented hinge to assist in maintaining the folded or upright position of the handle **1002** as shown in FIG. **18e**, and to ensure that in the folded position, the sharp component is completely inaccessible and the device assumes and maintains a low profile. To do so, a number of detents **1022** can be provided to gently secure the handle **1002** in either the up or down positions.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide a simple but effective connection method between the tube connection and set in some manner as desired by a user. As shown by way of example, a "squeeze-type" connection device **1100** is shown in FIGS. **19a-19c**. As shown in FIGS. **19a-19c**, a connection method is provided as a squeeze latch **1100** having one or more hinged and rotatable push buttons within the device that when pressed, release a shoulder, notch or other element of the set.

For example, as shown in FIG. **19a**, inside the body of the set **1104**, two push button latches **1102** are provided comprising rotatable L shaped elements, with pivot points **1108** located as shown in FIGS. **19a** and **19c**, and which can be used to secure the tube connection **1106** with the set **1104**. The latches **1102** can be urged into the securing position of FIG. **19a** using springs, material resilience or other means (not shown). Squeezing the exposed portions of the two latches **1102** toward the center of the device **1100** causes the internal arms of the latches **1102** to swing free of the connection with the set **1104** and release it as shown in FIG. **19c**. Once released, the tube connection **1106** and latches **1102** can be lifted from the set **1104**. In a similar manner, squeezing the exposed portions of the two latches **1102**

25

toward the center of the device **1100** can be used to place the tube connection **1106** back on the set **1104**. Further, such a latching embodiment allows the tube connection **1106** to rotate about the set **1104** prior to, during and after assembly.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to provide a connection tool and a set configured for use with such a connection tool in some manner as desired by a user. As shown by way of example, a connection tool **1154**, tubing connection **1152**, and set **1158** are shown in FIGS. **20a-20d**. That is, the device of FIGS. **20a-20d** illustrate an exemplary embodiment of a tool removable connection. Since user accessible connection mechanisms can be bulky or inadvertently actuated, a compromise can be to provide a release mechanism, but to require the use of a removal tool to trigger it as shown in FIG. **20a**.

As shown in FIGS. **20a-20c**, the tubing connection **1152** residing on top of the set **1158** will have a low profile, and the housing of the tubing connection **1152** comprises one or more small access openings **1162** as shown. While these openings **1162** are small enough to resist inadvertent disconnection, they permit insertion of specialized features on the external connection tool **1154**, which actuates the hinged and rotatable latches **1156** positioned inside the tubing connection **1152** to release it as shown in the cross-sectional views of FIGS. **20b-20d**.

Specifically, the low profile tube connection **1152** is secured to the set **1158** using the hinged and rotatable latches **1156** positioned inside the tubing connection **1152**. The tube connection **1152** comprises one or more rotatable brackets **1156** which are configured to rotate about pins **1170**. Each of the rotatable latches comprises a shoulder **1172** configured to capture and securely hold a similar shoulder **1174** provided upon the set **1158**. Accordingly, in an assembled positions as shown in FIG. **20b**, the brackets **1156** are rotated into a position that secures the set **1158**. That is, when the connection tool **1154** is not assembled with the low profile tube connection **1152** and set **1158**, the hinged and rotatable latches **1156** are in upright and secure positions as shown in FIG. **20b**. Any number of means can be used to maintain the latches **1156** in such an upright position such as springs, material elasticity, and/or contact with the skin surface. While in this position, the latches **1156** serve to capture and secure one or more of the shoulders **1174** of the set **1158**.

The connection tool **1154** comprises a substantially cylindrical body having one or more projections or pins **1160** extending from a lower surface. The lower surface can be contoured to more closely match the upper surface of the tube connection **1152**. Further, the number and arrangement of pins **1160** and/or openings **1162** can be provided based upon the set such that a particular connection tool can be used with only one or more sets, or a universal tool can be provided to work with all sets. The tool **1154** can be made of any lightweight, strong material such as plastic or metal, and the pins **1160** can be constructed of similar materials to simplify manufacture.

When the connection tool **1154** is pressed against an upper surface of the low profile tube connection **1152**, one or more of the pins **1160** extending from the tool **1154** enter the connection **1152** through one or more of the openings **1162** and contact the latches **1156**. A notch **1164** can be provided in the tool **1154** to allow clearance for the tube **1166**, and the tool can further comprise a contour **1168** and/or other surface or gripping features. Once engaged, the pins **1160** of the tool **1154** press the latches **1156** downward, from the upright and secured positions thereby releasing the

26

set **1158** as shown in FIG. **20c**. After the connection **1152** is released from the set **1158**, the tool **1154** and tube connection **1152** can be lifted free of the set **1158** as shown in FIG. **20d**.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements **1200** to manage the tubing of the device in some manner as desired by a user. As shown by way of example, tube management elements **1204** and **1206** are shown in FIGS. **21a-21c**. As shown in FIGS. **21a-21b**, one or more loop control elements **1204** and **1206** can be provided to manage a length of tubing **1202** as desired by the user. Specifically, such tubing management can comprise one or more tubing pull ties **1204** and **1206**. The two ties **1204** and **1206** can be constructed having integral pull tabs **1210** and **1212**, respectively, and can be installed on a loop **1208** of tubing **1202** as shown. Pulling on the pump and set ends of the tubing **1202** causes additional slack of the loop **1208** to be released, and the loop of FIG. **21b** to contract as shown in FIG. **21a**. Pulling outward on the pull tabs **1210** and **1212** can then be used to cause the loop **1208** to expand, and to reduce slack in the tubing as shown in FIG. **21b**.

As shown in FIG. **21c**, one or more of the pull ties can further comprise a clip **1214** to attach the loop of tubing to a user, and which can be pressed in the direction of arrow **A** to release the clip and loop of tubing from a user's belt or other clothing.

Further, one or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to manage tubing connections of the device in some manner as desired by a user. As shown by way of example, a "breakaway" coupling and/or connecting engagement between tubes **1226** and **1240** is shown in FIGS. **22a-22c**. An advantage of the exemplary embodiments illustrated in FIGS. **22a-22c** is the provision of a secure, leak-free connection, that is easily assembled with a push connection action by the user, and which can disengage automatically when tubing interference occurs, without creating great degrees of tension. That is, the exemplary embodiments illustrated in FIGS. **22a-22c** preferably allow disengagement with minimal "pulling" of opposite tube ends such that the set or pump is not pulled free of the user.

Each exemplary embodiment illustrated in FIGS. **22a-22c** preferably operates as a push connection. The tubing pieces can have male and female ends in which a selected feature provides the desired degree of engagement. As shown in FIG. **22a**, the male end **1228** of tube **1226** can have a tapered shape, with a diameter and surface characteristics to serve as an in-line, sealable, press-fit with the female end **1234** of tube **1240**. In FIG. **22b**, the male end **1230** can include one or more projecting detents **1242** to provide a snap-fit with similar recessed detents provided in the female end **1236**. In yet another exemplary embodiment, the male and female ends **1232** and **1238** can further include magnets **1244**, arranged to provide a magnetic attraction between tubes (i.e., north/south magnetic arrangements). In yet other embodiments, one of the magnets can be replaced with a metal element (not shown) which results in a substantially similar attraction with the remaining magnet.

In each embodiment, the coupling and/or connecting engagement between tubes allows for easy disconnect and/or safety breakaway in the event of an accidental snag of the tubing. Further, any one of the exemplary embodiments can be further configured to provide an audible indication of disconnections, such as a "pop" sound or other alert.

Although FIGS. 22a-22c show coupling and/or connecting engagements between tubes 1226 and 1240, in yet other embodiments of the present invention, the features can be used for connections between tube and set, tube and reservoir, and tube and pump.

Still further, one or more of the exemplary embodiments of the present invention described herein can be further provided with other additional features or elements to manage the tubing of the device in some manner as desired by a user. As shown by way of example, a tube management element 1250 is shown in FIGS. 23a-23b. As shown in FIGS. 23a-23b, an elastic accordion element 1254 is provided to manage a length of tubing 1252 as desired by the user. The length of the tubing 1252 can be controlled by a back-and-forth looped portion 1256 of the tubing 1252, wrapped in the elastic binding 1254, as shown. The elastic binding 1254 can be provided with one or more stitched or otherwise created seams 1258, to hold and/or direct an entering and exiting segment of tubing to maintain proper operation of the elastic binding during operation. This elastic array of tubing can be expanded and contracted as necessary to allow for the ideal management of the tubing.

Still further, one or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to minimize the tubing of the device in some manner as desired by a user. Great lengths of tubing can often result in a number of undesirable problems. Accordingly, where it is possible to do so, an exemplary embodiment of the present invention provides a system 1275 which can be safely implemented using a short tube 1278 as shown in FIGS. 24a-24c.

The short tube 1278 provided between the set 1276 and the pump 1280 can be minimal in length, including no slack that may pose a danger of tangling or fouling. The short tube 1278 can also be short enough to allow for disconnection at the pump 1280, such as through any of the disconnecting means described above, leaving the short length of tubing still attached to the site. In an exemplary embodiment of the present invention, the short tube 1278 can be between 2 and 12 inches in length, and preferably between 3 and 9 inches in length and more preferably between 4 and 6 inches in length. The connection 1282 can be a quick-disconnect type, and can be configured to be released by pressing one or more buttons 1284 of the pump 1280 as indicated by the arrows A.

Still further, one or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements 1285 to secure, contain, and/or conceal the tubing of the device in some manner as desired by a user. As shown by way of example, a tube containment pouch 1288 is shown in FIG. 25a. The kit can include the pouch 1288 that permits excess tubing 1290 to be bundled up and stowed in such a way as to allow only the required length 1286 to be freed for use. The pouch 1288 can be constructed in a manner substantially similar to that of fabric bag and having a Velcro enclosure, but is not limited thereto. For example, another exemplary embodiment is shown in FIG. 25b in which a flexible part 1292 can be provided that permits the tubing 1286 to be easily inserted and stored out of the way. The flexible part 1292 can be constructed of any suitable material, such as rubber, and can be provided with a number of slotted openings 1294 and 1296, with varying slot 1298 dimensions to facilitate both tubing sizes, and ease of tubing insertion, as desired by the user. The part 1292 can be provided as a half-sphere to maximize both tube contain-

ment spacing while maintaining a manageable shape configuration. The pouch 1288 or containment part 1292 can be discarded with the tube.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to conceal, decorate, or alter the appearance of the device in some manner as desired by a user. As shown by way of example, an assembled device 1300 of FIG. 26a illustrates a number of such additional features or elements to conceal, decorate, or alter the appearance of the device. In the exemplary embodiment shown, the additional elements are not provided to conceal the device but rather, decorate or alter the appearance in a desired manner. Specifically, instead of providing means to conceal the set 1302 and tube 1304, the inserted set is decorated or ornamented with customizable additional parts 1306 and 1308. For example, as shown in the exploded view of FIGS. 26b and 26c, the set can be provided with an image element 1306 that can be secured to the set 1302 either with a snap fit or other friction-type, adhesive-type, or other attachment, or can be trapped against the set 1302 by a cover piece 1308. In such a manner, the set 1302 can be personalized with the image element 1306 trapped, protected, and/or further customized by the clear, snap-on cover piece 1308.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements to simplify gripping and user positioning of the inserter. Such an exemplary feature is shown in element 1325 of FIGS. 27a-27b. In the exemplary embodiment shown in FIGS. 27a-27b, an inserter 1326 can be provided with one or more finger loops for use in the placement of the device. In the embodiment shown, the finger loops 1328 allow for two-finger placement as illustrated in FIG. 27b. The set can be placed using such an insertion device 1326 with the two finger loops 1328 as shown, which enhance stability and precision of set placement. Although two completely encircled finger loops are shown, embodiments of the present invention are not limited thereto. In yet other embodiments of the present invention, a combination of complete and partial loops (i.e., spurs) can be used.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements for the use of site preparation. Such an exemplary feature is shown in element 1350 of FIG. 28. In the exemplary embodiment shown in FIG. 28, an inserter 1352, such as those described above, can be provided with a spray mechanism 1354 at an upper most point of the inserter which contains a small amount of site preparation contents. The spray mechanism can be provided with a finger actuator 1356 that can be used by a user to spray the contents upon a site for site preparation. The contents can comprise, but are not limited to, anesthetic, disinfectant, or a combination of both, and can be applied by spraying from a small reservoir (not shown) and outlet incorporated into the insertion device.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements for the provision and use of an advanced design infusion set. Such an exemplary advanced design infusion set is shown comprising, in part, elements 1360 of FIGS. 29a-29c. FIG. 29a is a cross-sectional view of the hub 1362 of the infusion set, in which the hub 1362 provides an annular fluid reservoir and/or fluid path 1364 which can significantly minimize the penetration distance of the tubeset connector into the hub while maintaining a sufficiently large tubeset connector geometry.

The infusion set, as with others described herein, can be provided for the continuous subcutaneous infusion of insulin or other medications. The infusion set comprises the hub 1362, which is affixed to a skin surface 1366 of a body with adhesive as described above, and further comprises a tubeset 1368 which creates and maintains a fluid path from an infusion pump and/or reservoir (not shown) and the hub 1362. The exemplary hub 1362 illustrated in FIG. 29a comprises a polyurethane catheter 1370 extending from the hub 1362 through the skin surface 1366 and into the subcutaneous tissue. The hub can be constructed of materials having a viscoelastic property, and which are flexible, such as thermoplastic elastomer (TPE), thermoplastic urethane (TPU), silicone, or similar materials. As with many of the exemplary embodiments of the present invention described herein, the hub 1362 is configured to have a low profile to minimize interference, such as catching on clothing or fixed objects such as doorframes or cabinets. The hub 1362 can be further configured as described herein to have one or more clear or transparent features to enable site visualization and inspection.

The central portion of the hub 1362 can comprise the annular space 1364 in fluid communication with the catheter 1370 lumen. The annular space 1364 can be configured in any number of ways to minimize deadspace within the hub 1362, yet allow insertion needle penetration and subsequent closure of the penetration site, and provide tubeset needle penetration from any number of rotational positions of the tubeset as described in greater detail below. An advantage of providing such a hub 1362 with an annular interior is the ability to minimize the penetration distance of the tubeset connector into the hub while maintaining a sufficiently large tubeset connector geometry. Yet another advantage is the ability to minimize the volume of insulin or other medication left in the hub fluid cavity.

In an exemplary embodiment of the present invention, the annular space 1364 comprises a shallow or narrow central portion such that an upper surface, to be penetrated by the insertion needle 1372, is close to the catheter 1370 lumen. A peripheral area of the annular space 1364 is provided with a wider space extending along the inner wall of the hub 1362. In doing so, the engagement with the tubeset connector is simplified, and further allows tubeset engagement at any rotational position.

The hub 1362 can be inserted through the skin surface 1366 and into the subcutaneous tissue via the insertion needle 1372 that extends through the top surface of the hub 1362, through the annular space 1364 and through the catheter 1370 lumen. After insertion into the skin, the insertion needle 1372 (and needle hub) can be withdrawn and the resulting hole in the hub top surface can be configured to self-seal due to the elastomeric qualities of the hub materials after setting. That is, the annular fluid space 1364 can reseal any insertion openings generated by the insertion needle 1372, and allow penetration by a tubeset connector needle as described in greater detail below.

The exemplary tubeset connector 1368 also comprises, at least in part, the same or similar construction materials. The tubeset connector 1368 can be constructed of a viscoelastic material and utilizes the material characteristics to create and maintain a connection with the hub 1362. The tubeset connector 1368 comprises an elasticized ring 1374 with an inner diameter, shape and profile that matches a central, recessed or grooved portion 1375 of the hub 1362, and a tube 1378. As shown in FIG. 29a, the central portion 1375 of the hub 1362 can comprise a groove or contour for guiding, receiving and securing the elasticized ring 1374. To do so,

the elasticized ring 1374 can be gently expanded by the user and placed into the central portion 1375 such that when released, the elasticized ring 1374 relaxes into a size and shape that secures the tubeset 1368 with the hub 1362.

At one location of the elasticized ring 1374 inner surface, the tubeset needle or spike 1376 is provided and protrudes radially inward. Accordingly, when the elasticized ring 1374 is placed onto the hub 1362 as described above, the tubeset needle 1376 punctures the wall of the central portion 1375 of the hub 1362 and thereby creates a fluid path from the tubeset 1368 to the annular cavity 1364 of the hub 1362. The elastomeric qualities of the hub materials allow the needle punctures to self seal upon needle removal. In doing so, the tubeset 1368, using the elasticized ring 1374 connector, can be affixed to the hub 1362 in any rotational alignment position. Further, the elasticity of the materials allows the elasticized ring 1374 to be removed from one position and repositioned elsewhere as desired, and each unused insertion site is sealed.

One or more of the exemplary embodiments of the present invention described herein can be further provided with additional features or elements for the provision and use of an advanced design catheter. Such an exemplary advanced design catheter is shown in element 1380 of FIG. 30. FIG. 30 is a cross-sectional view of a catheter including one or more cross-drilled holes and a splined lumen in accordance with an exemplary embodiment of the present invention. In the exemplary embodiment shown, the catheter 1380 can be constructed of a body temperature softening polyurethane or similar material, such as a proprietary Vialon material, and include one or more features including a splined lumen and holes or openings along a body length, including cross-drilled holes. For example, the catheter can comprise a splined lumen 1382 to prevent kinking, and flow occlusion in the event of a kinked catheter or otherwise. The splines 1382 can be provided as members raised from the inner wall of the catheter 1380 and/or provided as members depressed into the inner wall of the catheter 1380, to be uniformly spaced and to extend along the axis of the catheter. For example, the splines can comprise a linear axis to run parallel to the axis of the catheter and/or can comprise a helical axis, either right-handed or left-handed, to form a helix curve about the axis of the catheter. Although four equally-spaced splines 1382 are shown in FIG. 30, embodiments of the present invention are not limited thereto.

Additionally, the catheter 1380 can comprise one or more holes or openings, such as the cross-drilled holes 1384 shown in FIG. 30. As shown, the holes can be aligned as resulting from drilling, punching or otherwise molding through holes at points along the body of the catheter. Any number of openings or holes can be provided, either uniformly over the body of the catheter or in arrangements of varying concentrations, and can comprise openings of approximately 1/4 of the inner diameter of the catheter, but each are not limited thereto. The holes can be provided to facilitate infusion of the contents, such as insulin or other medication, in the subcutaneous tissue adjacent to the catheter in addition to, or instead of, the tissue at the catheter tip.

To this point, a number advanced, improved, and novel new components, elements and packaging of current and future insulin infusion sets, have been described. Accordingly, a number of advantages and improvements over existing systems and methods include the feature of providing the set and some or all of its peripherals in one package. Many of the proposed insertion devices both insert and retract the needle and then cover it for safe disposal. The described fluid connections from the pump to the site require

31

no moving parts and allow for easy disconnection. Other features allow the user to inspect the insertion area for irritation after insertion via a magnifying window in the site, and tube management is obtained by using a tube reel or other embodiment that allows the user to manage excessive tube length conveniently and adjust length as needed. The described insulin reservoir and tube are pre-filled and do not require priming by the user, and the concealment and decorative patches allow the user to cover and protect the site either inconspicuously or with distinction. The described site targeting ring allows the user to more accurately locate the site prior to insertion, and the disposable wipes and sprays combine disinfectant and anesthetic application steps into one step.

Although only a few exemplary embodiments of the present invention have been described in detail above, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of this invention. Accordingly, all such modifications are intended to be included within the scope of this invention.

The invention claimed is:

1. An infusion set, comprising:

- a set, comprising a skin contacting surface and a catheter extending from the set;
- a flexible ring having an integral fluid path therethrough, wherein the flexible ring is configured to expand to a first diameter to allow placement over the set, and contract to a second diameter to securely circumscribe an outer diameter of the set;
- a tubing length secured to the flexible ring; and
- a fitting or spike disposed at an end of the tubing length and extending into the flexible ring, the fitting or spike configured to sealably mate with an opening of the set to permit communication of a medicament therebetween.

32

2. The infusion set of claim 1, wherein the set comprises a hub constructed of a viscoelastic material, wherein the fitting comprises a cannula configured to sealably pierce the hub.

3. The infusion set of claim 2, wherein the viscoelastic material comprises at least one of a thermoplastic elastomer (TPE), thermoplastic urethane (TPU), or silicone.

4. The infusion set of claim 1, further comprising at least one tubing connection, the at least one tubing connection comprising at least one of a tapered tubing end, a tubing end having one or more detents, and a magnetically attractable tubing end.

5. The infusion set of claim 4, wherein the at least one tubing connection is configured to generate an audible indication when disconnected.

6. The infusion set of claim 1, further comprising a medicament supply, the medicament supply comprising at least one medicament container, a tubing segment and a tubing connection connectable to a pump that is connectable to an opposing end of the tubing length, wherein the medicament container is configured to indicate a content quantity therein.

7. The infusion set of claim 1, wherein the catheter comprises a splined lumen, wherein the splined lumen comprises a helical axis.

8. The infusion set of claim 1, wherein the catheter comprises at least one hole or opening along a body length thereof.

9. The infusion set of claim 1, wherein the catheter is constructed of a body temperature softening polyurethane.

10. The infusion set of claim 1, wherein the set comprises at least one push button element, wherein operation of the at least one push button element is configured to release a tube connection from the set.

11. The infusion set of claim 1, wherein the catheter comprises a splined lumen, wherein the splined lumen comprises a linear axis.

12. The infusion set of claim 1, wherein the flexible ring is hollow.

* * * * *